

A PRISMATIC RIEMANN-HILBERT CORRESPONDENCE

DYLAN PENTLAND

ABSTRACT. Let p be an odd prime, let k be a perfect field of characteristic p , and let X be a smooth quasicompact p -adic formal scheme over $W(k)$ equipped with the log structure $\mathcal{M}$ induced by a horizontal normal crossings divisor D . We show that there is an equivalence of categories between constructible p -Kummer-étale $\mathbf{Z}_p$ -sheaves on the log adic generic fiber $(X, \mathcal{M})_\eta$ for the logarithmic stratification induced by D and certain finitely generated unit Laurent crystals on the log prismatic site. We verify p^+ -perverse t -exactness, give a local description of finitely generated unit crystals on a framed log affine in terms of (φ, Γ) -modules, and show pushforward compatibility for strict log smooth proper morphisms.

CONTENTS

1. Introduction	2
2. The Emerton-Kisin correspondence	8
3. Coherent Laurent crystals	13
3.1. Laurent crystals and Γ -modules	15
3.2. Roots in Laurent crystals	26
4. Finitely generated unit Laurent crystals	34
4.1. Basic constructions	34
4.2. Strict-étale descent	39
4.3. The Kummer-étale realization and constructibility	43
5. The main equivalence	50
5.1. The open stratum	51
5.2. Proof of the main theorem	61
6. Pushforward compatibility	75
6.1. Six functors and Laurent stacks	75
6.2. Pushforward compatibility	82
Appendix A. Log prismatic stacks	86
A.1. The log prismatic stack	86
A.2. Quasicoherent sheaves on the log prismatic stack	88
A.3. Solid Laurent crystals	94
A.4. Analytifications	100
References	103

1. INTRODUCTION

Let k be a perfect field of characteristic p . For smooth k -schemes, Emerton and Kisin gave a contravariant Riemann-Hilbert correspondence describing constructible $\mathbf{F}_p$ sheaves using finitely generated unit Frobenius modules in [EK04b]. Their result is a contravariant generalization of Katz's correspondence in [Kat73], which describes the full subcategory of étale local systems.

In mixed characteristic, Katz's theorem plays an important role in describing local systems on the adic generic fiber of a p -adic formal scheme using (φ, Γ) modules. This idea is originally due to Fontaine in [Fon90], and later generalized by Andreatta-Iovita in [AI08]. Thanks to Kedlaya-Liu in [KL16, Theorem 7.5.4], this theory also admits a logarithmic variant describing what are now called p -Kummer-étale local systems on the log adic generic fiber (see Definition 4.20).¹

In this article we give a Riemann-Hilbert correspondence in mixed characteristic extending these results to describe constructible sheaves on a fixed normal crossings stratification. Roughly speaking, our construction uses the Emerton-Kisin correspondence in the role played by Katz's correspondence in the theory of (φ, Γ) -modules.

For technical reasons, we must work logarithmically. In our setup, p is odd and $X/\mathrm{Spf} W(k)$ is a smooth quasicompact p -adic formal scheme, with log structure $\mathcal{M} = \mathcal{M}_D$ induced by a horizontal normal crossings divisor D . Letting $\rho(\bar{x}) = \mathrm{rk}_{\mathbf{Z}} \overline{\mathcal{M}}_{\bar{x}}^{\mathrm{gp}}$ for geometric points $\bar{x}$ of the adic generic fiber X_η , the induced logarithmic strata on the log adic generic fiber $(X, \mathcal{M})_\eta$ are the connected components of the locally closed loci $X_\eta^{(a)}$ where $\rho = a$.² We write $D_{\mathrm{cons}, D}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$ for Kummer-étale complexes constructible along these strata. As in the correspondences for Kummer-étale local systems of [IKY26] and [KL16, Theorem 7.5.4], objects $\mathcal{F}$ are p -Kummer when prime-to- p log inertia $I_{\bar{x}}^{\log, (p')}$ acts trivially on $\mathcal{F}_{\bar{x}}$ at every log geometric point $\bar{x}$ (Definition 4.20).

We let $(X, \mathcal{M})_\Delta$ denote the absolute log prismatic site of $(X, \mathcal{M})$, and $\mathcal{O}_\Delta[1/I_\Delta]_p^\wedge$ the sheaf assigning to a log prism $(A, I, M_{\mathrm{Spf} A})^a$ the ring $A[1/I]_p^\wedge$ as in [IKY26]. In Definition 4.3 we introduce the category $D_{\mathrm{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ of finitely generated unit Laurent crystals, which are certain crystals $\mathcal{E}$ over $\mathcal{O}_\Delta[1/I_\Delta]_p^\wedge$ equipped with an isomorphism $\varphi^* \mathcal{E} \simeq \mathcal{E}$ and a finiteness condition similar to [EK04b]. The main result of this article is then the following.

Theorem A (Theorem 5.26, Corollary 5.30). *Consider $(X, \mathcal{M}_D)$ with X smooth and quasicompact over $\mathrm{Spf} W(k)$ as above. There is a fully faithful symmetric monoidal functor*

$$T_{\mathrm{ét}} : D_{\mathrm{fgu}}((X, \mathcal{M}_D)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\mathrm{op}} \longrightarrow D_{\mathrm{cons}, D}^{(b)}((X, \mathcal{M}_D)_\eta, \mathbf{Z}_p)$$

¹See also [AB26].

²Strict-étale locally, with branches $D_1, \dots, D_r$, these are the connected components of $X_{\eta, \Lambda} = (\bigcap_{i \in \Lambda} (D_i)_\eta) \setminus (\bigcup_{j \notin \Lambda} (D_j)_\eta)$ for $\Lambda \subseteq \{1, \dots, r\}$, with $X_{\eta, \emptyset} = X_\eta \setminus D_\eta$.

which is an equivalence onto the full subcategory of p -Kummer complexes. If X_η is pure of dimension d , then $T_{\text{ét}}[d]$ identifies the standard t -structure on the source with the p^+ -perverse t -structure on its essential image.

One does not need to check on all prisms to verify the finitely generated unit condition. It in fact suffices to check on a covering family of logarithmic q -de Rham prisms, as in Remark 4.13. The theorem also holds over $W(k)[\zeta_{p^\ell}]$ for $\ell \geq 0$, as explained in Corollary 5.30.

One can also study ordinary constructible sheaves within this correspondence. Indeed there is a morphism of sites

$$\epsilon : (X, \mathcal{M})_{\eta, \text{prokét}} \longrightarrow X_{\eta, \text{proét}}$$

whose pullback ϵ^* is fully faithful on constructible sheaves and lands in the p -Kummer sheaves in the heart of $D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$. Thus ordinary constructible sheaves (for the stratification induced by D) form a full subcategory at the level of hearts. See Remark 5.29 for the corresponding condition on the crystal side.

Finitely generated unit Laurent crystals also admit an explicit local description in terms of (φ, Γ) -modules. We state the variant over $W(k)$, although we can handle $W(k)[\zeta_{p^\ell}]$ as well. In what follows a *framed log affine* is a $W(k)$ -algebra R that is p -completely étale over $W(k)\langle T_1, \dots, T_r, T_{r+1}^\pm, \dots, T_n^\pm \rangle$ with log structure $\mathcal{M}$ induced by $D = V(T_1 \dots T_r)$.

Theorem B (Variant of Theorem 4.6). *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine, with framing denoted by $\square$. Evaluation on the log q -de Rham prism $(R^\square[[q-1]], [p]_q, \mathbf{N}^r)^a$ over R induces an equivalence*

$$D_{\text{fgu}}((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \simeq D_{\text{fgu}}(R^\square((q-1)_p^\wedge)^{h\Gamma},$$

where $(a, b_1, \dots, b_n) \in \Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)^n$ acts by $q \mapsto q^a$ and $T_i \mapsto q^{b_i} T_i$, and “fgu” or finitely generated unit means that the complex has a unit Frobenius and is perfect over $R^\square((q-1)_p^\wedge)\langle F \rangle$.

We also use this to make $T_{\text{ét}}$ completely explicit locally in Lemma 4.27 (although in principle the definition in Definition 4.18 is already computable). In this formalism it is relatively straightforward to write down some simple examples. In the case of $\text{Spf } R = \text{Spf } W(k)\langle T \rangle$ with $\Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)$ and log structure $\mathcal{M}_{V(T)}$, we describe the objects corresponding to $j_! \mathbf{Z}_p$ and $i_* \mathbf{Z}_p$ under Theorem 4.6. Here j is the inclusion of the punctured disk $\mathbb{D}^\times \rightarrow (X, \mathcal{M}_{V(T)})_\eta$, and i the complementary inclusion of the origin (with its pullback log structure).

In the notation of Theorem B, $M = R^\square((q-1)_p^\wedge)[1/T]_p^\wedge$ corresponds to $j_! \mathbf{Z}_p$. We equip it with an action by $\Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)$ where $q \mapsto q^x$ for $x \in \mathbf{Z}_p^\times$ and $T \mapsto q^{p^x} T$ for $x \in \mathbf{Z}_p(1)$. Moreover it has a unit Frobenius, sending $T \mapsto T^p$ and $q \mapsto q^p$. The (φ, Γ) -module $R^\square((q-1)_p^\wedge)[1/T]_p^\wedge / R^\square((q-1)_p^\wedge)$ corresponds to $i_* \mathbf{Z}_p[-1]$ (for the unshifted version of $T_{\text{ét}}$). In Example 5.32, we give a more detailed calculation corresponding to a constructible sheaf (which has a nontrivial gluing datum between strata).

Our third main result is pushforward compatibility.

Theorem C (Theorem 6.11). *Let $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{N})$ be a strict log smooth proper morphism of pure relative dimension d between smooth quascompact p -adic formal schemes over $W(k)[\zeta_{p^\ell}]$ for some $\ell \geq 0$, and let D_X and D_Y be horizontal normal crossings divisors inducing $\mathcal{M}$ and $\mathcal{N}$. Suppose the adic generic fibers have pure dimensions d_X and d_Y . Thus $d = d_X - d_Y$. There is a commutative diagram*

$$\begin{array}{ccc} D_{\text{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\text{op}} & \xrightarrow{T_{\text{ét}, X}[d_X]} & D_{\text{cons}, D_X}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p) \\ \downarrow Rf_*\{d\}[d] & & \downarrow Rf_{\eta,*} \\ D_{\text{fgu}}((Y, \mathcal{N})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\text{op}} & \xrightarrow{T_{\text{ét}, Y}[d_Y]} & D_{\text{cons}, D_Y}^{(b)}((Y, \mathcal{N})_\eta, \mathbf{Z}_p). \end{array}$$

The shift and twist arise from a logarithmic Poincaré duality result for perfect Laurent crystals proved in Corollary 6.7. For the Laurent F-crystal equivalences in either [BS23] or [IKY26], the analogous pushforward compatibility theorem does not appear in the literature. The problem is that it is not obvious that the right adjoint Rf_* on Laurent crystals preserves $\text{Perf}(X_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1}$. For this reason existing pushforward compatibility results (such as the result stated in [GL25]) assume that the Laurent F-crystal arises from a prismatic F-crystal, as in that situation perfectness can be deduced from the Hodge-Tate comparison. As Theorem 6.11 is compatible with a contravariant version of [IKY26], we also obtain pushforward compatibility for this correspondence (as well as the one in [BS23, Corollary 3.7] as a degenerate case).

There are also other compatibilities we prove through more straightforward arguments. We show pullback compatibility in Corollary 4.16, and Theorem 5.4 gives compatibility with $j_!$ where j is the inclusion of $X_\eta \setminus D_\eta$. Locally, through Kashiwara's lemma (Lemma 5.18) we also give compatibility with i_* and $i^!$. Finally, due to the cohomological smoothness results in [Kip26a] we also have compatibility with $f^!$ for strict log smooth proper maps of pure relative dimension d by combining pushforward compatibility with $f^! \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge \simeq \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge \{d\}[2d]$ from Lemma 6.4.

Related work. Bhatt and Lurie construct a covariant p -adic Riemann-Hilbert functor, described in [Bha25, §1.1(4) and Theorem 6.3.2]. Let $K/\mathbf{Q}_p$ be a nonarchimedean field of finite p -cohomological dimension. For a p -adic formal scheme X locally of finite type over $\mathcal{O}_K$, it takes the form

$$\text{RH}_\Delta : D_{p\text{-comp}}((X_\eta)_{\text{ét}}, \mathbf{Z}_p)^{\text{oc}} \longrightarrow D(X^{\Delta, \text{pfd}})^{a, \varphi=1}.$$

Here the source consists of derived p -complete overconvergent sheaves, $X^{\Delta, \text{pfd}}$ is the perfect prismatization, and ' a ' denotes passage to almost coefficients. Their functor is fully faithful, symmetric monoidal, and identifies the dualizable subcategories. It is moreover compatible with the positive characteristic correspondence of [BL19] as explained in [Bha25, Theorem 6.2.3]. They also give an algebraic variant without almost coefficients in [Bha25, Variant 1.1.9]. When K is perfectoid, for Zariski-constructible inputs Bhatt and Lurie prove almost coherence and proper pushforward compatibility results, as well as perverse t -exactness with finite coefficients [Bha25, Theorem 7.3.6, Corollary 7.3.7, and Theorem 10.3.4].

Mann constructs in [Man22, Theorem 3.9.23] a related fully faithful functor from overconvergent étale $\mathbf{F}_p$ -complexes on small v -stacks to solid almost $\mathcal{O}^+/\pi$ -modules with unit Frobenius, where $\pi \mid p$ is a pseudouniformizer. It identifies the perfect subcategories and forms part of a six-functor formalism yielding proper pushforward comparisons (under suitable hypotheses) and Poincaré duality in [Man22, Corollary 3.9.24 and Theorem 3.10.20].

Our correspondence is less general, but has two advantages. The first is that we can locally formulate the crystal side of the correspondence as a generalization of (φ, Γ) -modules, which is then amenable to computation as we work over a deperfection base (see Example 5.32). The second is that the functor is an *equivalence* onto the full subcategory of p -Kummer constructible sheaves, so there is a clear and easily verifiable condition on the crystal side corresponding to constructibility. In [Bha25] there is not a clear (non-tautological) condition describing the essential image of constructible sheaves. In [Man22] it seems similarly difficult to extract such a condition, as the correspondence is obtained by hypercomplete v -descent from strictly totally disconnected perfectoid spaces.

Overview of the proofs. All arguments work first over $W(k)[\zeta_p]$, and then we extend to other bases. Once strict-étale descent is known, the argument for Theorem A is essentially module-theoretic as we can localize to a situation where we deal only with finitely generated unit (φ, Γ) modules as in Theorem B. We produce the equivalence $T_{\text{ét}}$ globally, but in Lemma 4.27 we give a computable local description. The main equivalence in Theorem A is built first on the open stratum $U_\eta = X_\eta \setminus D_\eta$. For essential surjectivity on the open stratum, one uses Lütkebohmert's theorem to extend a local system on U_η to a Kummer-étale local system $\mathbb{L}$ on X_η . Letting $j : U_\eta \rightarrow X_\eta$ be the inclusion, we construct a crystal $\mathcal{O}_\Delta(*D)$ corresponding to $j! \mathbf{Z}_p$. We show that after a finite prime-to- p Kummer cover $(X_N)_\eta$ we may make $\mathbb{L}$ a p -Kummer local system, and thus [IKY26] provides a logarithmic Laurent crystal $\mathcal{E}$. Keeping track of the descent data in the form of a μ_N^r -action, one may descend $\mathcal{E}(*D)$ to construct a preimage. Full faithfulness uses as its main component a more delicate decompletion statement that base change to the perfection is fully faithful; Remark 5.13 explains why this does not follow from Frobenius alone. We must also understand some full faithfulness results for analytification on perfectoid spaces, where we use results from [Bha25].

Closed strata with their pullback log structures are handled by a variant of Kashiwara's lemma (Lemma 5.18), which reduces to the corresponding result of [EK04b]. A dévissage argument using Kashiwara's lemma reduces the main theorem to the open and closed comparisons on strata, along with some (derived) full faithfulness results about maps between strata. The essential image is identified because extensions by zero of p -Kummer local systems on the strata generate the p -Kummer constructible subcategory. One must also verify $T_{\text{ét}}$ is constructible for the stratification along D ; again using the local formulation in Theorem B, a Fitting ideal argument controls the possible roots (and proves constructibility along D). Once all results are shown over $W(k)[\zeta_p]$, with some modifications we can work over $W(k)[\zeta_{p^\ell}]$ for $\ell \geq 1$ and we descend to $W(k)$.

To pass to the global version one must show it locally recovers the explicit description of Theorem 4.6 in terms of (φ, Γ) -modules. This is handled in §3. The group $\Gamma = (1 + p\mathbf{Z}_p) \rtimes \mathbf{Z}_p(1)^n$ acts on the log q -de Rham prism by $q \mapsto q^a$ and $T_i \mapsto q^{pb_i} T_i$. Proposition 3.12 shows that

evaluation on this prism yields a fully faithful functor into semilinear Γ -modules, which is the functor inducing the local equivalence. For essential surjectivity in Theorem 4.6 one uses roots to present finitely generated unit (φ, Γ) -modules in the target. A coherent root of a unit Frobenius module M is a coherent submodule $M_0 \subset M$ equipped with an injection $M_0 \rightarrow \varphi^* M_0$ so $M \simeq \operatorname{colim} \varphi^{*n} M_0$; it is minimal if it contains no proper root. Modulo p , [Bli08] gives a unique minimal coherent root in the sense of Definition 4.5 in our setting. Minimality forces this to be Γ -equivariant, and we use Theorem 3.5 to carry a Γ -equivariant root over $R_0^\square((q-1))_p^\wedge$ to a coherent Laurent crystal. This shows essential surjectivity modulo p , which is sufficient for the argument.

For strict-étale descent of $D_{\text{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$, it is convenient to have an ambient category satisfying strict-étale descent. We work with a solid variant $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ of Laurent crystals, explained in Definition A.11. The appendix proves that the logarithmic Breuil-Kisin atlases are completely descendable in the sense of [Kip26b, Definition 2.4.1.7]. Consequently, ordinary solid Laurent crystals are computed by the Čech nerve of the log q -de Rham prism (which is important for computation). We also show strict-étale descent using similar methods. One must then show that the finitely generated unit condition itself is strict-étale local in $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1}$. We do this by again using roots: the main point is to promote these minimal roots *as modules* from individual log q -de Rham evaluations to minimal roots in *coherent Laurent crystals* on framed log affines. Once this is done, their uniqueness as subcrystals makes them glue and proves the required descent (since ordinary coherent Laurent crystals do form a stack, using [Mat22]).

Finally, Theorem C first uses results in [Kip26b] and [Kip26a] to construct a stacky approach to these solid Laurent crystals and equip them with a suitable six functor formalism. The existence of this six functor formalism helps greatly in proving pushforward compatibility (for example by showing $Rf_* \simeq Rf_!$ is compatible with colimits, or showing strict log smooth proper maps preserve perfect Laurent crystals) but does not make the conclusion automatic. The idea is to show that the pushforward preserves the finitely generated unit condition by producing a variant of a root (using a perfect complex) and then formally deducing pushforward compatibility by adjunction.

Notation. Unless otherwise stated, all categories are ∞ -categories, $(X, \mathcal{M})$ denotes a smooth quasicompact log p -adic formal scheme over $\operatorname{Spf} W(k)[\zeta_{p^\ell}]$ for $\ell \geq 0$ equipped with the log structure coming from a horizontal normal crossings divisor D . We use the logarithmic stratification by the rank of $\overline{\mathcal{M}}^{\text{gp}}$. On a framed strict-étale chart, $X_{\eta, \Lambda}$ denotes the coordinate stratum on which exactly the boundary coordinates indexed by Λ vanish. We use the pro-Kummer-étale site of [DLLZ23, §5.1], and the corresponding convention for the ordinary pro-étale site. We write $D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$ for the locally bounded derived p -complete complexes on this site whose reductions modulo p^n , for every $n \geq 1$, are pulled back from Kummer-étale complexes constructible for the logarithmic stratification. We use the analogous convention for ordinary pro-étale constructible complexes. The p -Kummer sheaves form a full subcategory; see Definition 4.20.

We denote by $(X, \mathcal{M})_\Delta$ the absolute log prismatic site. In Remark A.10 we explain that the crystal categories we care about are agnostic to various modifications to this definition, but for the sake of choosing a convention we use the strict log prismatic site. Our convention is that we use the opposite category of log prisms over $(X, \mathcal{M})$ (in [BS22] this notation means the category of prisms, and they take opposites when regarding it as a site). We use the term ‘Laurent’ or ‘Laurentization’ to refer generally to the passage from a prism (or log prism) (A, I) to $A[1/I]_p^\wedge$.

We write $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ for the category of Laurent crystals with ordinary solid coefficients and derived p -complete coefficient rings. We write $D_{\text{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ for its subcategory of finitely generated unit Laurent crystals as in Definition 4.3; their evaluations are derived p -complete. The functor $T_{\text{ét}}$ is the (Kummer)-étale realization functor, while RSol denotes the crystalline Emerton-Kisin solution functor in characteristic p .

We use $(X, \mathcal{M})^\Delta$ for Olsson’s log prismatic stack (see Appendix A). We reserve $(X, \mathcal{M})^{\Delta, \blacksquare}$ and $(X, \mathcal{M})^{\mathcal{L}, \blacksquare}$ for the prismatic and Laurent analytic stacks derived from the construction of [Kip26b]; our conventions on analytic stacks are the same as [Kip26b], as analytic stacks only enter from Kipp’s construction of $X^{\Delta, \blacksquare}$. We will frequently use Spa for the corresponding analytic stack associated to a Huber pair, although when the analytic ring structure is clear from context we also use AnSpec .

On a framed log affine $(\text{Spf } R, \mathcal{M})$ over $W(k)[\zeta_{p^\ell}]$ as in Definition 3.2 with framing $\square$ and boundary coordinates $T_1, \dots, T_r$, let R_0 be the $W(k)$ -algebra in the framing, so that $R \simeq R_0 \widehat{\otimes}_{W(k)} W(k)[\zeta_{p^\ell}]$. For $\ell \geq 1$, the attached log q -de Rham prism is $(R_0^\square[[q-1]], \varphi^{\ell-1}([p]_q), \mathbf{N}^r)^a$, whose quotient by the prismatic ideal is R . The group $\Gamma = (1 + p^\ell \mathbf{Z}_p) \rtimes \mathbf{Z}_p(1)^n$ acts by $q \mapsto q^a$ and $T_i \mapsto q^{p^\ell b_i} T_i$. For $\ell = 0$, we take $R_0 = R$ and use the prism $(R^\square[[q-1]], [p]_q, \mathbf{N}^r)^a$ over R , whose quotient is $R \widehat{\otimes}_{W(k)} W(k)[\zeta_p]$; here $\Gamma = \mathbf{Z}_p^\times \rtimes \mathbf{Z}_p(1)^n$ acts by $q \mapsto q^a$ and $T_i \mapsto q^{p b_i} T_i$. In all cases, $R_0^\square[[q-1]] = R_0[[q-1]]$, and inverting the prismatic ideal and p -completing gives $A = R_0^\square((q-1))_p^\wedge$. The superscript $\square$ records the framing (inducing the Γ action), and $\bar{A}$ denotes reduction modulo p . For Γ -modules, we use $h\Gamma$ on a category to indicate a continuous cobar construction; see the start of §3.2 and the discussion preceding Theorem 4.6.

We write $\text{RHom}(-, -)$ for mapping complexes and $\underline{\text{RHom}}(-, -)$ for internal derived Hom.

We use $//$ for derived reduction: $M // f := \text{cofib}(f : M \rightarrow M)$, and $M // (f_1, \dots, f_r)$ denotes the iterated cofiber. Single slashes denote underived quotients.

Acknowledgements. The author would like to thank Mark Kisin for suggesting this problem to the author and for his guidance throughout the project. The author would also like to thank Oakley Edens, Daishi Kiyohara, and Kush Singhal for helpful conversations.

AI Use Disclosure. AI tools were used for literature search, proofreading this manuscript for typographical and mathematical errors, and assisting with some of the typesetting. AI tools assisted in technical details in several lemmas (not the overall strategies), particularly

in Lemma 3.11 and the helper lemmas for Proposition 5.9, and simplifying the author's first counterexample in Remark 5.13 (by finding a less complicated Artin-Schreier class).

2. THE EMERTON-KISIN CORRESPONDENCE

In this section, we review the Emerton-Kisin correspondence from [EK04b].

The story naturally begins with Katz's correspondence, which the Emerton-Kisin correspondence generalizes to constructible sheaves. The original statement of this result is [Kat73, Proposition 4.1.1], allowing the generality of a regular $\mathbf{F}_p$ -algebra R , but for our purposes we will be most interested in the case where R is a perfect $\mathbf{F}_p$ -algebra. In this setting the theorem appears in [KL15, Proposition 3.2.7] and [KL15, Theorem 8.5.3] (once we pass to the limit). However we will also need a derived variant, which appears to not have a direct reference but is a formal extension of existing results. We also take the opportunity to introduce a dévissage method for idempotent complete categories which we will use frequently in what follows.

We let $D_{\text{lis}}^{(b)}(\text{Spec } R, \mathbf{Z}_p)$ denote locally bounded derived p -complete complexes in $D((\text{Spec } R)_{\text{proét}}, \mathbf{Z}_p)$ such that each cohomology sheaf of the derived mod p reduction is lisse in the usual sense.

Theorem 2.1 (Katz). *Let R be a perfect $\mathbf{F}_p$ -algebra. Then there is an equivalence of categories*

$$\mathbf{Perf}(W(R))^{\varphi=1} \simeq D_{\text{lis}}^{(b)}(\text{Spec } R, \mathbf{Z}_p),$$

induced by the solution functor

$$M \longmapsto (M \otimes_{W(R)} W(\mathcal{O}_{\text{Spec } R, \text{proét}}))^{\varphi=1}.$$

Proof. We apply Lemma 2.3. We first verify the hypotheses. The solution functor is easily checked to be exact and $\mathbf{Perf}(\mathbf{Z}_p)$ -linear, as well as compatible with derived reduction modulo p^n (using perfectness). Both categories are idempotent complete, since their defining conditions are preserved under retracts. The source is presented as $\lim_n \mathbf{Perf}(W_n(R))^{\varphi=1}$ and the target as a limit of categories of lisse complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$, i.e. $D_{\text{lis}}^{(b)}(\text{Spec } R, \mathbf{Z}/p^n)$. The mod p functor is the equivalence of [BS23, Proposition 3.4], so Lemma 2.3 proves the result. $\square$

Remark 2.2. The Emerton-Kisin correspondence is contravariant. Since $\mathbf{Perf}$ is self-dual, the previous correspondence also admits a dual or contravariant version. The solution functor is $M \mapsto \mathbf{RHom}_{\mathbf{Perf}^\varphi(W(R)_{\text{proét}})}(M_{\text{proét}}, W(R)_{\text{proét}})$ (regarded as a proétale sheaf). Similarly other correspondences reducing to Katz's theorem, such as [BS23, Corollary 3.7] or [IKY26] admit contravariant versions.

The dévissage method we have just applied is a categorified version of Nakayama's lemma. For an object X of a $\mathbf{Perf}(\mathbf{Z}_p)$ -linear stable ∞ -category, write $X//p^n := \text{cofib}(p^n : X \rightarrow X)$. Recall a stable ∞ -category $\mathcal{C}$ is idempotent complete when $\mathcal{C}$ is closed under retracts in $\mathbf{Ind}(\mathcal{C})$.

Setup. Let $\mathcal{C}$ be an idempotent complete $\mathbf{Perf}(\mathbf{Z}_p)$ -linear stable ∞ -category. All tensor products of categories below are taken among idempotent complete stable ∞ -categories. Put

$$\mathcal{C}_n := \mathcal{C} \otimes_{\mathbf{Perf}(\mathbf{Z}_p)} \mathbf{Perf}(\mathbf{Z}/p^n).$$

We say that $\mathcal{C}$ is derived p -complete if the natural functor $\mathcal{C} \rightarrow \varprojlim_n \mathcal{C}_n$ is an equivalence. Write $\mathrm{red}_n : \mathcal{C} \rightarrow \mathcal{C}_n$ for reduction and $X_n = \mathrm{red}_n X$. Since $\mathbf{Perf}(\mathbf{Z}/p^n)$ is thickly generated by its unit, $\mathcal{C}_n$ is thickly generated by the objects X_n , and

$$\mathrm{Map}_{\mathcal{C}_n}(X_n, Y_n) \simeq \mathrm{Map}_{\mathcal{C}}(X, Y // p^n).$$

We ask that functors between derived p -complete categories $F : \mathcal{C} \rightarrow \mathcal{D}$ in this setup induce compatible functors $F_n : \mathcal{C}_n \rightarrow \mathcal{D}_n$.

Lemma 2.3. *Let $\mathcal{C}$ and $\mathcal{D}$ be derived p -complete idempotent complete $\mathbf{Perf}(\mathbf{Z}_p)$ -linear stable ∞ -categories. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact $\mathbf{Perf}(\mathbf{Z}_p)$ -linear functor in the setup above. If F_1 is an equivalence, then F is an equivalence.*

Proof. Full faithfulness is clear, using derived Nakayama on mapping complexes. Indeed,

$$\mathrm{Map}_{\mathcal{C}_1}(X_1, Y_1) \simeq \mathrm{Map}_{\mathcal{C}}(X, Y // p) \simeq \mathrm{Map}_{\mathcal{C}}(X, Y) \otimes_{\mathbf{Z}_p}^{\mathbf{L}} \mathbf{F}_p.$$

Note that by our setup $\mathrm{Map}_{\mathcal{C}}(X, Y)$ is derived p -complete. Note that similarly one gets full faithfulness of F_n as we will need later in the argument.

For essential surjectivity, it suffices to show the F_n are essentially surjective; we know this for $n = 1$. Let $Y_n \in \mathcal{D}_n$, and consider the fiber sequence

$$Y_n // p \longrightarrow Y_n // p^n \longrightarrow Y_n // p^{n-1}.$$

This shows $Y_n // p^n$ belongs to the thick subcategory generated by $Y_n // p$. One can show $Y_n // p$ is in the essential image of F_n because F_1 is essentially surjective, and the essential image is a thick subcategory. There is a minor issue here, which is that $Y_n // p \in \mathcal{D}_n$ and not $\mathcal{D}_1$ despite the notation. However since $\mathcal{D}_n$ is thickly generated by reductions it suffices to consider the case when Y_n is the reduction of $Y \in \mathcal{D}$. We may choose a preimage for $\mathrm{red}_1 Y \in \mathcal{D}_1$, and applying restriction of scalars to regard this preimage in $\mathcal{C}$ one obtains a preimage for $Y // p$; reducing we get the claim.

Thus $Y_n // p^n$ is in the essential image of F_n , so denote a choice of preimage by X_n .

Because we take the derived quotient by p^n , since $p^n = 0$ in $\mathcal{D}_n$ one has $Y_n // p^n = \mathrm{cofib}(0 : Y_n \rightarrow Y_n) = Y_n \oplus Y_n[1]$. So in particular Y_n is a retract of $Y_n // p^n$. But then idempotent completeness implies that Y_n itself is in the essential image of F_n . Indeed we have

$$Y_n \xrightarrow{i} Y_n // p^n \xrightarrow{r} Y_n$$

where ri is the identity; in particular ir is an idempotent in $\mathrm{End}_{\mathcal{D}_n}(F_n(X_n))$. By derived full faithfulness, it is an image of an idempotent e in $\mathrm{End}_{\mathcal{C}_n}(X_n)$ which splits as

$$X'_n \xrightarrow{i'} X_n \xrightarrow{r'} X'_n$$

where $r'i' = \text{id}_{X'_n}$ and $i'r' = e$. Then necessarily $F_n(X'_n) \simeq Y_n$, and the F_n are all essentially surjective. Thus every F_n is an equivalence, and derived p -completeness implies that F is an equivalence. $\square$

Katz's theorem is crucial in proving the equivalence

$$\text{Perf}(X_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1} \simeq D_{\text{lis}}^{(b)}(X_\eta^\diamond)$$

for bounded p -adic formal schemes X in [BS23]. Precisely, this means that when $X = \text{Spf } R$ is perfectoid, the result reduces to Katz's equivalence; their theorem is deduced by a reduction to this case. Our result for finitely generated unit Laurent crystals will be compatible in a similar way with the Emerton-Kisin correspondence for constructible sheaves.

We now extend the Emerton-Kisin correspondence of [EK04b] to $\mathbf{Z}_p$ coefficients, deduced from the mod p theorem by Lemma 2.3. We fix a perfect field k of characteristic p . If $\mathcal{X} / \text{Spf } W(k)$ is a smooth formal scheme, we let $D_{\mathcal{X}}$ denote the sheaf of infinitesimal differential operators. Locally, if $X \rightarrow \mathbf{A}_k^n$ is étale, this can be pulled back from the sheaf associated to $D_{\text{Spf } W(k)\langle T_1, \dots, T_n, \partial_1^{[i_1]}, \dots, \partial_n^{[i_n]} \rangle} = W(k)\langle T_1, \dots, T_n, \partial_1^{[i_1]}, \dots, \partial_n^{[i_n]} \rangle$, where all ∂_i are given full divided powers. We put

$$D_{F, \mathcal{X}} := \widehat{\bigoplus_{i \geq 0} F^{*i} D_{\mathcal{X}}}.$$

Modules over this noncommutative sheaf are infinitesimal crystals $\mathcal{E}$ with semilinear Frobenius $\varphi_{\mathcal{E}}$. A certain Taylor formula comparing Frobenius lifts makes $D_{F, \mathcal{X}}$ canonical and global; see [EK04a, §§6.1-6.2].

The Emerton-Kisin correspondence is compatible with a perverse t -structure on the étale side, which we now introduce.

Definition 2.4. Write $({}^p D^{\leq 0}, {}^p D^{\geq 0})$ for Gabber's perverse t -structure on $D_{\text{cons}}^{(b)}(X, \mathbf{Z}_p)$ [Gab04] (or rather its natural extension to $\mathbf{Z}_p$ coefficients), and write ${}^p H^i$ for perverse cohomology. We may then construct the p^+ -perverse t -structure

$${}^{p^+} D^{\leq 0} = \{K \in {}^p D^{\leq 1} \mid {}^p H^1(K) \text{ is locally bounded } p\text{-power torsion}\},$$

$${}^{p^+} D^{\geq 0} = \{K \mid K//p \in {}^p D^{\geq 0}(X, \mathbf{F}_p)\}.$$

Here locally bounded means that the torsion is killed by a fixed power of p after restriction to any quasicompact open. For every $n \geq 1$, derived reduction modulo p^n is right t -exact for p -perversity and left t -exact for p^+ -perversity. Moreover, $K \in {}^p D^{\leq 0}$ if and only if $K//p \in {}^p D^{\leq 0}(X, \mathbf{F}_p)$, while $K \in {}^{p^+} D^{\geq 0}$ if and only if $K//p \in {}^p D^{\geq 0}(X, \mathbf{F}_p)$. This is similar in spirit to [BH22, Construction 4.3 and Proposition 4.6].

One defines $D_{\text{fgu}}^b(D_{F, \mathcal{X}})$ as the full subcategory of $D_{\text{unit}}(D_{F, \mathcal{X}})$ (i.e. $F^*M \simeq M$) which are locally bounded and have cohomology sheaves finitely generated over $D_{F, \mathcal{X}}$. We will assume X

is quasicompact in this section for convenience (it is not strictly needed), so this amounts to being bounded.

Theorem 2.5. *Let $\mathcal{X} / \mathrm{Spf} W(k)$ be smooth and quasicompact. Then there is an equivalence*

$$\mathrm{RSol} : D_{\mathrm{fgu}}^b(D_{\mathrm{F}, \mathcal{X}})^{\mathrm{op}} \simeq D_{\mathrm{cons}}^b(X, \mathbf{Z}_p),$$

where X/k is the smooth special fiber of $\mathcal{X}$. Explicitly, the functor is given by $\mathrm{RSol}_X(M) = \mathrm{RHom}_{D_{\mathrm{F}, \mathcal{X}_{\mathrm{pro\acute{e}t}}}}(M_{\mathrm{pro\acute{e}t}}, \mathcal{O}_{\mathcal{X}, \mathrm{pro\acute{e}t}}) \in D(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$. If X is pure of dimension d , then $\mathrm{RSol}_X[d]$ identifies the standard t -structure on the source with the p^+ -perverse t -structure on the target.

Proof. We first note that one has

$$D_{\mathrm{fgu}}^b(D_{\mathrm{F}, \mathcal{X}}) \simeq \varprojlim_n D_{\mathrm{fgu}, \mathrm{tf}}^b(D_{\mathrm{F}, \mathcal{X}_{p^n=0}})$$

and similarly

$$D_{\mathrm{cons}}^b(X, \mathbf{Z}_p) \simeq \varprojlim_n D_{\mathrm{ctf}}^b(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n).$$

Here tf and ctf mean finite Tor dimension and constructible finite Tor dimension, respectively. These categories are closed under retracts in their ambient idempotent complete categories. They therefore satisfy the hypotheses of Lemma 2.3 (it is straightforward to verify these categories are what one obtains by taking $- \otimes \mathrm{Perf}(\mathbf{Z}/p^n)$). Moreover, [EK04b, Proposition 15.4.1 and Equation (16.1.3)] give

$$\mathrm{RSol}_X(M) // p^n \simeq \mathrm{RSol}_{X, \mathbf{Z}/p^n}(M // p^n).$$

Thus the solution functor induces the compatible functors on reductions, and Lemma 2.3 reduces the assertion to the case modulo p .

Since we have defined RSol using pro\acute{e}tale sheaves to handle $\mathbf{Z}_p$ coefficients well, after reducing modulo p in the target category we must use the equivalence

$$D_{\mathrm{cons}}^b(X_{\mathrm{pro\acute{e}t}}, \mathbf{F}_p) \simeq D_{\mathrm{cons}}^b(X_{\acute{e}t}, \mathbf{F}_p).$$

Let $\nu : X_{\mathrm{pro\acute{e}t}} \rightarrow X_{\acute{e}t}$ be the natural morphism of sites. Since X is smooth, it is locally Noetherian, and for each n pullback along ν identifies $D_{\mathrm{ctf}}^b(X_{\acute{e}t}, \mathbf{Z}/p^n)$ with $D_{\mathrm{ctf}}^b(X_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n)$ by the torsion comparison of [BS15, Corollary 5.1.6 and Proposition 5.2.6], applied on quasicompact opens. Combining this with [HRS23, Theorem 1.6(1)] gives the desired result.

Modulo p , [EK04b, Proposition 15.4.3] identifies finitely generated unit $D_{\mathrm{F}, \mathcal{X}}$ -complexes with the usual finitely generated unit $\mathcal{O}_{\mathrm{F}, X}$ -complexes, and [EK04b, §16.1.5] identifies their solution functors. Theorem 11.4.2 of [EK04b] therefore proves the mod p equivalence, while Theorem 11.5.4 gives the compatibility of its standard heart with the middle-perverse heart. This proves the equivalence.

Finally, assume X has pure dimension d , let M be any object of the source, and put $K = \mathrm{RSol}_X(M)[d]$. By derived Nakayama, $M \in D^{\leq 0}$ if and only if $M // p \in D^{\leq 0}$. The compatibility with derived reduction above and perverse t -exactness modulo p identify this condition with $K // p \in {}^p D^{\geq 0}$, which holds if and only if $K \in {}^{p^+} D^{\geq 0}$ by Definition 2.4. Thus the contravariant

equivalence RSol identifies these halves of the two t -structures. Orthogonality identifies the other halves and proves the claim. $\square$

We also record a variant for perfect rings, which we will use (in a mild way – we really only use existence of the functor and full faithfulness). In this setting we use the definition from [BL19] to make sense of the finitely generated unit condition: one asks for modules to be in $\text{Thick}(W(R)\langle F \rangle_p)$ and to be unit.

Theorem 2.6. *Let R be a perfect $\mathbf{F}_p$ -algebra. Then there is an equivalence*

$$\text{RSol} : D_{\text{fgu}}(W(R))^{\text{op}} \simeq D_{\text{cons}}^b(\text{Spec } R, \mathbf{Z}_p)$$

given by $\underline{\text{RHom}}_{D^\varphi(W(R)_{\text{proét}})}(M_{\text{proét}}, W(R)_{\text{proét}}) \in D((\text{Spec } R)_{\text{proét}}, \mathbf{Z}_p)$, with internal derived Hom taken on $(\text{Spec } R)_{\text{proét}}$. The equivalence is moreover symmetric monoidal.

Proof. The argument is similar to the previous case. Both sides are stable subcategories of their ambient derived categories closed under retracts, hence are idempotent complete and $\text{Perf}(\mathbf{Z}_p)$ -linear; the dual action of $\text{Perf}(\mathbf{Z}_p)$ on the source makes RSol linear. Their reductions (in the sense of the setting of Lemma 2.3) are the perfect unit categories over $W_n(R)$ and the constructible complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$.

Applying lemma 2.3 now reduces the claim to [BL19, Theorem 11.4.4], together with [BL19, Theorems 12.5.4 and 12.6.1].

For the final claim, one may essentially copy the argument in [BL19, Theorem 8.4.1]. The formula for RSol gives a lax symmetric monoidal structure:

$$\text{RSol}(M) \widehat{\otimes}_{\mathbf{Z}_p}^L \text{RSol}(N) \longrightarrow \text{RSol}(M \widehat{\otimes}_{W(R)}^L N).$$

Both sides are derived p -complete, so derived Nakayama reduces the assertion that this map is an equivalence to the case modulo p , where it may be further checked on geometric stalks. Over an algebraically closed perfect field κ , use the dual or contravariant form of Katz's theorem described in the remark following Theorem 2.1. The fact that this is symmetric monoidal can be checked in the same way as [BL19, Theorem 8.4.1]: the category is equivalent to $\text{Perf}(\mathbf{F}_p)$, and reducing to the heart (which is fine since it thickly generates) one can decompose objects as finite direct sums of the unit where the claim is obvious.

To check tensor closure, Remark 4.4 reduces us to characteristic p , where [BL19, Theorems 12.4.1 and 12.5.4, Proposition 12.3.8] express every finitely generated unit complex as $D(P)$ for a Frobenius complex P perfect over R , where D denotes a *weak dual*. By [BL19, Construction 11.2.7 and the proof of Proposition 12.5.1], $D(P) \simeq \text{colim}_n \varphi^{n*}(P^\vee)$, so cofinality of the diagonal $\mathbf{N} \rightarrow \mathbf{N}^2$ gives $D(P) \otimes_R^L D(Q) \simeq D(P \otimes_R^L Q)$. Since $P \otimes_R^L Q$ is again R -perfect, its weak dual is finitely generated unit by [BL19, Proposition 12.3.11]. $\square$

3. COHERENT LAURENT CRYSTALS

In this section, we prove several important results about coherent Laurent crystals on the framed log affines of Definition 3.2. These will be used to understand the local structure of finitely generated unit crystals in terms of (φ, Γ) -modules.

Assumption 3.1. Let $X/\mathcal{O}_K$ be smooth and quasicompact, let D be a horizontal normal crossings divisor³, and let $\mathcal{M} = \mathcal{M}_D$ be the log structure induced by D .

Fix a compatible system $(\zeta_{p^m})_{m \geq 1}$ extending ζ_p , and use it to identify $\mathbf{Z}_p(1)$ with $\mathbf{Z}_p$.

In this section we assume p is odd and work over $W(k)[\zeta_p]$ (mostly for expository reasons). At the end of this section in Corollary 3.19, we explain how to make modifications to obtain the result over $W(k)$ (and also with more roots of unity).

Definition 3.2. A framing of $(\mathrm{Spf} R, \mathcal{M})$ consists of a p -completely étale $W(k)$ -algebra R_0 over $W(k)\langle T_1, \dots, T_r, T_{r+1}^\pm, \dots, T_n^\pm \rangle$ and an identification $R \simeq R_0 \widehat{\otimes}_{W(k)} W(k)[\zeta_p]$ under which $\mathcal{M}$ is the pullback of the log structure with chart $\mathbf{N}^r$ whose generators map to $T_1, \dots, T_r$. We denote the resulting log formal scheme $\mathrm{Spf} W(k)[\zeta_p]\langle T_1, \dots, T_r, T_{r+1}^\pm, \dots, T_n^\pm \rangle$ by $\mathbb{T}_r^n$, the induced strict morphism to it by $\square$, and call $(\mathrm{Spf} R, \mathcal{M})$ a framed log affine.

There is also a variant over $\mathcal{O}_K$ for general $K/W(k)[1/p]$, obtained by replacing $W(k)[\zeta_p]$ with $\mathcal{O}_K$ and requiring $R \simeq R_0 \widehat{\otimes}_{W(k)} \mathcal{O}_K$.

Remark 3.3. For a framed log affine, we use n and r as the corresponding integers in Definition 3.2. We call the étale lifts of $T_1, \dots, T_r$ “boundary coordinates”.

Once we have later shown descent results, the following lemma will frequently be used to reduce to the situation of a framed log affine.

Lemma 3.4. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. Then $(X, \mathcal{M})$ admits a finite strict-étale cover by framed log affines over $\mathcal{O}_K$.*

Proof. By the normal crossings hypothesis, a strict-étale cover separates the local branches of D and makes them a simple normal crossings divisor. Fix $x \in X$. Let $D_1, \dots, D_r$ be the branches of D through x , and pick a small affine Zariski neighborhood $\mathrm{Spf} R$ around x so that no other component of D meets the chosen neighborhood. Since D is simple normal crossings relative to $\mathrm{Spf} \mathcal{O}_K$ and $X/\mathrm{Spf} \mathcal{O}_K$ is smooth, there are functions $t_1, \dots, t_r, s_{r+1}, \dots, s_n \in R$ such that $D_i = V(t_i)$ and

$$dt_1, \dots, dt_r, ds_{r+1}, \dots, ds_n$$

is a basis of $\widehat{\Omega}_{R/\mathcal{O}_K}^1$. Choose constants $c_j \in \mathcal{O}_K$ such that $u_j = s_j + c_j$ has nonzero image in $\kappa(x)$; thus after Zariski localizing once more, each u_j is a unit. The functions t_i and u_j define a

³By this we mean that $D \cup X_s$ is a normal crossings divisor and no irreducible component of D is contained in the special fiber; equivalently, strict-étale locally the pair has the form of Definition 3.2

morphism

$$\square : \mathrm{Spf} R \rightarrow \mathbb{T}_r^n.$$

It is induced by the ring map sending T_i to t_i for $i \leq r$ and T_j to u_j for $j > r$ (the latter are invertible, which is why we localized to make u_j units).

The pullback of $V(T_1 \cdots T_r)$ is $V(t_1 \cdots t_r) = D \cap \mathrm{Spf} R$, hence $\square$ is strict as a morphism of log formal schemes. The Jacobian criterion shows that the underlying morphism is p -completely étale.

Let π be a uniformizer of $\mathcal{O}_K$ with Eisenstein polynomial $E(u) \in W(k)[u]$ of degree e . Since $\mathcal{O}_K/p \simeq k[\pi]/(\pi^e)$, invariance of étale algebras under nilpotent thickenings descends R/p uniquely to an étale algebra $\bar{R}_0$ over the reduction modulo (p, π) of the target coordinate ring. Its unique p -completely étale lift R_0 over the corresponding $W(k)$ -algebra satisfies $R \simeq R_0 \hat{\otimes}_{W(k)} \mathcal{O}_K$, compatibly with the framing. $\square$

Now let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine with framing $\square$. Let R_0 be the algebra in the framing, and put $A^+ = R_0^\square[[q-1]] := R_0[[q-1]]$ and $A = R_0^\square((q-1))_p^\wedge := (A^+[1/[p]_q])_p^\wedge$. The identification $A^+/(p)_q \simeq R$ sends q to ζ_p . The log q -de Rham prism $(A^+, [p]_q, \mathbf{N}^r)^a$ (in the sense of [Kos22]) uses this to witness it as an object in $(\mathrm{Spf} R, \mathcal{M})_\Delta$, and its log chart sends e_i to T_i for $i \leq r$. We write $\bar{R} = R_0/p$, $\bar{A}^+ = A^+/p$, and $\bar{A} = A/p$.

Put $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$. For $\gamma = (a, b_1, \dots, b_n) \in \Gamma$, regard b_i as an element of $\mathbf{Z}_p$ through the identification fixed above. The formulas $q \mapsto q^a$ and $T_i \mapsto q^{pb_i}T_i$ for $i \leq n$ reduce to the identity modulo $[p]_q$. They therefore lift uniquely along the framing to log prism automorphisms over $(\mathrm{Spf} R, \mathcal{M})$; see Lemma A.17. Thus the crystal property gives an action of Γ on every evaluation in $D_\bullet(A)$ for a solid Laurent crystal as defined in Definition A.11. We use the symbol $\square$ to indicate that the Γ action on the prism depends on the framing.

Our first goal is to prove that evaluation on the log q -de Rham prism induces a t -exact conservative functor

$$\rho_A^* : D_\bullet((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \longrightarrow D_\bullet(A)^{h\Gamma},$$

which is fully faithful. On objects with p -completely nuclear evaluation (p -complete with derived mod p reduction nuclear), the target admits an interpretation in terms of continuous semilinear Γ -modules.

We also establish a result about roots. This will be crucial in giving a workable local description of finitely generated unit crystals. Its proof uses the full faithfulness of Proposition 3.12 for coherent Laurent crystals.

Theorem 3.5. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine and put $\bar{A} = A/p$. Let M be a finite $\bar{A}$ -module equipped with a continuous semilinear Γ -action as above. Suppose there is a Γ -equivariant injection $M \hookrightarrow \varphi^*M$. Then there is a coherent Laurent crystal $\mathcal{E}$ killed by p , with evaluation M , and an injection $\mathcal{E} \hookrightarrow \varphi^*\mathcal{E}$ whose evaluation is the given map.*

3.1. Laurent crystals and Γ -modules. Fix a framed log affine $(\mathrm{Spf} R, \mathcal{M})$ over $\mathrm{Spf} W(k)[\zeta_p]$. We let $A^+ = R_0^\square[[q-1]]$ denote the q -de Rham prism attached to the framing $\square$. In this subsection we will define a functor

$$\rho_A^{\Gamma,*} : D_\bullet((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \longrightarrow D_\bullet(A)^{h\Gamma}$$

and check that it is fully faithful. Here $(-)^{h\Gamma}$ denotes the continuous bar construction. It will turn out that only its p -completely nuclear subcategory is naturally thought of as continuous semilinear representations, but we will only use these in applications.

We explain some of the content of Appendix A.3 here, as we will be working in this formalism. Throughout we work with analytic rings as defined by Clausen and Scholze in [CS23] – see also [RC24] and [Kip26b] for more background material. If $(B, (d))$ is a prism, we give $B[1/d]_p^\wedge$ the stack-level p -complete analytic structure defined by the tower of Tate-Huber pairs $((B//p^n)[1/d], B//p^n)$ from [Kip26a, Example 2.1.5 and Remark 2.1.7]. For dealing with semilinear Γ -modules we will need $C(\Gamma, -)$ as well, with its natural induced analytic ring structure (as a topological ring one uses the compact-open topology). The precise definition of $D_\bullet((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ is given in Definition A.11 using these analytic structures, as a limit over all log prisms A in $(\mathrm{Spf} R, \mathcal{M})_\Delta^{\mathrm{op}}$.

We will show that actually there is a natural stacky interpretation of this functor, using the results in Appendix A.1 and Appendix A.3. For the reader not wishing to wade through the technical details needed to properly set up this theory, the essential point is that $(\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}$ is an analytic stack such that $D_\bullet((\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}) \simeq D_\bullet((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$. We postpone explaining details about this Laurent stack until §6.1 where it is explained in detail as we will only use relatively formal properties of it here. We will mostly use some properties of this stack established in the appendix, citing when necessary.

It is possible to carry the full faithfulness argument out entirely site-theoretically (indeed this was the first proof the author obtained) but the strategy is far easier to understand at a high level from a stacky perspective. We will realize $\rho_A^{\Gamma,*}$ as a pullback functor along

$$\rho_A^\Gamma : [\mathrm{AnSpec} A/\Gamma] \rightarrow (\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}.$$

The stacky interpretation then gives a natural strategy to attack the problem. In such a situation to verify full faithfulness what one needs to do is to check that the morphism of stacks is cohomologically proper, and moreover that the canonical map $\mathcal{O} \rightarrow R\rho_{A,*}^\Gamma \mathcal{O}$ is an isomorphism. Once this is established full faithfulness is a purely formal consequence of the projection formula.

The proof will be a reduction to the cohomology calculation in Lemma 3.11 (used to establish that $\mathcal{O} \simeq R\rho_{A,*}^\Gamma \mathcal{O}$), for which we will need a very explicit description of $A^{(1)}$ and how it sits inside the algebra of continuous functions $C(\Gamma, A)$. The following lemma computes $A^{+, (1)}$ as a log prismatic envelope.

Lemma 3.6. *Let $A^+ = R_0^\square[[q-1]]$, and let $(A^+)^{(1)}$ be the ring of the self-coproduct of $(A^+, [p]_q, \mathcal{M})^a$ in the category of log prisms $(\mathrm{Spf} R, \mathcal{M})_\Delta^{\mathrm{op}}$. Write q_0, q_1 for the two copies of q , put $u_0 = q_1/q_0$, and*

let $u_i = T_{i,1}/T_{i,0}$ for $1 \leq i \leq n$. Then

$$(A^+)^{(1)} \simeq R_0^\square[[q_0 - 1]][u_0^{\pm 1}, u_1^{\pm 1}, \dots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}} \right\}^{\wedge}_{\delta, (p, [p]_{q_0})}.$$

Here $\delta(u_j) = 0$ for $0 \leq j \leq n$.

Proof. This follows by verifying universal properties. For simplicity write $(A^+)^{(1)}$ for $R_0^\square[[q_0 - 1]][u_0^{\pm 1}, u_1^{\pm 1}, \dots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}} \right\}^{\wedge}_{\delta, (p, [p]_{q_0})}$ with its natural log prism structure (with prismatic ideal $[p]_{q_0}$ and log structure pulled back from $\mathrm{Spf} A^+$); we will verify that it has the required universal property.

Let us recall what this universal property should be for the self-coproduct of the log q -de Rham prism. We ask, for a test log prism $(B, [p]_{q_0}, \mathcal{N})^a \in (\mathrm{Spf} R, \mathcal{M})_\Delta$ that in the diagram

$$\begin{array}{ccc} (B, [p]_{q_0}, \mathcal{N})^a & \xrightarrow{f_0} & ((A^+)^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a \xrightarrow{p_0} (\mathrm{Spf} A_0^+, ([p]_{q_0}), \mathcal{M}_0)^a \\ & \searrow g \quad \downarrow p_1 & \downarrow \\ & (\mathrm{Spf} A_1^+, ([p]_{q_1}), \mathcal{M}_1)^a \longrightarrow & * \end{array}$$

(Note: The diagram shows a curved arrow f_1 from $(B, [p]_{q_0}, \mathcal{N})^a$ to $(\mathrm{Spf} A_1^+, ([p]_{q_1}), \mathcal{M}_1)^a$ and a curved arrow f_0 from $(B, [p]_{q_0}, \mathcal{N})^a$ to $((A^+)^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a$. A dashed arrow g points from $(B, [p]_{q_0}, \mathcal{N})^a$ to $((A^+)^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a$. A vertical arrow p_1 points from $((A^+)^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a$ to $(\mathrm{Spf} A_1^+, ([p]_{q_1}), \mathcal{M}_1)^a$. A horizontal arrow p_0 points from $((A^+)^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a$ to $(\mathrm{Spf} A_0^+, ([p]_{q_0}), \mathcal{M}_0)^a$. A vertical arrow points from $(\mathrm{Spf} A_0^+, ([p]_{q_0}), \mathcal{M}_0)^a$ to $*$. A horizontal arrow points from $(\mathrm{Spf} A_1^+, ([p]_{q_1}), \mathcal{M}_1)^a$ to $*$.)

there is a unique dashed arrow g (we can assume the ideal is $[p]_{q_0}$ by rigidity). Here the diagram is as sheaves in the strict log prismatic site, where the prisms are regarded this way via the Yoneda embedding.⁴

The dashed arrow corresponds to a map of log prisms $g : (A^+)^{(1)} \rightarrow B$ restricting to f_0 and f_1 . After fixing $f_0 : A_0^+ \rightarrow B$, the choices of f_1 are

$$\{(v_j)_{j=0}^n \in (B^\times)^{n+1} : \delta(v_j) = 0 \text{ and } v_j \equiv 1 \pmod{[p]_{q_0}}\}.$$

We produce these v_j by noting $f_1(q) = v_0 f_0(q)$ and $f_1(T_i) = v_i f_0(T_i)$ are forced to differ by such units (as we must have a strict map of log prisms over $(\mathrm{Spf} R, \mathcal{M})$).

By the universal property of log prismatic envelopes [Kos22, Propositions 3.6 and 3.7], after fixing f_0 the maps g are classified by the same data, through $g(u_j) = v_j$ on the underlying rings. Indeed, the divided generators map to $(v_j - 1)/[p]_{q_0}$, while the second chart maps e_i to (e_i, v_i) for $i \leq r$. Conversely, these assignments give the unique map g , and its restriction to A_1^+ is f_1 by the unique étale lift along the framing. Hence the two universal properties agree. $\square$

⁴If we did not work in the strict site, the result is the same: we can replace the log structure on B with a common pullback log structure where both maps f_i are strict. For this one can use the same method as [DLMS24, Corollary 2.2 and proof of Lemma 2.10].

Our next task will be to understand $A^{(1)} := A^{+, (1)}[1/[p]_q]^\wedge$ as a subring of $C(\Gamma, A)$.

Once this is defined, as in Definition A.11 one sets

$$D_\bullet(B[1/d]_p^\wedge) := \varprojlim_n \text{Mod}_{(B//p^n)[1/d]}(D_\bullet(\text{Spa}(B, B))),$$

with transition functors given by derived reduction.

Definition 3.7. For $\gamma = (a, b_1, \dots, b_n) \in \Gamma$, put $x_0(\gamma) = (a - 1)/p$ and $x_i(\gamma) = b_i$ for $1 \leq i \leq n$. Set $\eta_{0,m}(x_0) = q^{m+p\binom{m}{2}}(q - 1)^m \binom{x_0}{m}_{q^p}$ and $\eta_{i,m}(x_i) = q^{p\binom{m}{2}}(q - 1)^m \binom{x_i}{m}_{q^p}$ for $i > 0$. Write η_α for the products of these functions in the $(n + 1)$ variables and set

$$C_q(\Gamma, A) = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} A^+ \eta_\alpha \right)_{(p, q-1)}^\wedge [1/[p]_q]_p^\wedge.$$

We define $C_q(\Gamma^s, A)$ similarly, over indices $\alpha \in \mathbf{N}^{s(n+1)}$.

There is also an alternative formula which will be of use. Pullback along (id, γ) then sends $u_i \in A^{+, (1)}$ to $(q^p)^{x_i(\gamma)}$ for $0 \leq i \leq n$. Set

$$[m]_{q^p}! \gamma_{m, q^p}(x) = \prod_{j=0}^{m-1} ((q^p)^x - q^{pj}),$$

where $[m]_{q^p}! = \prod_{j=1}^m [j]_{q^p}$. Then $\eta_{0,m}(x_0) = q^m [p]_q^{-m} \gamma_{m, q^p}(x_0)$ and $\eta_{i,m}(x_i) = [p]_q^{-m} \gamma_{m, q^p}(x_i)$ for $i > 0$.

The formula for η_α in Definition 3.7 shows that these functions are uniformly bounded. Consequently, an expansion whose coefficients tend to zero $(p, q - 1)$ -adically converges uniformly, and evaluation defines a continuous map $C_q(\Gamma^s, A) \rightarrow C(\Gamma^s, A)$, where the target carries the compact-open topology. Here we give $C_q(\Gamma^s, A)$ the natural topology induced by the completion in Definition 3.7, followed by Laurentization and p -adic completion. Lemma 3.8 shows that this map is injective, but its topology on the source is not the subspace topology.

For $s \geq 0$, let $(A^+)^{(s)}$ be the s th term of the Čech nerve of the log q -de Rham prism in $(\text{Spf } R, \mathcal{M})_\Delta$, and put $A^{(s)} = ((A^+)^{(s)}[1/[p]_q])_p^\wedge$.

We will show evaluation on the log q -de Rham prism induces a functor

$$\rho_A^{\Gamma, *}: D_\bullet((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \longrightarrow D_\bullet(A)^{h\Gamma}.$$

The crystal property does give an abstract Γ action, but does not obviously ensure continuity and higher coherences which is why we make the following more delicate construction. Lemma A.17 writes the source $D_\bullet((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ as the limit

$$\lim \left(D_\bullet(A) \rightrightarrows D_\bullet(A^{(1)}) \rightrightarrows D_\bullet(A^{(2)}) \rightrightarrows \dots \right)$$

The target can be thought of as solid A -modules with continuous semilinear Γ -action, although Lemma 3.9 tells us this is only reasonable under a nuclearity hypothesis (which will be satisfied

in all applications). Modulo p^n , the rings in the preceding diagram and in the continuous bar construction carry their natural analytic structures $\mathrm{Spa}(B, B)$. The Γ -action on $A^+ = R_0^\square[[q-1]]$, after p -completely inverting $[p]_q$ induces maps

$$\theta_s : A^{(s)} \longrightarrow C(\Gamma^s, A).$$

A priori these land only in all functions, but we will show they land in continuous functions. This then induces $\rho_A^{\Gamma,*}$ as

$$D_\bullet((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \xrightarrow{\tilde{\rho}_A^*} \lim_{[s] \in \Delta} D_\bullet(C(\Gamma^s, A)) =: D_\bullet(A)^{h\Gamma}.$$

We point out that because A is induced from a topological ring that is an inverse limit of Banach rings, one can write $C(\Gamma^s, A) = \lim_n C(\Gamma^s, A/p^n)$. We also note that in the situation the quotient topology on A/p^n and the Tate topology given by $(A^+/p^n)[1/(q-1)]$ agree, so there is no ambiguity (indeed this holds for any prism). Finally we also note that $C(\Gamma^s, B)[1/d]_p^\wedge \simeq C(\Gamma^s, B[1/d]_p^\wedge)$ for any prism $(B, (d))$ because Γ^s is profinite (combined with the previous fact about inverse limits). Thus we freely exchange these in what follows.

These maps θ_s are in fact injective as ordinary ring maps, and we show the previously defined $C_q(\Gamma^s, A)$ are their images. In what follows we give $A^{(s)}$ the analytic ring structure induced from the log prism $A^{+, (s)}$ in the logarithmic Čech nerve, using $\lim_n ((A^{+, (s)}/p^n)[1/d], (A^{+, (s)}/p^n))$ with their Huber pair analytic ring structures (as in the convention set in the discussion after Lemma 3.6). For $C_q(\Gamma^s, A)$ we make a similar definition, using the completed direct sum presentation as we did for its topological ring structure.

Lemma 3.8. *The map θ_s identifies $A^{(s)}$ with the subring $C_q(\Gamma^s, A) \subset C(\Gamma^s, A)$:*

$$A^{(s)} \xrightarrow{\sim} C_q(\Gamma^s, A).$$

This is further an isomorphism of analytic rings.

The map $C_q(\Gamma^s, A) \rightarrow C(\Gamma^s, A)$ is continuous as topological rings where the target is equipped with the compact-open topology; the same assertion holds when regarded as solid modules.

Proof. As above, we use $A = R_0^\square((q-1))_p^\wedge$. We first treat $s = 1$. We actually will show a stronger claim, which is that there is a natural isomorphism of adic rings

$$\theta_1 : A^{+, (1)} \simeq \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} A^+ \eta_\alpha \right)_{(p, q-1)}^\wedge.$$

After p -completely inverting $[p]_q$ or equivalently $q-1$, this isomorphism of ordinary adic rings implies the desired isomorphism of analytic rings (as in this setting passing from topological rings to analytic rings is fully faithful).

We do this by using the explicit log prismatic envelope presentation of $A^{+, (1)}$. Recall $(A^+)^{(1)}$ from Lemma 3.6 is the $(p, q-1)$ -adic (equivalently $(p, [p]_q)$) completion of

$$A^+[u_0^{\pm 1}, \dots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}} \right\}_\delta.$$

We used $u_0 = q_1/q_0$ and $u_i = T_{i,1}/T_{i,0}$ as shorthand. The map θ_1 can be made explicit integrally. For $g \in \Gamma$, using the coordinates $g = (x_0, \dots, x_n)$ of Definition 3.7, the evaluation of $\theta_1(a) \in \text{Map}(\Gamma, A^+)^5$ for $a \in A^{+, (1)}$ at g is obtained by applying the A^+ -algebra map determined by $u_i \mapsto (q^p)^{x_i(g)}$ for $0 \leq i \leq n$.

Using [GLSQ23, Theorem 4.4], the cyclotomic factor $A^+[u_0^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}} \right\}_\delta^\wedge$ is, up to a change of variables, the $(p, q-1)$ -completion of the free A^+ -module on their Δ -divided powers $\omega_{0,\Delta}^{\{n\}}$ (see [GLSQ23, Definition 1.7]) as a $(p, q-1)$ -adic ring. Indeed from the discussion preceding Theorem 4.4. in *loc. cit.* they identify $\widehat{\bigoplus_{n \geq 0} A^+ \omega_{0,\Delta}^{\{n\}}}$ with the prismatic envelope of the δ -pair $(A^+[[\xi]], (\xi))$, which checking universal properties is indeed our ring (one has $\xi \mapsto q_0(u_0 - 1)$).

Now we see where θ_1 sends this basis. Using [GLSQ23, Remark 1.8(5)] provides a “blowing up” map Bl, yielding

$$A^+ \langle \xi \rangle_{q_0^p} \xrightarrow{\text{Bl}} \widehat{\bigoplus_{n \geq 0} A^+ \omega_{0,\Delta}^{\{n\}}} \simeq A^+[u_0^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}} \right\}_\delta^\wedge \xrightarrow{\theta_1} \text{Map}(\Gamma, A^+)$$

such that the q -divided power $\xi^{[m]_{q_0^p}}$ is sent to $[p]_{q_0}^m \omega_{0,\Delta}^{\{m\}}$ under Bl. This is helpful, because we know where $\xi^{[m]_{q_0^p}}$ is sent to under the composite map $\theta_1 \circ \text{Bl}$: ξ is sent by θ_1 to

$$q((q^p)^{x_0} - 1) \in \text{Map}(\Gamma, A^+).$$

Thus we may deduce

$$[p]_{q_0}^m \theta_1(\omega_{0,\Delta}^{\{m\}}) = q^m \gamma_{m,q^p}(x_0),$$

so in particular using the alternative formula after Definition 3.7 we see $\theta_1(\omega_{0,\Delta}^{\{m\}}) = \eta_{0,m}$ using $[p]_{q_0}$ -torsionfreeness.

Next, we handle the remaining factors corresponding to u_i for $i > 0$. The argument is more or less the same. Using [GLSQ22, Definition 1.6 and Corollary 5.6], the corresponding factor $A^+[u_i^{\pm 1}] \left\{ \frac{u_i - 1}{[p]_{q_0}} \right\}_\delta^\wedge$ is the completion of the free A^+ -module on its negative-level divided powers $\omega_{i,\Delta}^{\{m\}}$. The analogous map in [GLSQ22, Proposition 1.7] let us deduce the formula

$$[p]_q^m \theta_1(\omega_{i,\Delta}^{\{m\}}) = \gamma_{m,q^p}(x_i).$$

Similarly we find $\theta_1(\omega_{i,\Delta}^{\{m\}}) = \eta_{i,m}$. Thus θ_1 induces a surjection onto

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} A^+ \eta_\alpha \right)_{(p,q-1)}^\wedge.$$

⁵We haven't yet checked the output is a continuous function.

It is injective: evaluation at nonnegative integral values of each x_i is triangular, with $\eta_{i,m}(j) = 0$ for $m > j$ and $\eta_{i,m}(m)$ a unit times $(q-1)^m$. Only finitely many terms contribute at each such value, and A^+ is $(q-1)$ -torsion-free. Hence the integral map is, as desired, an isomorphism onto $(\bigoplus_{\alpha} A^+ \eta_{\alpha})_{(p,q-1)}^{\wedge}$; the isomorphism is of $(p, q-1)$ -adic rings, so the claim about analytic ring structures also follows. One sees θ_1 lands in $C(\Gamma, A)$ by the discussion after Definition 3.7.

For general s , $A^{+, (s)} = (A^{+(1)})^{\widehat{\otimes}_{A^+} s}$. Using this one may check

$$(A^+)^{(s)} \simeq \left(\bigoplus_{\alpha \in \mathbf{N}^{s(n+1)}} A^+ \eta_{\alpha} \right)_{(p,q-1)}^{\wedge},$$

so by the same argument the general claim follows. $\square$

By definition, $D_{\blacksquare}(A)^{h\Gamma} = \lim_{[s] \in \Delta} D_{\blacksquare}(C(\Gamma^s, A))$. The next lemma identifies its p -completely nuclear subcategory with continuous semilinear representations.

We first fix notation. Let G be a profinite group and let B be a derived p -complete analytic ring with continuous G -action. Write $\mathbf{Rep}_{B^t_{\blacksquare}}(G)$ for the category of solid modules over the twisted solid group algebra $B^t_{\blacksquare}[G]$; equivalently, this is the category of solid B -modules with continuous semilinear G -action. Here $\mathbf{Z}_{\blacksquare}[G]$ denotes the solid group algebra. We set

$$R\Gamma_{\blacksquare}(G, M) := R\mathrm{Hom}_{\mathbf{Z}_{\blacksquare}[G]}(\mathbf{Z}_{\blacksquare}, M) \simeq R\mathrm{Hom}_{B^t_{\blacksquare}[G]}(B, M).$$

For $M \in D_{\blacksquare}(B)$, we write

$$C(G, M) := \underline{R\mathrm{Hom}}_B(B[G], M).$$

Recall that $D_{\blacksquare}(B)^{\mathrm{nuc}}$ consists of the objects M for which, for every compact $P \in D_{\blacksquare}(B)^{\omega}$, the canonical map

$$P^{\vee} \otimes_B^{L, \blacksquare} M \longrightarrow \underline{R\mathrm{Hom}}_B(P, M)$$

is an equivalence. For a discrete analytic ring this condition is automatic; for example, $D_{\blacksquare}(\mathbf{Z}_{\mathrm{disc}})^{\mathrm{nuc}} = D(\mathbf{Z})$.

Lemma 3.9. *Let B be a derived p -complete solid analytic ring with a continuous action of a compact p -adic Lie group G . Give $C(G^s, B)$ its natural analytic ring structure, and assume $C(G^s, B)$ is p -completely nuclear over B . The natural functor*

$$\lim_{[s] \in \Delta} D_{\blacksquare}(C(G^s, B)) \longrightarrow \mathbf{Rep}_{B^t_{\blacksquare}}(G)$$

restricts to an equivalence from objects whose evaluation at $[0]$ is p -completely nuclear⁶ to representations whose underlying B -module is p -completely nuclear. If $M \in D_{\blacksquare}(B)$ is the underlying solid B -module under this equivalence,

$$\varprojlim_{[s] \in \Delta} C(G^s, B) \otimes_B^{L, \blacksquare} M \simeq \varprojlim_{[s] \in \Delta} C(G^s, M) \simeq R\Gamma_{\blacksquare}(G, M).$$

⁶By this we mean it is derived p -complete and satisfies the nuclearity property for derived p -complete objects; equivalently, it is derived p -complete and nuclear mod p .

Proof. We use a similar argument to [Mik26, Corollary 1.50]. The Barr-Beck-Lurie theorem, as in [Mik26, Remark 1.49, Corollary 1.50] (which is a similar argument), identifies the source and target with coalgebras in $D_{\blacksquare}(B)$ for the following comonads. The first is induced by the endofunctor

$$\mathbb{T} : D_{\blacksquare}(B) \xrightarrow{C(G,B) \otimes_B^{L,\blacksquare} -} D_{\blacksquare}(C(G,B)) \longrightarrow D_{\blacksquare}(B).$$

Here the first morphism uses the map $B \rightarrow C(G,B)$ sending $b \in B$ to the constant function (valued at b). The second morphism is pushforward along the action map $\text{act} : B \rightarrow C(G,B)$ (i.e. b is sent to the function sending g to $g(b)$). The comonad structure is the obvious one (the comultiplication is induced by the multiplication on G)

The second is induced by the endofunctor $\mathbb{S} : D_{\blacksquare}(B) \rightarrow D_{\blacksquare}(B)$ given by

$$\mathbb{S}(M) = C(G, M),$$

where $(bf)(g) = g(b)f(g)$ on $\mathbb{S}(M)$. Thus

$$\varprojlim_{[s] \in \Delta} D_{\blacksquare}(C(G^s, B)) \simeq \text{CoAlg}_{\mathbb{T}}(D_{\blacksquare}(B)).$$

Similarly, $\text{Rep}_{B^t}(\mathbb{G}) \simeq \text{CoAlg}_{\mathbb{S}}(D_{\blacksquare}(B))$.

The map $f \otimes m \mapsto (g \mapsto f(g)m)$ defines a natural transformation $\mathbb{T} \rightarrow \mathbb{S}$ compatible with the counits and comultiplications, hence a map of comonads.

Suppose now that M is p -completely nuclear. For the compact object $P = B[\underline{G}]$, nuclearity and the identifications $P^\vee \simeq C(G, B)$ and $\underline{\text{RHom}}_B(P, M) \simeq C(G, M)$ give

$$\mathbb{T}(M) = C(G, B) \otimes_B^{L,\blacksquare} M \simeq P^\vee \otimes_B^{L,\blacksquare} M \simeq \underline{\text{RHom}}_B(P, M) \simeq C(G, M) = \mathbb{S}(M).$$

By the assumed p -complete nuclearity of $C(G, B)$ and closure of nuclear objects under tensor products, both comonads restrict to the full subcategory of p -completely nuclear modules in $D_{\blacksquare}(B)$. The preceding calculation identifies their restrictions as comonads. Their coalgebra categories are therefore equivalent, proving the claim.

The cohomological comparison follows by taking RHom with the unit. $\square$

For the Laurent coefficient rings actually used in this section, these hypotheses can be checked on B^+ with $B = (B^+[1/d])_p^\wedge$. The module $C(G^s, B^+)$ is the (p, d) -adic completion of the locally constant functions; it has discrete derived reduction modulo (p, d) and is nuclear by [AM24, Lemma 2.18(i)]. Modulo each p^n , compactness of G^s gives a uniform bound on denominators, so Laurent base change recovers the continuous functions with values in $(B^+/p^n)[1/d]$. Passing to the derived p -complete limit gives the required p -complete nuclearity. Indeed one has

$$C(G, B) \otimes_B^{L,\blacksquare} s \simeq C(G^s, B).$$

To compute solid group cohomology, we use the resolution of [Bos23, Proposition B.3 and Lemma B.4]. If $G \simeq \mathbf{Z}_p^r$, let Λ_G be the solid algebra associated with $\mathbf{Z}_p[[G]]$. For topological generators $\gamma_1, \dots, \gamma_r$, there is a finite projective resolution in solid Λ_G -modules

$$(1) \quad 0 \longrightarrow \Lambda_G^{\binom{r}{1}} \longrightarrow \dots \longrightarrow \Lambda_G^{\binom{r}{1}} \xrightarrow{(\gamma_1-1, \dots, \gamma_r-1)} \Lambda_G \longrightarrow (\mathbf{Z}_p)_{\blacksquare} \longrightarrow 0.$$

The omitted arrows are the Koszul differentials. Thus solid group cohomology of a complex of solid $\mathbf{Z}_p$ -modules with continuous G -action is computed by the Koszul complex of the operators $\gamma_i - 1$. The topology of Definition 3.7 ensures that the translations and homotopies below are morphisms of solid modules.

One can carry out the needed cohomology calculation in the $\eta_{i,m}$ basis of Definition 3.7, but it is actually slightly easier to work modulo p where a more natural characterization is available. Via the coordinates $x = (x_0, \dots, x_n) : \Gamma \rightarrow \mathbf{Z}_p^{n+1}$ of Definition 3.7⁷, a continuous function $f : \Gamma \rightarrow M$ with values in a complete separated module has a unique *Mahler expansion* $f(x) = \sum_{\alpha} a_{\alpha} \binom{x}{\alpha}$. Its Mahler coefficient of degree α is $a_{\alpha} = (\Delta_0^{\alpha_0} \cdots \Delta_n^{\alpha_n} f)(0)$, where $\Delta_i f(x) = f(x + e_i) - f(x)$ and e_i is the i th standard basis vector of $\mathbf{Z}_p^{n+1}$. We will use this in the following to give an alternative characterization of C_q using Mahler expansions, at least modulo p , that we use in Lemma 3.11.

Lemma 3.10. *Let $\bar{A} = A/p$ (similarly for $\bar{A}^+$) and consider $f \in C(\Gamma, \bar{A})$. We take its Mahler expansion*

$$f(x) = \sum_{\alpha \in \mathbf{N}^{n+1}} a_{\alpha} \binom{x}{\alpha},$$

where $\binom{x}{\alpha} = \prod_i \binom{x_i}{\alpha_i}$. Suppose there are constants $\sigma > 1$ and C for which

$$a_{\alpha} \in (q-1)^{[\sigma|\alpha|-C]} \bar{A}^+$$

for every multi-index α . Then $f \in C_q(\Gamma, \bar{A})$.

Proof. We will prove the stronger assertion

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \bar{A}^+ \eta_{\alpha} \right)_{(q-1)}^{\wedge} = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \bar{A}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)_{(q-1)}^{\wedge}.$$

In one variable x , set $\tilde{\eta}_m = q^{p\binom{m}{2}} \binom{x}{m}_{q^p}$, a renormalized version of $\eta_{i,m}$ from Definition 3.7 (by a factor of $(q-1)^m$). For $m \geq 1$, the q -Pascal recurrence

$$\tilde{\eta}_m(x+1) - \tilde{\eta}_m(x) = (q^{pm} - 1)\tilde{\eta}_m(x) + q^{p(m-1)}\tilde{\eta}_{m-1}(x)$$

shows that the coefficient of $\binom{x}{j}$ in $\tilde{\eta}_m$ vanishes for $j < m$, is a unit for $j = m$, and belongs to $(q^p - 1)^{j-m} \bar{A}^+$ for $j > m$. Here we use that the finite difference operator $\Delta(f) = f(x+1) - f(x)$ extracts the coefficient of $\binom{x}{j}$ in its Mahler expansion as $\Delta^j f(0)$. The first claim is simply because $\tilde{\eta}_m(j) = 0$ when $j < m$, the second because we obtain $q^{p\binom{m}{2}}$ at $j = m$, and the third follows by induction using the q -Pascal recurrence.

⁷The map x is only a homeomorphism, and does not respect the group structures.

In $\overline{A}^+$ one has $q^p - 1 = (q - 1)^p$, and hence for all $0 \leq i \leq n$ (rescaling by the unit q^m at $i = 0$) one has

$$\eta_{i,m} = u_{i,m}(q - 1)^m \binom{x_i}{m} + \sum_{j>m} c_{i,m,j}(q - 1)^j \binom{x_i}{j}.$$

Here $u_{i,m} \in (\overline{A}^+)^\times$, and $c_{i,m,j} \in (q - 1)^{(p-1)(j-m)} \overline{A}^+$. This shows the $(q - 1)^{|\alpha|} \binom{x}{\alpha}$ and the η_α bases are related by an operator $D(1 + N)$ where D is diagonal and invertible and

$$N^r \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{A}^+ (q - 1)^{|\alpha|} \binom{x}{\alpha} \right)_{(q-1)}^\wedge \subset (q - 1)^{(p-1)r} \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{A}^+ (q - 1)^{|\alpha|} \binom{x}{\alpha} \right)_{(q-1)}^\wedge,$$

hence N is topologically nilpotent. Thus we deduce the claimed isomorphism. $\square$

We now carry out the needed cohomology calculation.

Lemma 3.11. *Right translation defines a continuous Γ -action on $C_q(\Gamma, A)$ by $(\gamma f)(g) = f(g\gamma)$. Then there is a natural equivalence*

$$A \xrightarrow{\sim} R\Gamma_\bullet(\Gamma, C_q(\Gamma, A)).$$

Proof. By derived Nakayama it is enough to work modulo p . The proof of Lemma 3.10 shows there is a lattice

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{A}^+ (q - 1)^{|\alpha|} \binom{x}{\alpha} \right)_{q-1}^\wedge \subset C_q(\Gamma, \overline{A})$$

Let $\text{Fil}^\bullet$ be the $q - 1$ -adic filtration on this lattice; this filtration is complete and separated.

Let $\gamma_0, \dots, \gamma_n$ be the usual generators of Γ , in the coordinates chosen in Lemma 3.10. Right translation by γ_i sends x_i to $x_i + 1 + px_0$, while γ_0 sends x_0 to $x_0 + 1 + px_0$. In characteristic p , one has the standard binomial identity $\binom{x+pz}{m} = \sum_{\ell \geq 0} \binom{x}{m-p\ell} \binom{z}{\ell}$. We claim that, for every $0 \leq i \leq n$, the two-term complex

$$\left[C_q(\Gamma, \overline{A}) \xrightarrow{\gamma_i - 1} C_q(\Gamma, \overline{A}) \right]$$

retracts onto the functions independent of x_i in degree 0. Indeed, let π_i be the projection onto the Mahler terms with $\alpha_i = 0$. We define

$$h_i \left((q - 1)^{|\alpha|} \binom{x}{\alpha} \right) = (q - 1)^{|\alpha|} \binom{x_i}{\alpha_i + 1} \prod_{j \neq i} \binom{x_j}{\alpha_j}.$$

This gives a continuous endomorphism of $C_q(\Gamma, \overline{A})$ such that $\Delta_i h_i = 1$, $h_i \Delta_i = 1 - \pi_i$, and $h_i \text{Fil}^r \subset \text{Fil}^{r-1}$. Here Δ_i is the finite difference operator used earlier; it simply lowers the i th component of α by one. The terms with $\ell > 0$ in $\binom{x+pz}{m} = \sum_{\ell \geq 0} \binom{x}{m-p\ell} \binom{z}{\ell}$ lower Mahler

degree by at least $p - 1$. It follows that $1 - (\gamma_i - 1)h_i$ carries Fil^r into Fil^{r+p-2} . Since $p > 2$, the series in

$$h'_i = h_i \sum_{k \geq 0} (1 - (\gamma_i - 1)h_i)^k$$

converges, and $(\gamma_i - 1)h'_i = 1$. Moreover, $\gamma_i - 1$ vanishes on the image of π_i . The identity $1 = h_i\Delta_i + \pi_i$ therefore gives $\gamma_i - 1 = (\gamma_i - 1)h_i\Delta_i$, and hence $h'_i(\gamma_i - 1) = h_i\Delta_i = 1 - \pi_i$. This proves the claim.

Applying this successively to the factors of $\mathbf{Z}_p(1)^n$ in the resolution (1) shows that the natural $1 + p\mathbf{Z}_p$ -equivariant map

$$C_q(1 + p\mathbf{Z}_p, \overline{A}) \longrightarrow \mathrm{R}\Gamma_{\bullet}(\mathbf{Z}_p(1)^n, C_q(\Gamma, \overline{A}))$$

is an equivalence after forgetting the action, hence is an equivalence. Applying the claim with $i = 0$ and the Hochschild-Serre equivalence gives

$$\overline{A} \xrightarrow{\sim} \mathrm{R}\Gamma_{\bullet}(\Gamma, C_q(\Gamma, \overline{A})).$$

The finite resolutions above show that solid group cohomology preserves derived p -completeness and commutes with reduction modulo p . Derived Nakayama proves the claim. $\square$

Proposition 3.12. *Evaluation on the log q -de Rham prism attached to $A^+ = R_0^{\square}[q - 1]$ induces a fully faithful functor*

$$\rho_A^{\Gamma,*} : D_{\bullet}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \longrightarrow D_{\bullet}(A)^{h\Gamma}.$$

When the evaluation on the log q -de Rham prism is p -completely nuclear, the target can be thought of as semilinear Γ -modules by the equivalence of Lemma 3.9.

Proof. Under the stack/site equivalence of Proposition 6.5, $\rho_A^{\Gamma,*}$ is induced on $D_{\bullet}(-)$ by pullback along

$$\rho_A^{\Gamma} : [\mathrm{AnSpec} A/\Gamma] \rightarrow (\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \bullet}.$$

Indeed, taking the presentation $\mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathrm{AnSpec} A^{(n)}$ of the target given by Proposition A.22 after base change to $(X, \mathcal{M})^{\mathcal{L}, \bullet}$ and the natural presentation $\mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathrm{AnSpec} C(\Gamma^n, A)$ of the source one obtains the description of the functor appearing before Lemma 3.8. Here, both presentations are obtained by taking the Čech nerve of the cover by $\mathrm{AnSpec} A$. We note that one should not construct this map by claiming only that the map $\mathrm{AnSpec} A \rightarrow (\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \bullet}$ obtained from the fpqc epimorphism of formal stacks obtained in Lemma A.4 is Γ -equivariant, because a priori the action might not be continuous. But once one uses presentations of both stacks and Lemma 3.8 to check the action is continuous, as in the site-theoretic construction it becomes well-defined.

Now we explain why the assertion reduces to properness of ρ_A^{Γ} in the sense of [Kip26b, Definition 2.1.3.1]. We use the same conventions as [Kip26b], and work with the six functor formalism on analytic stacks. In our case the analytic stacks all lie over $\mathrm{AnSpec} \mathbf{Z}_{p, \bullet}$ with its solid analytic ring structure. We have a diagram

$$\begin{array}{ccc}
[\mathrm{AnSpec} A^{(1)}/\Gamma] & \longrightarrow & [\mathrm{AnSpec} A/\Gamma] \\
\downarrow f & & \downarrow \rho_A^\Gamma \\
\mathrm{AnSpec} A & \xrightarrow{\rho_A} & (\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}.
\end{array}$$

This square is cartesian, using that $\mathrm{AnSpec} A \times_{(\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}} \mathrm{AnSpec} A = \mathrm{AnSpec} A^{(1)}$ from the Čech nerve presentation of $(\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}$. In particular, note that the Γ action is nontrivial in $[\mathrm{AnSpec} A^{(1)}/\Gamma]$ as it is acting by right translation. Assuming ρ_A^Γ is cohomologically proper for $D_\blacksquare$ we obtain base change for this diagram: in particular for the unit $\mathcal{O} \in D_\blacksquare([\mathrm{AnSpec} A/\Gamma])$ there is a base change isomorphism

$$(2) \quad \rho_A^* R\rho_{A,*}^\Gamma \mathcal{O} \xrightarrow{\sim} Rf_* \mathcal{O} \simeq R\Gamma_{\mathrm{solid}}(\Gamma, C_q(\Gamma, A))$$

by p -complete nuclearity, Lemma 3.9 and Lemma 3.8 to identify $A^{(1)}$ with $C_q(\Gamma, A)$. We use the presentation in Lemma 3.6 and [AM24, Lemma 2.18(i)] to show that $A^{(1)}$ is p -completely nuclear over A (through either map $A \rightarrow A^{(1)}$ in the Čech nerve).

Now using Lemma 3.11, on the right hand side of (2) the canonical map $A \rightarrow Rf_* \mathcal{O} \simeq R\Gamma_{\mathrm{solid}}(\Gamma, C_q(\Gamma, A))$ is an isomorphism. This shows that the canonical map

$$\mathcal{O}_{(\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}} \longrightarrow R\rho_{A,*}^\Gamma \mathcal{O}$$

is an isomorphism, because applying the conservative evaluation ρ_A^* identifies it with the canonical isomorphism $A \rightarrow R\Gamma_{\mathrm{solid}}(\Gamma, C_q(\Gamma, A)) \simeq A$. Again using cohomological properness, full faithfulness of the pullback $\rho_{A,*}^\Gamma$ is formal by the projection formula. We can compute

$$R\rho_{A,*}^\Gamma(\rho_A^{\Gamma,*} \mathcal{E}) \simeq R\rho_{A,!}^\Gamma(\mathcal{O} \otimes^{\mathcal{L}, \blacksquare} \rho_A^{\Gamma,*} \mathcal{E}) \simeq R\rho_{A,!}^\Gamma \mathcal{O} \otimes^{\mathcal{L}, \blacksquare} \mathcal{E} \simeq \mathcal{E}$$

for $\mathcal{E} \in D_\blacksquare((\mathrm{Spf} R, \mathcal{M})^{\mathcal{L}, \blacksquare}) \simeq D_\blacksquare((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$, showing that the adjunction unit is an equivalence, hence that $\rho_{A,*}^\Gamma$ is fully faithful.

Thus, all that remains is properness. Proposition A.22 and base change to the Laurent stack $(X, \mathcal{M})^{\mathcal{L}, \blacksquare}$ show that ρ_A is a representable proper effective epimorphism. Since properness is local on the target for such covers by [Kip26b, Remark 1.1.3.7], it suffices to check the claim for

$$[\mathrm{AnSpec} A^{(1)}/\Gamma] \rightarrow \mathrm{AnSpec} A$$

The property is stable under base change, so this can be checked using A^+ and $A^{+,(1)}$. The right translation action fixes the image of A^+ , giving a factorization of the map as

$$[\mathrm{AnSpec} A^{+,(1)}/\Gamma] \longrightarrow \mathrm{AnSpec} A^+ \times B\Gamma \longrightarrow \mathrm{AnSpec} A^+.$$

After base change, pulling the first map back along $\mathrm{AnSpec} A^+ \rightarrow \mathrm{AnSpec} A^+ \times B\Gamma$ gives $\mathrm{AnSpec} A^{+,(1)} \rightarrow \mathrm{AnSpec} A^+$, which is proper by Lemma A.13. It remains to show that

$$B\Gamma \rightarrow \mathrm{AnSpec} \mathbf{Z}_{p,\blacksquare}$$

is proper for the six functor formalism on analytic stacks. One may use an argument similar to [HM24, Proposition 5.3.2]. The map $g : \mathrm{AnSpec} \mathbf{Z}_{p,\blacksquare} \rightarrow B\Gamma$ is representable proper, while $B\Gamma \rightarrow \mathrm{AnSpec} \mathbf{Z}_{p,\blacksquare}$ is 1-truncated and its composite with g is the identity. By [HM24,

Corollary 4.7.5(iii)], it remains only to verify that $g_*\mathcal{O} = C(\Gamma, \mathbf{Z}_p)$ is descendable. The finite resolutions in (1), applied to the normal subgroup $\mathbf{Z}_p(1)^n$ and quotient $1 + p\mathbf{Z}_p \simeq \mathbf{Z}_p$, show that the trivial representation $\mathbf{Z}_p$ is perfect over $\mathbf{Z}_p[[\Gamma]]$. Taking derived solid $\mathbf{Z}_p$ -duals and using $\mathbf{Z}_p[[\Gamma]]^\vee \simeq C(\Gamma, \mathbf{Z}_p)$ gives

$$\mathbf{Z}_p \in \text{Thick}(C(\Gamma, \mathbf{Z}_p)) \subset D_\bullet(B\Gamma),$$

as desired. $\square$

3.2. Roots in Laurent crystals. We now turn our attention to proving Theorem 3.5. As we now impose a perfectness condition, condensed mathematics is not relevant in this subsection. In Appendix A.1 we show in Corollary A.19 that perfect Laurent crystals have a canonical t -structure in the normal crossings setting, allowing us to define $\text{Coh}((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$.

For a framed log affine, put $A = R_0^\square((q-1))_p^\wedge$ as before. We use A to mean this throughout this subsection, and use $\bar{A} := A/p$. One can understand the category of perfect A -complexes with continuous Γ action or $\text{Perf}(A)^{h\Gamma}$ as the totalization

$$(3) \quad \lim \left(\text{Perf}(A) \xrightleftharpoons[p_1]{p_0} \text{Perf}(C(\Gamma, A)) \xrightleftharpoons{\quad} \text{Perf}(C(\Gamma^2, A)) \xrightleftharpoons{\quad} \dots \right)$$

where A has the topology specified after Lemma 3.6. This is compatible with the previous very general description $\lim_{[s] \in \Delta} D_\bullet(C(\Gamma^s, A))$ by taking dualizable objects, which will only depend on the underlying ring (i.e. evaluation on $*$ is an equivalence with ordinary perfect complexes). These are automatically p -completely nuclear, so by Lemma 3.9 there is no subtlety about the relation to modules with semilinear Γ -action.

For coherent objects, to write down the data it is enough to keep the terms through $C(\Gamma^2, \bar{A})$ in (3). The descent data can naturally be encoded as a comodule over the Hopf algebroid $(\bar{A}, C(\Gamma, \bar{A}))$. Explicitly, an object of the three-term totalization is a finite $\bar{A}$ -module M together with an isomorphism

$$\alpha : C(\Gamma, \bar{A}) \otimes_{p_1, \bar{A}}^\bullet M \xrightarrow{\sim} C(\Gamma, \bar{A}) \otimes_{p_0, \bar{A}}^\bullet M$$

satisfying the identity and cocycle conditions. Writing $p_0(a)(g) = a$ and $p_1(a)(g) = g(a)$, the rule $\delta(m) = \alpha(1 \otimes m)$ identifies α with a p_1 -semilinear coaction $\delta : M \rightarrow C(\Gamma, \bar{A}) \otimes_{p_0, \bar{A}}^\bullet M$. The identity and cocycle conditions become the counit and coassociativity axioms, respectively, giving the equivalence with finite comodules over the Hopf algebroid.

Upon restricting to Coh the description is also compatible with classical theory describing Γ -modules, where one equips finite modules with their weak topology as in [SV11, Lemmas 8.2 and 8.22] (which discusses the case of a point). In particular, one may check that there is an equivalence between $\text{Coh}(A)^{h\Gamma}$ (the heart of the cobar description described above) and the category of finite A -modules with a continuous Γ -action, with the A -module given its weak topology. For clarity, over a topological ring A the weak topology on a finitely presented module M is the quotient topology induced by

$$A^n \twoheadrightarrow M$$

which is independent of the choice of generators. The following lemma clarifies the situation.

Lemma 3.13. *Let A be a topological ring, and let Γ be a profinite group acting continuously on A . We make two topological assumptions on A :*

- *The topology on A is complete Hausdorff metrizable.*
- *The ring A has the property that a finitely generated left module $M \subseteq A^n$ has its weak topology agree with its subspace topology.*

Then $\mathbf{Mod}^{\text{fp}}(A)^{h\Gamma}$ is equivalent to the category of finitely presented A -modules M equipped with a semilinear action for which $\Gamma \times M \rightarrow M$ is continuous for the weak topology on M .

The functor F inducing the equivalence sends an object in $\mathbf{Mod}^{\text{fp}}(A)^{h\Gamma}$ to the corresponding semilinear Γ -module via the cobar description.

Proof. The main content of the argument is actually just checking F is well-defined. A priori, the Γ action could be discontinuous for the weak topology. What we must check is that the coaction map

$$M \rightarrow C(\Gamma, A) \otimes_{p_0, A} M \simeq C(\Gamma, M)$$

is continuous for the weak topology on M . This is really two claims: that the second isomorphism is an isomorphism of topological modules where $C(\Gamma, A)$ has the compact-open topology and M has the weak topology, and that the map is continuous. Once this is established the equivalence is clear, since the inverse to F simply forgets all the topological data.

We will use the following general property of $C(\Gamma, -)$ with respect to exact sequences.

Lemma 3.14. *Let A be complete Hausdorff metrizable, and let*

$$0 \longrightarrow K \longrightarrow A^n \longrightarrow M \longrightarrow 0$$

be an exact sequence of left modules where A^n has its natural topology, K has the subspace topology and is closed, and M the quotient topology. Then

$$0 \longrightarrow C(\Gamma, K) \longrightarrow C(\Gamma, A^n) \longrightarrow C(\Gamma, M) \longrightarrow 0$$

is a strictly exact sequence of topological modules over $C(\Gamma, A)$ where each is equipped with the compact-open topology.

Here the term “strict exact” means that the leftmost term has the subspace topology and the rightmost term has the quotient topology.

Proof. We first check $C(\Gamma, K) \longrightarrow C(\Gamma, A^n)$ is injective and acquires the subspace topology. For injectivity we may test this at the level of sets, where it is immediate since $K \rightarrow A^n$ is injective any function which is zero on A^n is necessarily zero on K . Necessarily $C(\Gamma, K)$ acquires the subspace topology, as a subbasis of opens are the collections $V_{\Gamma', U} = \{f \in C(\Gamma, K) : f(\Gamma') \subset U\}$ where $\Gamma' \subseteq \Gamma$ is compact and U open. The subspace topology on the other hand asks for the same opens but where $U \subseteq A^n$ is open and we intersect with K . Since K has the subspace topology we can match the subbases for the topology.

Middle exactness is also clear, as the functions sent to 0 under $C(\Gamma, A^n) \rightarrow C(\Gamma, M)$ are precisely those valued in K . Thus the nontrivial content is surjectivity of $C(\Gamma, A^n) \rightarrow C(\Gamma, M)$, and that the target acquires the quotient topology.

Using [Wig73, Theorem M, p. 86], every continuous $f : \Gamma \rightarrow M$ lifts to $f : \Gamma \rightarrow A^n$ viewing both as additive topological groups. This proves set-theoretic surjectivity. We must also show that $C(\Gamma, M)$ acquires the quotient topology, which is what we need for strictness. The map is continuous, so we must check $C(\Gamma, A^n) \rightarrow C(\Gamma, M)$ is an open map.

Letting $\pi : A^n \rightarrow M$ denote the quotient map, by abuse of notation we also use π for the map $C(\Gamma, A^n) \rightarrow C(\Gamma, M)$. By virtue of being a homomorphism we must check $\pi_*(V_{\Gamma, U}) = V_{\Gamma, \pi(U)}$ (in our previous notation for the subbasis of the compact-open topology; this is a basis around zero). More precisely we must check $f : \Gamma \rightarrow \pi(U) \subset M$ lifts to $\tilde{f} : \Gamma \rightarrow U \subset A^n$. Viewing f as a map to M the previous result gives a lift, which need not land in U . Around $\gamma \in \Gamma$ we may always correct this lift in an open neighborhood of γ by taking $\tilde{f} - k_\gamma$, where $k_\gamma \in K$. Since Γ is profinite we can take a finite clopen partition into Γ_i (not as subgroups!) so Γ_i are contained in these opens, allowing us to correct $\tilde{f}$ piecewise by k_{γ_i} to land in U . This shows the claim. $\square$

Granting the lemma the proof is straightforward. When $M = A^n$ the claim that F is well-defined is clear. For general M , consider the presentation $0 \rightarrow K \rightarrow A^n \twoheadrightarrow M$. Since we assume the weak and subspace topologies on K are the same, it follows that $K \subset A^n$ is a complete additive subgroup hence closed (completeness follows from giving K the quotient topology from $A^m \twoheadrightarrow K$).

By the lemma we therefore get an exact sequence

$$0 \rightarrow C(\Gamma, K) \rightarrow C(\Gamma, A^n) \rightarrow C(\Gamma, M) \rightarrow 0$$

and applying it again with $A^m \twoheadrightarrow K$ we get a right exact sequence

$$C(\Gamma, A^m) \rightarrow C(\Gamma, A^n) \rightarrow C(\Gamma, M) \rightarrow 0.$$

For the first two we have $C(\Gamma, A) \otimes_{p_0, A} A^n \simeq C(\Gamma, A^n)$ and similarly for A^m , so keeping track of topologies we get an isomorphism of topological modules $C(\Gamma, A) \otimes_{p_0, A} M \simeq C(\Gamma, M)$ where M has its weak topology.

Finally, continuity of $M \rightarrow C(\Gamma, A) \otimes_{p_0, A} M$ is automatic from continuity of $A \rightarrow C(\Gamma, A)$. The claim then follows. $\square$

Both assumptions are easy to verify for $R_0^\square((q-1))_p^\wedge$.

Lemmas A.17 and A.16 give the same comodule description for Laurent crystals that we just discussed for semilinear Γ -modules. Put $\overline{A}^{(m)} = A^{(m)}/p$. In the coaction formulation, coherent Laurent crystals are described by replacing $C(\Gamma, \overline{A})$ and $C(\Gamma^2, \overline{A})$ with $\overline{A}^{(1)}$ and $\overline{A}^{(2)}$. For the following lemma, we note there is a natural identification $C(\Gamma^s, N) \simeq C(\Gamma^s, \overline{A}) \otimes_{\overline{A}, p_0}^\square N$ for a finite $\overline{A}$ -module N equipped with its weak topology, as we showed in Lemma 3.13. We keep the solid decoration in what follows to remind the reader that we are working with modules with

their weak topology or canonical solid structure as dualizable objects; in the coherent setting the distinction does not matter as they are identified after passing to the associated condensed module.

Lemma 3.15. *Let $\overline{A} = A/p$, let $s = 1, 2$, and let $p_0 : \overline{A} \rightarrow \overline{A}^{(s)}$ be the map in the Čech nerve corresponding to the canonical inclusion $[0] \hookrightarrow [s]$. For every finite $\overline{A}$ -module N , evaluation induces an injection*

$$\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} N \longrightarrow C(\Gamma^s, N).$$

If $M \subset N$ is finite, then inside $C(\Gamma^s, N)$ one has

$$\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} M = (\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} N) \cap C(\Gamma^s, M).$$

Proof. By Definition 3.7 and Lemma 3.8, an element of $\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} N$ has an expansion

$$f = \sum_{\alpha \in \mathbf{N}^{s(n+1)}} m_{\alpha} \eta_{\alpha}.$$

One may verify that for $\alpha, \ell \in \mathbf{N}^{s(n+1)}$, one has $\eta_{\alpha}(\ell) = 0$ unless $\alpha \leq \ell$ coordinatewise, while $\eta_{\ell}(\ell) \in \overline{A}^{\times}$. If f evaluates to zero, then

$$0 = f(\ell) = \sum_{\alpha \leq \ell} m_{\alpha} \eta_{\alpha}(\ell).$$

The sum is finite. Induction on $|\ell|$ therefore gives $m_{\ell} = 0$ for every ℓ , proving injectivity.

The map $p_0 : A \rightarrow A^{(s)}$ is p -completely flat: before localizing p -completely at $[p]_q$ this follows from [Kos22, Proposition 4.13], and the property is preserved by inverting $[p]_q$ and applying p -adic completion. Since A is Noetherian, [Bha20, Lemma 5.15] makes $p_0 : A \rightarrow A^{(s)}$ flat. Its reduction $p_0 : \overline{A} \rightarrow \overline{A}^{(s)}$ is then flat. Consider the commutative square

$$\begin{array}{ccc} \overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} N & \longrightarrow & C(\Gamma^s, N) \\ \downarrow & & \downarrow \\ \overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} (N/M) & \longrightarrow & C(\Gamma^s, N/M). \end{array}$$

The lower horizontal map is injective by the first part. Flatness identifies the kernel of the left vertical map with $\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} M$. If an element of the upper left term maps into $C(\Gamma^s, M)$, its image in the lower right term vanishes. It therefore vanishes in the lower left term and belongs to $\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} M$ (using injectivity of the lower horizontal map). The reverse inclusion follows from naturality, proving

$$\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} M = (\overline{A}^{(s)} \otimes_{\overline{A}, p_0}^{\blacksquare} N) \cap C(\Gamma^s, M).$$

□

Lemma 3.16. *Let M be a finite $\bar{A}$ -module with continuous semilinear Γ -action. For $k \gg 0$, the coaction of $\varphi^{*k}M$ factors through*

$$C_q(\Gamma, \bar{A}) \otimes_{\bar{A}, p_0}^{\blacksquare} \varphi^{*k}M \subset C(\Gamma, \varphi^{*k}M).$$

The same holds with Γ^2 as needed for the cocycle condition.

Proof. Put $\bar{A}^+ = A^+/p$, and $\bar{A} = A/p$. We first show that M has a lattice $M^+ \subset M$ which is finite over $\bar{A}^+$ and Γ -stable. This follows by continuity of the Γ action for the weak topology on M . Given a finite set of generators S for any lattice M^+ , we observe that by continuity $\Gamma S \subset (q-1)^{-N}M^+$ (here we also use compactness of Γ). Now we simply replace M^+ by $\bar{A}^+ \cdot \Gamma S \subset (q-1)^{-N}M^+$. Since $\bar{A}^+$ is Noetherian it follows that this module is finite, and by construction it is Γ -stable.

In what follows we will also equip M with a complete separated valuation $\nu_{q-1} : m \mapsto \sup\{r \in \mathbf{Z} : m \in (q-1)^r M^+\}$. The point of picking a Γ -stable lattice is that this makes the action of Γ isometric, which will be useful in applying results in [BR25] later in the proof.

The key idea will be to observe that in Lemma 3.10 we can immediately adapt it with coefficients to give a criterion for membership in $C_q(\Gamma, \bar{A}) \otimes_{\bar{A}, p_0}^{\blacksquare} M \subset C(\Gamma, M)$ in terms of how rapidly Mahler coefficients tend to zero. Indeed, taking the existing criterion for $\bar{A}$ and choosing a finite free surjection $\bar{A}^{+\oplus n} \twoheadrightarrow M^+$ and inverting $q-1$ proves the criterion for finite coefficients. For $f \in C(\Gamma, M)$, take its Mahler expansion

$$f(x) = \sum_{\alpha \in \mathbf{N}^{n+1}} m_{\alpha} \binom{x}{\alpha},$$

where $\binom{x}{\alpha} = \prod_i \binom{x_i}{\alpha_i}$. The criterion is that if there are $\sigma > 1$ and C for which

$$m_{\alpha} \in (q-1)^{\lceil \sigma|\alpha| - C \rceil} M^+$$

for every multi-index α then $f \in C_q(\Gamma, M) = C_q(\Gamma, \bar{A}) \otimes_{\bar{A}, p_0}^{\blacksquare} M$.

Crucially, taking Frobenius pullbacks *improves* this condition – we will use [BR25] to argue that automatically the Γ action satisfies a similar condition but possibly with a worse rate of convergence, using their notion of “super-Hölder”. The notion of “super-Hölder” can be thought of as a version of locally analytic vectors which works over $\mathbf{Z}_p$ or modulo p , as observed in [Por25, Example 2.14].

For $j \geq 0$, put $\Gamma_j = \{g^{p^j} : g \in \Gamma\}$. Then Γ is uniform and

$$\Gamma_j = (1 + p^{j+1}\mathbf{Z}_p) \ltimes p^j\mathbf{Z}_p(1)^n.$$

Moreover, $gg'^{-1} \in \Gamma_j$ if and only if $x_i(g) - x_i(g') \in p^j\mathbf{Z}_p$ for every i . Thus $x = (x_0, \dots, x_n) : \Gamma \rightarrow \mathbf{Z}_p^{n+1}$ is a coordinate in the sense of [BR25, Proposition 1.1.1]. Continuity on a finite set

of generators gives $\ell \geq 0$ and $d > 0$ such that

$$(g-1)(q-1)^r M^+ \subset (q-1)^{r+d} M^+$$

for $g \in \Gamma_\ell$ and $r \in \mathbf{Z}$. If $g \in \Gamma_{\ell+j}$, write $g = g_0^{p^j}$ with $g_0 \in \Gamma_\ell$. Since $g-1 = (g_0-1)^{p^j}$ in characteristic p , we obtain

$$(g-1)(q-1)^r M^+ \subset (q-1)^{r+dp^j} M^+.$$

It follows from [BR25, Lemmas 1.2.8 and 1.2.10] that each value $\delta(m)$ of the coaction is super-Hölder with $e = 1$ and $\lambda = \log_p(d) - \ell$ in the sense of [BR25, Definition 1.2.1]. Applying [BR25, Theorem 4.3.4] in the coordinate x , the Mahler coefficients of $\delta(m)$ satisfy

$$a_\alpha \in (q-1)^{\lceil c \max_i \alpha_i - C \rceil} M^+ \subset (q-1)^{\lceil c|\alpha|/(n+1) - C \rceil} M^+.$$

Here $c > 0$ and C are independent of α .

At this point we are nearly done, as all that remains is to verify that Frobenius pullbacks can improve the constant c . Since $\varphi^k(q-1) = (q-1)^{p^k}$ modulo p , the coaction of $\varphi^{*k}M$, evaluated at $1 \otimes m$ for $m \in M^+$, has Mahler coefficients in

$$(q-1)^{\lceil p^k c |\alpha|/(n+1) - p^k C \rceil} (\overline{A}^+ \otimes_{\varphi^k, \overline{A}^+} M^+).$$

Choose k so that $p^k c / (2(n+1)) > 1$. The coefficient variant of the criterion in Lemma 3.10 now shows that the coaction on $1 \otimes m$ belongs to $C_q(\Gamma, \overline{A}) \otimes_{\overline{A}, p_0}^{\blacksquare} \varphi^{*k}M$.

The same argument in the $2(n+1)$ product coordinates proves the assertion over Γ^2 . $\square$

Proposition 3.17. *Let M be a finite $\overline{A}$ -module with continuous semilinear Γ -action. For $k \gg 0$, the module $\varphi^{*k}M$ arises from a coherent Laurent crystal killed by p .*

Proof. Choose k large enough for Lemma 3.16 in degrees one and two, and put $N = \varphi^{*k}M$. Lemmas 3.8 and 3.15 give injective vertical maps in the commutative diagram

$$\begin{array}{ccccc} N & \rightrightarrows & C(\Gamma, N) & \rightrightarrows & C(\Gamma^2, N) \\ \parallel & & \uparrow & & \uparrow \\ N & \rightrightarrows & \overline{A}^{(1)} \otimes_{\overline{A}, p_0}^{\blacksquare} N & \rightrightarrows & \overline{A}^{(2)} \otimes_{\overline{A}, p_0}^{\blacksquare} N \end{array}$$

The coaction $m \mapsto (g \mapsto g(m))$ factors uniquely through the lower degree one term by Lemma 3.16. Compatibility of the evaluation maps with the Čech maps makes the diagram commute. The counit and coassociativity identities hold in the upper row, hence in the lower row by injectivity in degrees one and two from Lemma 3.15. Thus the lower row defines a descent datum on N as a coherent Laurent crystal. $\square$

Lemma 3.18. *Let $M \subset N$ be finite $\overline{A}$ -modules with continuous semilinear Γ -action. If N is the evaluation of a coherent Laurent crystal killed by p and M is Γ -stable, then M also arises from a coherent Laurent crystal.*

Proof. One can identify the crystal's coaction with

$$N \longrightarrow \overline{A}^{(1)} \otimes_{\overline{A}, p_0}^{\blacksquare} N \hookrightarrow C(\Gamma, N).$$

We are given $M \subset N$ equipped with a (coaction) map $M \rightarrow C(\Gamma, M)$, such that

$$\begin{array}{ccc} N & \longrightarrow & C(\Gamma, N) \\ \uparrow & & \uparrow \\ M & \longrightarrow & C(\Gamma, M) \end{array}$$

commutes. By Lemma 3.15 we see the coaction restricts to M giving

$$M \longrightarrow \overline{A}^{(1)} \otimes_{\overline{A}, p_0}^{\blacksquare} M.$$

The cocycle and counit conditions follow by a similar argument. $\square$

Proof of Theorem 3.5. Let $\beta : M \hookrightarrow \varphi^* M$ be the given map. Since $\overline{A}$ is regular, Frobenius is flat by Kunz's theorem. Iterating β therefore embeds both M and $\varphi^* M$ as Γ -stable submodules of $N = \varphi^{*k} M$ for every $k \geq 1$.

Now choose k large enough for Proposition 3.17, and let $\mathcal{F}$ be the resulting coherent crystal with evaluation N . Lemma 3.18 gives coherent subcrystals $\mathcal{E}, \varphi^* \mathcal{E} \subset \mathcal{F}$ with evaluations M and $\varphi^* M$ (full faithfulness in Proposition 3.12 shows the second crystal is really the Frobenius pullback; nuclearity is automatic for coherent crystals). Proposition 3.12 also lifts β uniquely to a morphism

$$\tilde{\beta} : \mathcal{E} \longrightarrow \varphi^* \mathcal{E}.$$

Its kernel has evaluation 0 on the log q de Rham prism, so t -exactness and conservativity from Lemma A.17 and Corollary A.19 make $\tilde{\beta}$ injective. $\square$

As promised we now investigate other bases.

Corollary 3.19. *Both Proposition 3.12 and Theorem 3.5 hold over $W(k)[\zeta_{p^\ell}]$ for $\ell > 0$, by changing to $\Gamma = (1 + p^\ell \mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ and keeping the same base $A = R_0^\square((q-1))_p^\wedge$.*

For $\ell = 0$, again keeping the same base $A = R_0^\square((q-1))_p^\wedge$ but using $\Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)^n$ Proposition 3.12 holds and Theorem 3.5 also holds.

Proof. For $\ell > 0$ the modification is very easy, and can be done by simply setting the prismatic ideal in the q -de Rham prism to be $\varphi^{\ell-1}([p]_q)$. The toric factor acts by $T_i \mapsto q^{p^\ell b_i} T_i$, so this action is trivial modulo $\varphi^{\ell-1}([p]_q)$.

For $\ell = 0$ and p odd, one can evade redoing the calculations in Proposition 3.12 in favor of a simpler calculation. We will first show that the pullback along

$$f : [W(k)[\zeta_p]^{\mathcal{L}, \blacksquare} / \mu_{p-1}] \longrightarrow W(k)^{\mathcal{L}, \blacksquare}$$

is fully faithful on $D_{\blacksquare}(-)$. Here the μ_{p-1} action comes from functoriality and the Galois action on $W(k)[\zeta_p]/W(k)$. One can use the same argument as in Proposition 3.12 to reduce to checking if the pushforward of the structure sheaf on the source is $\mathcal{O}_{W(k)^{\mathcal{L}, \blacksquare}} = \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}$ via the projection formula after we verify f is prim (for the six functor formalism on analytic stacks). If we know this, then it is formal to deduce the claim in the general case for $[(R_0[\zeta_p], \mathcal{M})^{\mathcal{L}, \blacksquare}/\mu_{p-1}] \rightarrow (R_0, \mathcal{M})^{\mathcal{L}, \blacksquare}$: one may use base change for prim maps to deduce the required isomorphism. Verifying f is prim turns out to be formal from our results so far. One can use the diagram

$$\begin{array}{ccc}
 [\mathrm{AnSpec} B/\mathbf{Z}_p^{\times}] & \longrightarrow & [\mathrm{AnSpec} W(k)((q-1))_p^{\wedge}/\mathbf{Z}_p^{\times}] \\
 \downarrow & & \downarrow \rho_{W(k)((q-1))_p^{\wedge}}^{\Gamma} \\
 & & [W(k)[\zeta_p]^{\mathcal{L}, \blacksquare}/\mu_{p-1}] \\
 & & \downarrow f \\
 \mathrm{AnSpec} W(k)((\tilde{p}))_p^{\wedge} & \longrightarrow & W(k)^{\mathcal{L}, \blacksquare}
 \end{array}$$

where the outer rectangle is cartesian. To see this, we compute B as a prismatic envelope using Lemma A.6; the analytic realization $(-)^{\blacksquare}$ of Definition 6.1 and passage to Laurent stacks preserve finite limits.

The upper right arrow is proper, as in the argument of Proposition 3.12; moreover the argument also shows $\rho_{W(k)((q-1))_p^{\wedge}}^{\Gamma} \mathcal{O} \simeq \mathcal{O}$. The left vertical map is proper by the same argument as in Proposition 3.12, using Lemma A.13 and properness of $B\mathbf{Z}_p^{\times}$. Hence the composite of the right vertical maps is proper, since the bottom map is an effective epimorphism of analytic stacks by Proposition A.20. Now [HM24, Corollary 4.7.5(ii)] shows that f is prim, which gives the required projection formula and base change properties for the argument by [HM24, Corollary 4.5.11(ii) and Lemma 4.5.13(ii)].

Thus we check $\mathcal{O}_{W(k)^{\mathcal{L}, \blacksquare}} \simeq Rf_* \mathcal{O}_{[W(k)[\zeta_p]^{\mathcal{L}, \blacksquare}/\mu_{p-1}]}$. One can use the same argument as in Proposition 3.12 to reduce to checking if the pushforward of the structure sheaf on the source is $\mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}$. Using Proposition 6.5, we may work site-theoretically, so one may reduce to checking the stronger claim

$$W(k)[[\tilde{p}]] \xrightarrow{\sim} \Delta_{W(k)[\zeta_p]/W(k)[[\tilde{p}]]}^{h\mu_{p-1}}.$$

Here we use conservativity of evaluation on $W(k)[[\tilde{p}]]$ to make this reduction. This can be checked directly, using the Hodge-Tate comparison. Indeed, $L_{W(k)[\zeta_p]/W(k)} \simeq \Omega_{W(k)[\zeta_p]/W(k)}^1$ compatibly with the Galois action on $W(k)[\zeta_p]/W(k)$, but the target has no ordinary μ_{p-1} invariants; higher exterior powers also follow, as in this case they are just shifts of $\Omega_{W(k)[\zeta_p]/W(k)}^1$. The μ_{p-1} invariants are exact so the desired result follows from

$$W(k)[[\tilde{p}]] = W(k)[[q-1]]^{\mu_{p-1}}.$$

We now explain the second assertion, the proof of Theorem 3.5. The Čech calculation carries over verbatim for the log prism $(R_0^\square[[\tilde{p}]], (\tilde{p}), \mathcal{M})$ – this prism is constructed analogously to the log q -de Rham prism via a framing, but we note that one must be cautious as it does not possess a full Γ action. Its first Čech term is

$$(R_0^\square[[\tilde{p}]])^{(1)} \simeq R_0^\square[[\tilde{p}_0, \tilde{p}_1]][u_1^{\pm 1}, \dots, u_n^{\pm 1}] \left\{ \frac{\tilde{p}_1}{\tilde{p}_0}, \frac{u_1 - 1}{\tilde{p}_0}, \dots, \frac{u_n - 1}{\tilde{p}_0} \right\}_{\delta, (p, \tilde{p}_0)}^\wedge,$$

where $u_i = T_{i,1}/T_{i,0}$ and $\delta(u_i) = 0$.

Recall that we write $A^+ = R_0^\square[q - 1]$ and $A = R_0^\square((q - 1))_p^\wedge$. The proof of Theorem 3.5 needs two nontrivial results: high Frobenius pullbacks become crystals, and evaluation is injective with finite coefficients. For the latter, the inclusion $\iota : R_0^\square[[\tilde{p}]] \hookrightarrow A^+$ gives evaluation along $(\iota, g\iota)$ into $C(\Gamma, A)$ after Laurentization. The triangular argument of Lemma 3.15 applies to this nerve as well.

For the first result, work modulo p . Regard the first Čech term as an $R_0^\square[[\tilde{p}]]/p$ -algebra via p_0 , and extend scalars along $\iota \bmod p$ to $\bar{A}^+ = A^+/p$. Before Laurentization, evaluation on the representatives $g = ((1 + p)^b, x)$ of Γ/μ_{p-1} identifies this base change with

$$\left(\bigoplus_{m \geq 0, \nu \in \mathbb{N}^n} \bar{A}^+ \tilde{p}^m (q - 1)^{|\nu|} \binom{b}{m} \binom{x}{\nu} \right)_{(q-1)}^\wedge.$$

The corresponding term for A^+ is $\left(\bigoplus_{m \geq 0, \nu \in \mathbb{N}^n} \bar{A}^+ (q - 1)^{m+|\nu|} \binom{b}{m} \binom{x}{\nu} \right)_{(q-1)}^\wedge$. Coefficients in these expansions tend to zero $(q - 1)$ -adically; Laurentization then inverts $q - 1$. Since $\tilde{p} = (q - 1)^{p-1} \cdot \text{unit}$, we also have $\frac{(q-1)^{p(m+|\nu|)}}{\tilde{p}^m (q-1)^{|\nu|}} = (q - 1)^{m+(p-1)|\nu|} \cdot \text{unit}$.

Apply Lemma 3.16 for the $(1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ -action. One further Frobenius pullback puts the coaction in $(R_0^\square[[\tilde{p}]])^{(1)}[1/\tilde{p}]_p^\wedge$, giving a crystal. Its μ_{p-1} -equivariance allows descent by invariants; the same calculation in degree two gives the cocycle condition by injectivity. The proof of Theorem 3.5 now applies unchanged. $\square$

4. FINITELY GENERATED UNIT LAURENT CRYSTALS

4.1. Basic constructions. Following [EK04b] and the more general version in [BL19] (which explains how to remove finiteness hypotheses), we first generalize the notion of finitely generated unit modules.

Definition 4.1. Let (A^+, I) be a prism with $I = (d)$. Equip $A = A^+[1/d]_p^\wedge$ with the analytic ring structure in Definition A.11, and we use $D_\bullet(A)$ as defined there. Note A has a natural Frobenius φ induced by the Frobenius on A^+ (since $\varphi(d) \equiv d^p \pmod{p}$).

Write $A\langle F \rangle$ for

$$\widehat{\bigoplus_{i \geq 0} A F^i},$$

where $Fx = \varphi(x)F$. We put $D_{\blacksquare}(A\langle F \rangle)$ for solid modules over this noncommutative analytic ring, with the analytic ring structure induced from A .

For $M \in D_{\blacksquare}(A\langle F \rangle)$, we say M is *unit* if

$$\varphi_A^* M := A \otimes_{A, \varphi_A}^{L, \blacksquare} M \longrightarrow M$$

given by $a \otimes m \mapsto aF_M(m)$ is an equivalence. A unit object is *finitely generated unit* if it is perfect over $A\langle F \rangle$; these objects form the full subcategory $D_{\text{fgu}}(A)$. Here, we mean perfect in the sense that it lies in the thick subcategory generated by $A\langle F \rangle$ as we work over a noncommutative base.

We remark that we can naturally view these as certain unit Frobenius modules, so $D_{\text{fgu}}(A) \subset D_{\blacksquare}(A)^{\varphi=1}$ as a full subcategory. In fact one can show the category on a fixed prism is insensitive to the solid structure, but we will still remember it so that our crystals live in an ambient category which is a stack.

Lemma 4.2. *Let $(A^+, (d))$ be an orientable bounded prism, and let $A = A^+[1/d]_p^\wedge$ with the analytic ring structure from Definition A.11. Then*

$$\text{Thick}_{A\langle F \rangle} D(A\langle F \rangle) \simeq \text{Thick}_{A\langle F \rangle} D_{\blacksquare}(A\langle F \rangle)$$

with inverse given by $M \mapsto M()$. The solid modules in the target thick subcategory are all p -completely nuclear.*

Proof. The comparison is exact and carries $A\langle F \rangle$ to the corresponding solid module. We will show the natural functor induces an equivalence on their mapping spectra. It is then fully faithful on the thick closure of the generator, and its essential image is a thick subcategory containing the target generator.

We can compute

$$\text{Map}_{D_{\blacksquare}(A\langle F \rangle)}(A\langle F \rangle, A\langle F \rangle) \simeq \text{Map}_{D_{\blacksquare}(A)}(A, A\langle F \rangle) \simeq A\langle F \rangle.$$

The first equivalence uses the free-forgetful adjunction. For the second, the localization and completion adjunction identifies mapping out of A here with mapping out of A^+ in the integral analytic category $D_{\blacksquare}((A^+, A^+))$. The equivalence can then be checked modulo p by derived Nakayama. It follows from compactness of $A^+//p$: modulo p localization at d is a filtered colimit and $(A\langle F \rangle)//p = \bigoplus_{i \geq 0} (A//p)F^i$.

The unit A^+ of the integral analytic category is nuclear. Since A is the filtered colimit of multiplication by d in derived p -complete solid modules, it is p -completely nuclear. The p -completely nuclear objects are closed under colimits, so the underlying module of $A\langle F \rangle = \widehat{\bigoplus_{i \geq 0} AF^i}$ is p -completely nuclear. The forgetful functor from Frobenius modules preserves finite colimits and retracts, so the same holds throughout the thick subcategory generated by the free $A\langle F \rangle$ -module. $\square$

Definition 4.3. Let $(X, \mathcal{M})$ be a bounded fs log p -adic formal scheme. A crystal

$$\mathcal{E} \in D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$$

is finitely generated unit if for every object $(A, I, M_{\mathrm{Spf} A})^a$ of $(X, \mathcal{M})_{\Delta}$, its evaluation belongs to $D_{\mathrm{fgu}}(A[1/I]_p^{\wedge})$ in the sense of Definition 4.1.

We write $D_{\mathrm{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ for the full subcategory of finitely generated unit crystals, and $\mathbf{Mod}_{\mathrm{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ for its intersection with the heart of $D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Remark 4.4. Let B be a p -adically complete associative ring in which p is central, and let M be a derived p -complete left B -module. Then

$$M \in \mathbf{Perf}(B) \iff (B/p) \otimes_B^L M \in \mathbf{Perf}(B/p).$$

Indeed, use the effectivity statement of [Sta26, Lemma 15.99.4, Tag 09AW] for the tower $\{(B/p^n) \otimes_B^L M\}_{n \geq 1}$; its associative analogue follows by lifting compatible bounded complexes of finite projective modules along the central quotients B/p^n . The same holds in the solid setting; the unit Frobenius condition can also be checked modulo p by derived Nakayama.

When $(X, \mathcal{M})$ satisfies Assumption 3.1 over $W(k)[\zeta_p]$, Remark 4.13 shows that it suffices to test Definition 4.3 on the log q -de Rham prisms associated to framed log affines.

Our first main goals will be to give an explicit local description of this category, as well as deduce strict-étale descent. We note that while $D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$ is a strict-étale stack (by Corollary A.18), it is not immediate that the finitely generated unit subcategory is also a strict-étale stack. While ordinary dualizable objects can be detected termwise in a limit, because we work over $A\langle F \rangle$ we use the thick subcategory generated by the unit (which satisfies no such property). Instead, descent of the fgu condition will use its characterization by coherent roots, which we now study.

Definition 4.5. Let A be as in 4.1, and assume that A is Noetherian regular. If M is a unit Frobenius module over A , a *root* of M is a coherent A -submodule $M_0 \subset M$ together with an injective map $\beta : M_0 \rightarrow \varphi_A^* M_0$ whose iterates identify

$$M \simeq \mathrm{colim}_n \varphi_A^{*n} M_0.$$

It is *minimal* if it has no proper subroot. A distinguished minimal root is a minimal root contained in every other root of M as a submodule of the same ambient unit module.

One can make the analogous definitions for any unit object in the heart of $D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. When referring to a root of a module we mean the former, and when referring to a root of a crystal we mean the latter.

The essential idea of the proof of strict-étale descent is now the following: Proposition 4.8 characterizes the finitely generated unit condition on framed log affines by t -boundedness and the existence of a distinguished minimal coherent root for every $H^i(\mathcal{E}/p)$. Distinguished

minimal roots are unique as subcrystals and therefore glue on intersections, once we know strict-étale descent for the ambient category $D_{\bullet}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$. This reduces the descent problem to descent of coherent Laurent crystals, to which the main results of [Mat22] apply. There are of course many technical details to dispense of, but the core argument is very simple.

Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine in the sense of Definition 3.2 and put $A = R_0^{\square}((q-1))_p^{\wedge}$, the Laurent version of its logarithmic q -de Rham prism. The following theorem identifies finitely generated unit crystals with (φ, Γ) -modules and constructs distinguished minimal roots. A key aspect is that we may ignore solid modules locally and work with continuous semilinear Γ -actions on topological modules. By $D_{\mathrm{fgu}}(A)^{h\Gamma}$, we mean unit objects in

$$\lim_{[s] \in \Delta} \mathrm{Thick}_{C(\Gamma^s, A\langle F \rangle)} D(C(\Gamma^s, A\langle F \rangle))$$

with no solid structures involved. Here $A\langle F \rangle$ has its natural topology (induced from A and the completed direct sum). Note $C(\Gamma^s, A) \widehat{\otimes}_A A\langle F \rangle \simeq C(\Gamma^s, A\langle F \rangle)$ as topological rings.

The condition of lying in these thick subcategories is stable under the pullbacks $C(\Gamma^s, A) \otimes_A^L -$ by nuclearity of $A\langle F \rangle$ (passing to the associated solid condensed module recovers the solid tensor product). For a unit object in the heart, membership in the thick subcategory at $s = 0$ is equivalent to finite generation over $A\langle F \rangle$; the unit hypothesis is essential here. Indeed, the nontrivial implication may be checked modulo p by Remark 4.4: [Bli08, Corollary 2.26] supplies a coherent root, whose two-term induced resolution from [EK04b, Proposition 5.3.3] makes the module perfect.

Theorem 4.6. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine in the sense of Definition 3.2 and put $A = R_0^{\square}((q-1))_p^{\wedge}$. Evaluation on the log q -de Rham prism $(R_0^{\square}[q-1], [p]_q, \mathcal{M})^a$ induces an equivalence*

$$\rho_A^* : D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} D_{\mathrm{fgu}}(A)^{h\Gamma}.$$

This is t -exact for the t -structure on $D_{\bullet}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Proof. Lemma 4.2 shows that the evaluation of every finitely generated unit crystal is p -completely nuclear. Proposition 3.12 therefore gives derived full faithfulness, where the target is solid finitely generated unit modules. Lemma 4.2 identifies the target with ordinary finitely generated unit complexes with a continuous Γ -action, as defined before the theorem as unit objects in $\lim_{[s] \in \Delta} \mathrm{Thick}_{C(\Gamma^s, A\langle F \rangle)} D(C(\Gamma^s, A\langle F \rangle))$. Strictly speaking it applies directly to only the first term $s = 0$, but an identical argument covers all terms.

We first prove essential surjectivity onto the heart of the target modulo p . Put $\overline{A} = A/p$, and let M be a finitely generated unit $\overline{A}$ -module with continuous semilinear Γ -action. Since $\overline{A}$ is regular and F -finite, [Bli08, Corollary 2.26] gives a distinguished minimal root

$$\beta : M_{\min} \hookrightarrow \varphi^* M_{\min}.$$

Every Γ -translate is again a root. Moreover distinguished containment (i.e. its defining property of being a distinguished minimal root) gives $\gamma M_{\min} = M_{\min}$, so β is Γ -equivariant. Theorem 3.5 lifts it to a coherent Laurent crystal $\mathcal{E}_0$. Its unitalization $\mathcal{E}$ satisfies $\rho_A^* \mathcal{E} \simeq M$.

The existence of a root $\mathcal{E}_0$ so that $\mathcal{E} \simeq \operatorname{colim}_k \varphi^{*k} \mathcal{E}_0$ shows that $\mathcal{E}$ is finitely generated unit, as this presentation applied to evaluation on any log prism induces a two-term resolution showing it lies in the thick subcategory of Definition 4.1. Since evaluation is compatible with shifts and fibers, [Hau24, Lemma 5.4.3] upgrades essential surjectivity on the heart modulo p to essential surjectivity on the derived categories. Thus evaluation is an equivalence modulo p . Lemma 2.3 now gives the claim.

We note that in this proof of essential surjectivity implicitly the preimages constructed are all in the heart of $D_{\blacksquare}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. Indeed mod p the preimage is a unitalization of a coherent root, and coherent crystals are in the heart of the solid t -structure by Corollary A.19. The analogous procedure, instead run for exact sequences in the heart of $D_{\operatorname{fgu}}(A)^{h\Gamma}$ to iteratively write heart objects as extensions of p -torsion objects (and taking a limit), constructs preimages in the heart by full faithfulness.

For t -exactness, we note that if $\mathcal{E} \in D_{\operatorname{fgu}}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ is in

$$D_{\operatorname{fgu}}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \cap D_{\blacksquare}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\leq 0}$$

it is automatic that it lands in the ≤ 0 part on the target. For the ≥ 0 part, suppose it fails and pick the smallest r so $N = H^r(\rho_A^* \mathcal{E}) \neq 0$. This is a finitely generated unit Γ -module, so by the previous paragraph's observation has a preimage $\mathcal{E}_N$ which is in $D_{\blacksquare}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\heartsuit}$. By full faithfulness we get a nonzero map $\mathcal{E}_N[-r] \rightarrow \mathcal{E}$, but by the orthogonality defining the canonical t -structure on solid Laurent crystals in Corollary A.19 this must be the zero map. $\square$

Remark 4.7. On the heart one may again work classically, asking that the Γ action is continuous for the weak topology when viewed as a finitely generated unit $A\langle F \rangle$ -module. Indeed as in the previous argument, modulo p there is a distinguished Γ -equivariant minimal root. But then Lemma 3.13 applies, and writing the module via unitalization we see the Γ action is continuous for the weak topology modulo p . In fact we need not work mod p and can run a similar argument to Lemma 3.13, but we warn one must use the unit condition for the second hypothesis on K to work.

By applying Theorem 3.5 we can prove a root criterion for the finitely generated unit condition.

Proposition 4.8. *Let $(\operatorname{Spf} R, \mathcal{M})$ be a framed log affine. A crystal*

$$\mathcal{E} \in D_{\blacksquare}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$$

is finitely generated unit if and only if it is t -bounded and each crystal $H^i(\mathcal{E} // p)$ admits a distinguished minimal root in $\operatorname{Coh}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})_{p=0}$.

Proof. Put $A = R_0^{\square}((q-1))_p^{\wedge}$ and $\overline{A} = A/p$. Suppose first that the condition on roots holds. By Remark 4.4, it is enough to work modulo p . A coherent root is a perfect Laurent crystal, so its evaluation on every log prism is perfect. On a log prism $(B, J, \mathcal{N})^a$, the two-term induced resolution of [EK04b, Proposition 5.3.3] makes its unitalization perfect over $(B[1/J]_p^{\wedge} // p)[F]$. Thus every cohomology module of the evaluation of $\mathcal{E} // p$ on the log q -de Rham prism is perfect over $\overline{A}[F]$. The unit condition is detected on cohomology by [EK04b, 5.4.1]. Since $\mathcal{E}$

is t -bounded and perfect objects are closed under finite extensions, its evaluation is finitely generated unit. Theorem 4.6 and Proposition 3.12 show that $\mathcal{E} // p$ is finitely generated unit.

Conversely, Proposition 3.12 makes evaluation on the log q -de Rham prism t -exact and conservative. After forgetting the solid structure, Lemma 4.2 makes the evaluated complex perfect over $\overline{A}[F]$. Proposition 11.3.9 of [BL19] shows that the evaluation M of $H^i(\mathcal{E} // p)$ is an ordinary finitely generated unit module. As in the proof of Theorem 4.6, its distinguished minimal root $M_{\min}$ is Γ -stable and its root map is Γ -equivariant. Theorem 3.5 lifts this map to a coherent Laurent crystal $\mathcal{E}_{\min}$. Proposition 3.12 identifies its unitalization with $H^i(\mathcal{E} // p)$, since both evaluate to M . Transporting the canonical inclusion into the unitalization gives

$$\mathcal{E}_{\min} \hookrightarrow H^i(\mathcal{E} // p).$$

It is injective by the t -exactness and conservativity of Proposition 3.12, and its iterates unitalize to the ambient crystal. Thus it is a root.

Let $\mathcal{E}'_0$ be any other coherent root, with evaluation M'_0 . Distinguished containment gives a Γ -equivariant inclusion $M_{\min} \hookrightarrow M'_0$. Proposition 3.12 lifts it uniquely to $\mathcal{E}_{\min} \rightarrow \mathcal{E}'_0$ and makes the lift injective by t -exact conservativity. The two composites to $\varphi^* \mathcal{E}'_0$ evaluate to the same map, so the same proposition makes the inclusion compatible with the root maps. Hence $\mathcal{E}_{\min}$ is contained in every root. Any subroot evaluates to a subroot of $M_{\min}$, so minimality and conservativity in Proposition 3.12 make it equal to $\mathcal{E}_{\min}$. Thus $\mathcal{E}_{\min}$ is the distinguished minimal root as a crystal. $\square$

4.2. Strict-étale descent. Proposition 4.8 reduces strict-étale descent to descent of coherent Laurent crystals on $(X, \mathcal{M})_{\Delta}$. The following lemmas give the required compatibility of minimal roots with completion and étale base change.

Lemma 4.9. *Let A be a regular local F -finite $\mathbf{F}_p$ -algebra, let $\widehat{A}$ be its completion, and let $N_0 \subset N$ be a root of a finitely generated unit $A[F]$ -module. Then N_0 is minimal if and only if $\widehat{A} \otimes_A N_0 \subset \widehat{A} \otimes_A N$ is minimal.*

Proof. The forward implication is [Bli04, Corollary 2.11]. Conversely, assume the completed root is minimal and let $N_1 \subset N_0$ be a subroot. Its completion is a subroot, hence $\widehat{A} \otimes_A N_1 = \widehat{A} \otimes_A N_0$.

Since $A \rightarrow \widehat{A}$ is faithfully flat, it follows that $N_1 = N_0$. Thus N_0 has no proper subroot, hence is minimal. $\square$

Lemma 4.10. *Let R be a $(q-1)$ -adically complete Noetherian $k[[q-1]]$ -algebra, topologically of finite type and formally smooth over $k[[q-1]]$, and let S be another such algebra. Suppose $R((q-1)) \rightarrow S((q-1))$ induces an étale morphism of $k((q-1))$ -affinoid spaces. Then scalar extension sends minimal roots of finitely generated unit modules to minimal roots.*

Proof. Put $A = R((q-1))$ and $B = S((q-1))$, and let $N_0 \subset N$ be a minimal root over A , with root map $\beta : N_0 \rightarrow \varphi_A^* N_0$. Both rings are regular F -finite, so the hypotheses in [Bli08] for

the results we will use are satisfied. Flatness of $A \rightarrow B$ and naturality of Frobenius pullback show that $B \otimes_A N_0$ is a root of $B \otimes_A N$, so we only concern ourselves with minimality. More generally, flat pullbacks send roots to roots.

Following [Bli08, §2.4], let $\sigma_\beta^{-1}(N_0) \subset N_0$ be the smallest submodule N'_0 such that $\beta(N_0) \subset \varphi_A^* N'_0$. By [Bli08, Lemma 2.17 and Proposition 2.18] this construction commutes with localization, and a root is minimal precisely when $\sigma_\beta^{-1}(N_0) = N_0$. It follows that a root over A is minimal if and only if its localization at every maximal ideal is minimal, and similarly over B . Together with Lemma 4.9, this shows that minimality may be checked on the *completed* local rings at the classical points of the rigid spaces.

Fix a maximal ideal $\mathfrak{m} \subset B$, let $x \in \text{Spa}(B)$ be the corresponding rigid point, and put $\mathfrak{n} = \mathfrak{m} \cap A$. By [KL15, Definition 8.2.16] and the affinoid basis following [KL15, Definition 8.2.19], the morphism factors near x into affinoid open immersions and a finite étale morphism. Since affinoid open immersions induce isomorphisms on completed local rings by [BGR84, 7.3.2/3], the map

$$\widehat{A_{\mathfrak{n}}} \longrightarrow \widehat{B_{\mathfrak{m}}}$$

is finite étale. Both rings are again regular and F-finite. It therefore remains to show that finite étale scalar extension preserves minimal roots.

Let $A \rightarrow B$ now be a finite étale map between regular F-finite rings. Put $D_A^{(1)} = \text{End}_A(\varphi_{A,*} A)$, and define $D_B^{(1)}$ similarly. The relative Frobenius is an isomorphism by [Sta26, Lemma 41.14.3, Tag 0EBS], and F-finiteness together with Kunz's theorem gives

$$B \otimes_A D_A^{(1)} \simeq D_B^{(1)}.$$

Blickle in [Bli08, §2.4] gives a description of $\sigma_\beta^{-1}(M)$ as $\sigma^{-1}(D_A^{(1)} \beta(M))$, where σ denotes base change $A^{(1)} \otimes_A - : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_{D_A^{(1)}}$ viewing the Frobenius twist $A^{(1)}$ as a left $D_A^{(1)}$ -module and a right A -module via Frobenius.

Writing β_B for the base change to B of a root map β , this gives

$$\sigma_{\beta_B}^{-1}(B \otimes_A N_0) = \sigma^{-1}(D_B^{(1)} \beta(B \otimes_A N_0)).$$

Now substitute $B \otimes_A D_A^{(1)}$ for $D_B^{(1)}$. It follows, using that σ^{-1} commutes with étale base change, that

$$\sigma_{\beta_B}^{-1}(B \otimes_A N_0) = B \otimes_A \sigma_\beta^{-1}(N_0).$$

Thus finite étale pullbacks preserve minimal roots, applying the criterion that $\sigma_\beta^{-1}(N_0) = N_0$. Applying this to the completed local rings above and using the previous criterion proves the claim. $\square$

Corollary 4.11. *Let $f : (\text{Spf } S, \mathcal{M}_{\text{Spf } S}) \rightarrow (\text{Spf } R, \mathcal{M}_{\text{Spf } R})$ be a strict-étale map of framed log affines compatible with the framings. If $\mathcal{E}$ is a finitely generated unit crystal killed by p with distinguished minimal root $\mathcal{E}_{\min}$, then $f^* \mathcal{E}$ is finitely generated unit and $f^* \mathcal{E}_{\min}$ is its distinguished minimal root.*

Proof. The pullback is t -exact because f is strict-étale. Applying Lemma 4.10 to the evaluations on the logarithmic q -de Rham prisms for the compatible framings shows that $f^*\mathcal{E}_{\min}$ is minimal. By [Bli08, Corollary 2.26], the minimal root is unique and contained in every root, so it is distinguished. The claim for crystals now follows from Theorem 4.6 and the argument of Proposition 4.8. $\square$

We can now deduce strict-étale descent.

Theorem 4.12. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $W(k)[\zeta_p]$. Then the assignment*

$$(X, \mathcal{M}) \mapsto D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$$

is a stack for the strict-étale topology on such $(X, \mathcal{M})$.

Proof. By Corollary A.18, $D_{\bullet}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$ satisfies strict-étale descent. The pullback is t -exact, and the cover may be taken finite, so boundedness is local. Let $\{\rho_i : U_i \rightarrow (X, \mathcal{M})\}_{i \in I}$ be a strict-étale cover such that every $\rho_i^*\mathcal{E}$ is finitely generated unit.

Using Remark 4.4 and Proposition 4.8 we may reduce the claim to showing

$$\{\rho_i^*\mathcal{E} \in \text{Mod}_{\text{fgu}}((U_i)_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})_{p=0}\} \implies \{\mathcal{E} \in \text{Mod}_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})_{p=0}\}$$

for $\mathcal{E} \in \text{QCoh}_{\bullet}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$ killed by p , or that the condition can be detected locally on a strict-étale cover. Note that the converse is obvious, because any test log prism for the U_i is also one for $(X, \mathcal{M})$, so the finitely generated unit condition on the right side already includes the conditions imposed on the left hand side.

By Lemma 3.4, after a finite strict-étale refinement we may assume without loss of generality that every $U_i = \text{Spf } R_i$ is a framed log affine. Put $U_{ij} = U_i \times_{(X, \mathcal{M})} U_j$ with projections p_i, p_j ; each affine open of U_{ij} inherits a framing through either projection.

Proposition 4.8 gives distinguished minimal roots $\mathcal{R}_i \subset \rho_i^*\mathcal{E}$. Corollary 4.11, applied with each framing, makes $p_i^*\mathcal{R}_i$ and $p_j^*\mathcal{R}_j$ distinguished minimal roots on any open affine in $\mathcal{E}|_{U_{ij}}$. Since U_{ij} is quasicompact by virtue of X being quasiseparated, we may cover U_{ij} by finitely many such framed log affines.

Their intrinsic uniqueness in Proposition 4.8 gives

$$p_i^*\mathcal{R}_i = p_j^*\mathcal{R}_j \subset \mathcal{E}|_{U_{ij}}.$$

Indeed by the remark after Corollary A.19 one has descent for coherent Laurent crystals $\text{Coh}^{\varphi}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ with a lax Frobenius. Thus we can check agreement locally on the chosen finite open cover of U_{ij} by framed log affines, which also shows agreement of the root maps. These equalities also satisfy the cocycle condition on triple overlaps.

Strict-étale descent for $\text{Coh}^{\varphi}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})_{p=0}$ and $D_{\bullet}^{\varphi}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\heartsuit}_{p=0}$ produces a global coherent root. Restricting any subroot or any other root to the cover shows that the descended root is again distinguished minimal. Injectivity and unitalization are local, so the unitalization of the descended coherent root is $\mathcal{E}$. On a log prism $(A, I, M)^a$, its two-term induced resolution is perfect over $(A[1/I]_p^{\wedge}/p)[F]$. Hence $\mathcal{E}$ is finitely generated unit. $\square$

Remark 4.13. Combined with Theorem 4.6, this shows that the finitely generated unit condition can equivalently be tested upon evaluation by a covering family of log q -de Rham prisms as claimed in the introduction.

Let us record an important fact we showed in the proof of the theorem.

Corollary 4.14. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $W(k)[\zeta_p]$. A crystal*

$$\mathcal{E} \in D_{\bullet}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\varphi=1}$$

is finitely generated unit if and only if it is t -bounded and, for every ℓ , the unit Laurent crystal $H^{\ell}(\mathcal{E}/p)$ admits a unique global minimal coherent root.

Proof. The forward implication is the construction in the proof of Theorem 4.12, applied to the cohomology sheaves of $\mathcal{E}/p$ using Proposition 4.8 locally. The converse follows from the first paragraph of the proof of Proposition 4.8 and Theorem 4.12. $\square$

Remark 4.15. This is the Laurent analogue of the root criterion for finitely generated unit modules on a k -scheme X . By [EK04b, Theorem 6.1.3], a finitely generated unit $\mathcal{O}_{F,X}$ -module admits a coherent root on every quasicompact open; as X is quasicompact, the root is global. Such a root need not carry a crystal structure like in the previous corollary, but it turns out there always exists a root carrying such a structure. Cartier descent gives a fully faithful functor

$$\mathrm{QCoh}(X^{(1)}) \rightarrow \mathrm{QCoh}((X/k)_{\mathrm{cris}})$$

with essential image the crystals with vanishing p -curvature. If $\mathrm{Fr} : X \rightarrow X^{(1)}$ is relative Frobenius, it sends $\mathcal{F}$ to $(\mathrm{Fr}^* \mathcal{F}, \nabla_{\mathrm{can}})$, where $(\mathrm{Fr}^* \mathcal{F})^{\nabla_{\mathrm{can}}=0} \simeq \mathcal{F}$.

Write the absolute Frobenius as $F_X = \pi \circ \mathrm{Fr}$, with $\pi : X^{(1)} \rightarrow X$. Given a coherent root $\beta : M_0 \rightarrow F_X^* M_0$, replace it by

$$F_X^* \beta : F_X^* M_0 \longrightarrow F_X^{*2} M_0.$$

Since F_X is finite flat, this map is injective and has the same unitalization. Moreover, $F_X^* M_0 = \mathrm{Fr}^* \pi^* M_0$ and $F_X^* \beta = \mathrm{Fr}^* (\pi^* \beta)$, so the map is horizontal for the canonical Cartier connections. It is therefore a coherent crystalline root. Uniqueness in Cartier descent identifies its unitalization with the crystal structure on M (similarly) induced by being an $\mathcal{O}_X$ -module with unit Frobenius.

Corollary 4.16. *Let $f : (Y, \mathcal{N}) \rightarrow (X, \mathcal{M})$ be a morphism between pairs satisfying Assumption 3.1 over $W(k)[\zeta_p]$. Then Lf^* preserves finitely generated unit crystals.*

Proof. Let $\mathcal{E}$ be a finitely generated unit crystal on $(X, \mathcal{M})_{\Delta}$, and let $(B, I, M_{\mathrm{Spf} B})^a$ be a log prism over $(Y, \mathcal{N})$. Composition with f makes it a log prism over $(X, \mathcal{M})$, and pullback of crystals gives a Frobenius-equivariant equivalence

$$(Lf^* \mathcal{E})(B, I, M_{\mathrm{Spf} B}) \simeq \mathcal{E}(B, I, M_{\mathrm{Spf} B}).$$

Thus every evaluation of $Lf^* \mathcal{E}$ is perfect over $B[1/I]_p^{\wedge} \langle F \rangle$ by Definition 4.3. $\square$

Theorem 4.12 and Lemma 3.4 permit strict-étale localization to framed log affines, where Theorem 4.6 describes finitely generated unit crystals explicitly.

4.3. The Kummer-étale realization and constructibility. We construct a functor

$$T_{\text{ét}} : D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}} \rightarrow D((X, \mathcal{M})_{\eta, \text{prokét}}, \mathbf{Z}_p).$$

Its essential image consists of p -Kummer complexes constructible for the logarithmic stratification induced by D . We first recall a basis for the target site.

Lemma 4.17. *Let $(X, \mathcal{M})$ be a locally Noetherian fs log p -adic formal scheme. Then the log affinoid perfectoid objects form a basis B of $(X, \mathcal{M})_{\eta, \text{prokét}}$ stable under fiber products.*

Proof. This is [DLLZ23, Propositions 5.3.11 and 5.3.12]. $\square$

Thus it suffices to define $T_{\text{ét}}$ on this basis. For a basis object $U = \varprojlim_i U_i$, we write $\widehat{U} = \text{Spa}(S, S^+)$ for its associated perfectoid space, equipped with the saturated log structure induced by the tower; see [DLLZ23, Remark 5.3.5].

Definition 4.18. Let $(X, \mathcal{M})$ be a locally Noetherian fs log p -adic formal scheme, and let $\mathcal{E}$ be a finitely generated unit Laurent crystal. For $U \in B$ with associated log perfectoid space $(\text{Spa}(S, S^+), \mathcal{M}_S)$, let Δ_{S^+} be the perfect saturated log prism with underlying prism $(W(S^{+,b}), \ker \theta)$ and log structure induced by the tilted log structure on S^b . Set

$$T_{\text{ét}}(\mathcal{E})(U) := \text{RHom}_{D_{\blacksquare}(W(S^b))}(\mathcal{E}(\Delta_{S^+}), W(S^b)) \in \widehat{D}(\mathbf{Z}_p).$$

These assignments define a presheaf on B . We denote its right Kan extension to $(X, \mathcal{M})_{\eta, \text{prokét}}$ by $T_{\text{ét}}(\mathcal{E})$.

Here $D_{\blacksquare}(W(S^b))$ is the inverse limit over n of the categories of $W(S^b)/p^n$ -modules in the ordinary solid category of the analytic ring $\text{Spa}(\mathbb{A}_{\text{inf}}(S^+)/p^n, \mathbb{A}_{\text{inf}}(S^+)/p^n)$. We remark that the perfect (log) prism we use has underlying ring

$$W(S^{+,b}) = \mathbb{A}_{\text{inf}}(S^+),$$

whereas its Laurent variant is

$$W(S^b) = \mathbb{A}_{\text{inf}}(S^+)[1/\xi]_p^{\wedge},$$

with $(\xi) = \ker(\theta)$ the prismatic ideal in $\mathbb{A}_{\text{inf}}(S^+)$ (one can see these are equal using that they are both strict p -rings).

Lemma 4.19. *Let $(X, \mathcal{M})$ be a locally Noetherian fs log p -adic formal scheme. Definition 4.18 yields a well-defined functor*

$$T_{\text{ét}} : D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}} \rightarrow D((X, \mathcal{M})_{\eta, \text{prokét}}, \mathbf{Z}_p).$$

Proof. It is enough to check hyperdescent on B . For $U \in B$, [DLLZ23, Lemma 5.3.8] identifies the localized topos $((X, \mathcal{M})_{\eta, \text{prokét}})_{/U}$ with $\widehat{U}_{\text{proét}}$. Let $U_{\bullet} \rightarrow U$ be a hypercover by basis objects, with associated perfectoid spaces $\widehat{U} = \text{Spa}(S, S^+)$ and $\widehat{U}_m = \text{Spa}(S_m, S_m^+)$. Passage to associated perfectoid spaces preserves fiber products and carries covering families to v -covers, so this gives a v -hypercover of $\widehat{U}$.

Put $A = W(S^b)$ and $A_m = W(S_m^b)$. The v -acyclicity of the tilted structure sheaf on affinoid perfectoid spaces [Sch26a, Theorem 1.2] identifies the derived reductions modulo p of A and $R\varprojlim_m A_m$ (which agree with their ordinary mod p reductions). Both are derived p -complete, so derived Nakayama gives

$$A \simeq R\varprojlim_{[m] \in \Delta} A_m$$

in $D_{\blacksquare}^{\varphi}(A)$, where Frobenius compatibility follows from functoriality of Witt vectors.

Put $M = \mathcal{E}(\Delta_{S^+})$ and $M_m = \mathcal{E}(\Delta_{S_m^+})$. The crystal property shows M_m is simply a base change of M . Using $A \simeq R\varprojlim_{[m] \in \Delta} A_m$, we obtain

$$R\varprojlim_{[m] \in \Delta} \text{RHom}_{D_{\blacksquare}^{\varphi}(A_m)}(M_m, A_m) \simeq \text{RHom}_{D_{\blacksquare}^{\varphi}(A)}(M, A).$$

Thus $T_{\text{ét}}(\mathcal{E})$ is a hypercomplete sheaf. $\square$

It is easy to see that for every morphism $f : (Y, \mathcal{N}) \rightarrow (X, \mathcal{M})$ between pairs satisfying Assumption 3.1, the construction is compatible with pullback:

$$T_{\text{ét}, Y}(Lf^* \mathcal{E}) \simeq Lf_{\eta}^* T_{\text{ét}, X}(\mathcal{E}).$$

The functor $T_{\text{ét}}$ actually lands in the full subcategory of p -Kummer sheaves, which we now define.

Definition 4.20. Let $(X, \mathcal{M})$ be a locally Noetherian fs log p -adic formal scheme. We call $K \in D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$ p -Kummer if prime-to- p logarithmic inertia acts trivially on every cohomology stalk of $K//p^n$ at every log geometric point, for every $n \geq 1$. For a bounded derived lisse sheaf, this agrees with [IKY26, Lemma 1, Definition 2, and Proposition 5]. Lemma 4.22 gives the equivalent description on the perfectoid p -power Kummer tower.

We next prove that $T_{\text{ét}}(\mathcal{E})$ is constructible for the stratification induced by D . We first need the following descent result.

Lemma 4.21. *On pairs satisfying Assumption 3.1 over $\mathcal{O}_K$, the full subcategories of p -Kummer sheaves in*

$$D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

form a stack for the strict-étale topology.

Proof. Let $g : (U, \mathcal{N}) \rightarrow (X, \mathcal{M})$ be a strict-étale cover, and let $U^{(\bullet)}$ be its Čech nerve. Passage to the log adic generic fiber preserves fiber products and carries g to a strict-étale cover. Descent for pro-Kummer-étale sheaves gives

$$D^{(b)}((X, \mathcal{M})_{\eta, \text{prokét}}, \mathbf{Z}_p) \simeq \varprojlim_{[m] \in \Delta} D^{(b)}((U^{(m)}, \mathcal{N}^{(m)})_{\eta, \text{prokét}}, \mathbf{Z}_p).$$

Here local boundedness descends because it is strict-étale local. By strictness, if $X_\eta^{(a)}$ and $U_\eta^{(a)}$ denote the loci where $\text{rk}_{\mathbf{Z}} \overline{\mathcal{M}}_x^{\text{gp}} = a$ (respectively for $\mathcal{N}$), then

$$U_\eta^{(a)} = U_\eta \times_{X_\eta} X_\eta^{(a)}.$$

The logarithmic strata are therefore compatible. Kummer-étale origin of the finite-coefficient reductions descends by Kummer-étale descent and the full faithfulness of pullback [DLLZ23, Proposition 5.1.7]; lissity on the strata also descends.

It remains to check the claim still holds when we restrict to p -Kummer sheaves. Strictness canonically identifies the logarithmic inertia groups, so the condition in Definition 4.20 is strict-étale local. $\square$

By Lemmas 3.4 and 4.21, constructibility for the logarithmic stratification induced by D may be checked on a framed log affine $\text{Spf } R$. Let $R_0^\square[[q-1]]_{\text{perf}}$ be the completed perfection of the framed log q -de Rham prism. If $\text{Spf } R_m$ is the p -completed pullback of the framing along the Kummer map $T_i \mapsto T_i^{p^m}$ over $W(k)[\zeta_{p^m}]$, then

$$(R_0^\square)_\infty := R_0^\square[[q-1]]_{\text{perf}}/([p]_q) = \left(\varinjlim_m R_m \right)_p^\wedge.$$

Its log structure $\mathcal{M}_\infty$ is induced by the saturated chart $\mathbf{N}[1/p]^r \rightarrow (R_0^\square)_\infty$ sending e_i/p^m to T_i^{1/p^m} for $1 \leq i \leq r$. Write $X_{\infty, \eta} = \text{Spa}((R_0^\square)_\infty[1/p], (R_0^\square)_\infty)$. The notation $(X_{\infty, \eta}, \mathcal{M}_\infty)_{\text{prokét}}$ means the localization of $(X, \mathcal{M})_{\eta, \text{prokét}}$ at the pro-object defined by this tower; its Kummer-étale site is understood by finite-level approximation. We use ϖ for the tilted pseudouniformizer, and write $(R_0^\square)_{\infty, \eta}^b = ((R_0^\square)_\infty)^b[1/\varpi]$.

For $m \geq 1$, the Γ -action extends to this tower, for $1 \leq i \leq n$, by

$$\gamma(T_i^{1/p^m}) = q^{pb_i/p^m} T_i^{1/p^m} = \zeta_{p^m}^{b_i} T_i^{1/p^m}.$$

Lemma 4.22. *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine and let $K \in D^{(b)}((\text{Spf } R, \mathcal{M})_{\eta, \text{prokét}}, \mathbf{Z}_p)$ be derived p -complete. For $\Lambda \subseteq \{1, \dots, r\}$, put*

$$X_{\infty, \eta, \Lambda} = X_{\infty, \eta} \times_{X_\eta} X_{\eta, \Lambda},$$

and let $\epsilon_\infty : (X_{\infty, \eta}, \mathcal{M}_\infty)_{\text{prokét}} \rightarrow (X_{\infty, \eta})_{\text{proét}}$ be the natural morphism of sites. The following are equivalent:

- (1) K is constructible for the stratification induced by D and is p -Kummer;
- (2) its pullback to $X_{\infty, \eta}$ is $\epsilon_\infty^* \mathcal{L}_\infty$ for some $\mathcal{L}_\infty \in D^{(b)}((X_{\infty, \eta})_{\text{proét}}, \mathbf{Z}_p)$ which is constructible for the finite stratification $X_{\infty, \eta} = \bigsqcup_\Lambda X_{\infty, \eta, \Lambda}$.

Proof. We first work with $\mathbf{Z}/p^n$ -coefficients. For a log geometric point $\bar{y}^{\log}$ over an ordinary geometric point $\bar{y}$ of $X_{\infty, \eta}$, put

$$I_{\bar{y}} = \mathrm{Hom} \left(\overline{\mathcal{M}}_{\infty, \bar{y}}^{\mathrm{gp}}, \widehat{\mathbf{Z}}(1) \right).$$

The characteristic monoid is p -divisible, so $I_{\bar{y}}$ is pro-(prime-to- p). For a bounded constructible complex $\mathcal{F}$, our convention on finite coefficients allows us to apply the Kummer-étale stalk calculation of [DLLZ23, Lemma 4.4.27]:

$$(\mathrm{R}\epsilon_{\infty, *} \mathcal{F})_{\bar{y}} \simeq \mathrm{R}\Gamma_{\mathrm{cont}}(I_{\bar{y}}, \mathcal{F}_{\bar{y}^{\log}})$$

Indeed, take pointed étale neighborhoods and adjoin the remaining prime-to- p roots of the chart. The same strict-local calculation computes cohomology by the log geometric stalk, with descent group $I_{\bar{y}}$. Propositions 5.1.6, 5.1.7, and 5.2.1 of [DLLZ23] justify passage from the finite-level Kummer-étale calculation to the tower and its pro-Kummer-étale site.

Taking $I_{\bar{y}}$ -invariants is exact on discrete continuous $\mathbf{Z}/p^n$ -representations, since it is pro-(prime-to- p). Hence the adjunction unit

$$\mathcal{F} \rightarrow \mathrm{R}\epsilon_{\infty, *} \epsilon_{\infty}^* \mathcal{F}$$

is an equivalence for every bounded constructible $\mathcal{F}$. Moreover, the counit $\epsilon_{\infty}^* \mathrm{R}\epsilon_{\infty, *} \mathcal{F} \rightarrow \mathcal{F}$ is an equivalence if and only if $I_{\bar{y}}$ acts trivially on every $H^a(\mathcal{F}_{\bar{y}^{\log}})$. Here the calculation commutes with strict restriction to strata by [DLLZ23, Lemma 4.5.4]; on each stratum, taking these invariants preserves finite local systems. Thus $\mathrm{R}\epsilon_{\infty, *} \mathcal{F}$ is constructible, and the counit can be tested on Kummer-étale stalks. This proves that ϵ_{∞}^* is fully faithful on constructible complexes, with essential image characterized by trivial logarithmic inertia on cohomology stalks. The assertions with $\mathbf{Z}_p$ coefficients follow by derived p -completion.

We may now prove the claim. Write $\pi : (X_{\infty, \eta}, \mathcal{M}_{\infty}) \rightarrow (X, \mathcal{M})_{\eta}$ for the tower map. If (1) holds, Definition 4.20 makes logarithmic inertia act trivially on the cohomology stalks of $\pi^* K$. The preceding calculation gives (2), with $\mathcal{L}_{\infty} = \mathrm{R}\epsilon_{\infty, *} \pi^* K$.

Conversely, suppose (2) holds. Kummer-étale origin of the finite-coefficient reductions descends along π by the equalizer argument of [BS15, Lemma 5.1.4], using [DLLZ23, Proposition 5.1.6] for Kummer-étale presentations. The bounded derived assertion follows by truncation and full faithfulness. Derived lissity also descends on each stratum, so K is constructible. At a point of $X_{\eta, \Lambda}$, the map on characteristic groups is $\mathbf{Z}^{|\Lambda|} \rightarrow \mathbf{Z}[1/p]^{|\Lambda|}$, and hence

$$I_{\bar{y}}^{\log} \simeq \mathrm{Hom} \left(\mathbf{Z}[1/p]^{|\Lambda|}, \widehat{\mathbf{Z}}(1) \right) \simeq I_{\pi(\bar{y})}^{\log, (p')}.$$

Thus prime-to- p logarithmic inertia acts trivially on the cohomology stalks of every finite-coefficient reduction of K , so K is p -Kummer by Definition 4.20. $\square$

Let $T_{\mathrm{ét}}^{\mathrm{perf}}$ denote the scheme-theoretic solution functor on $\mathrm{Spec}(\mathrm{R}_0^{\square})_{\infty, \eta}^b$ of Theorem 2.6. For a finitely generated unit module M over $\mathrm{R}_0^{\square}((q-1))_p^{\wedge}$, we use the convention

$$T_{\mathrm{ét}}^{\mathrm{perf}}(M) := T_{\mathrm{ét}}^{\mathrm{perf}} \left(M \otimes_{\mathrm{R}_0^{\square}((q-1))_p^{\wedge}}^{L, \blacksquare} W((\mathrm{R}_0^{\square})_{\infty, \eta}^b) \right),$$

For $M \in D_{\text{fgu}}(W((R_0^\square)_{\infty, \eta}^b))$, its value after base change to $\text{Spec } S^b$ is

$$\text{RHom}_{\widehat{D}^\varphi(W(S^b))} \left(\left(M \otimes_{W((R_0^\square)_{\infty, \eta}^b)}^L W(S^b) \right)_p^\wedge, W(S^b) \right) \in D(\mathbf{Z}_p).$$

The following lemma controls the Fitting ideals on each stratum. Using the minimal root characterization of being a finitely generated unit module, this will be crucial in sufficiently constraining $\text{Mod}_{\text{fgu}}(R_0^\square((q-1))_p^\wedge)^{h\Gamma}$ enough to show the essential image of $T_{\text{ét}}$ is constructible.

Lemma 4.23. *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine with $\overline{R} = R_0/p$. If $\mathfrak{a} \subset \overline{R}^\square((q-1))$ is stable under $\mathbf{Z}_p(1)^n \subset \Gamma$, then, on each connected component of*

$$\text{Spa } \overline{R}^\square((q-1)) \setminus V(T_1 \cdots T_r)$$

the image of $\mathfrak{a}[1/T_1, \dots, 1/T_r]$ is either zero or the unit ideal.

Proof. Put $X = \text{Spa } \overline{R}^\square((q-1))$ and $X^\circ = X \setminus V(T_1 \cdots T_r)$. The framing gives an étale map

$$X \longrightarrow \text{Spa } k((q-1))\langle T_1, \dots, T_r, T_{r+1}^{\pm 1}, \dots, T_n^{\pm 1} \rangle.$$

The target is reduced, hence so is X , and every T_i is a unit on X° . The i th copy of $\mathbf{Z}_p(1)$ acts by $T_i \mapsto q^{pb_i} T_i$ and fixes the other coordinates.

Put $\mathcal{I} = \mathfrak{a}\mathcal{O}_{X^\circ}$. Let x be a classical point at which $\mathcal{I}_x$ is proper. After replacing $k((q-1))$ by a finite extension containing the residue field of x , choose a rational lift $\tilde{x}$ of x . The rigid analytic inverse function theorem identifies a neighborhood of $\tilde{x}$ with a polydisk. We note vanishing may be checked after passage to this extension. Taking the elements of $\mathbf{Z}_p(1)^n \leq \Gamma$ with coordinates $b_i = p^{m_i-1}$ for integers m_i , stability gives zeros of every $f \in \mathfrak{a}$ at the points with coordinates $q^{p^{m_i}} T_i(\tilde{x})$ for large $m_i \geq 1$ lying in this polydisk. Since

$$q^{p^m} - 1 = (q-1)^{p^m} \longrightarrow 0.$$

these points form a product of convergent sequences.

Applying the single variable rigid analytic identity theorem repeatedly, as in [Pen25, Lemma 3.9], show that a finite set of generators of $\mathfrak{a}$ vanishes on a neighborhood of $\tilde{x}$. Faithful flatness gives $\mathcal{I}_x = 0$, so if $\mathcal{I}_x$ is not proper ideal then $\mathcal{I}_x = \mathcal{O}_{X^\circ, x}$.

The closed analytic locus

$$\text{Supp}(\mathcal{I}) \cap \text{Supp}(\mathcal{O}_{X^\circ}/\mathcal{I})$$

has no classical points. If it were nonempty, a point in it would admit an affinoid neighborhood in X° . Its intersection with this neighborhood would be a nonempty closed analytic locus and hence would contain a classical point by the affinoid Nullstellensatz [BGR84, 5.2.6/3 and 7.1.3/1], a contradiction. The zero and unit loci of $\mathcal{I}$ are therefore complementary open and closed subsets of X° . On each connected component, $\mathcal{I}$ is zero or the unit ideal. $\square$

Lemma 4.24. *Let $A = \overline{R}^\square((q-1))$ for a framed log affine, and let $J = (T_i \mid i \in \Lambda)$ for some $\Lambda \subset \{1, \dots, r\}$. If $M \in D_{\text{fgu}}(A)^{h\Gamma}$, then*

$$(A/J) \otimes_A^L M$$

has bounded finitely generated unit cohomology over A/J and carries the induced Γ -action.

Proof. Frobenius preserves J , and extension of scalars gives

$$(A/J)[F] \otimes_{A[F]}^L M \simeq (A/J) \otimes_A^L M.$$

The resulting complex is unit because Frobenius pullback commutes with extension of scalars. It is perfect over $(A/J)[F]$, so its underlying A/J -complex is bounded. Since A/J is regular, Noetherian, and F -finite, [BL19, Proposition 11.3.9] makes its cohomology modules finitely generated unit. Finally, J is Γ -stable and Γ commutes with Frobenius, so the quotient map is Γ -equivariant. $\square$

Proposition 4.25. *Let $(\mathrm{Spf} R, \mathcal{M}_{\mathrm{Spf} R})$ be a framed log affine with framing $\square$, and let $\mathcal{E}$ be an object of $D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M}_{\mathrm{Spf} R})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. Then $T_{\mathrm{\acute{e}t}}^{\mathrm{perf}}(\mathcal{E}(R_0^{\square}((q-1))_p^{\wedge}))$ is constructible for the stratification of $\mathrm{Spec}(R_0^{\square})_{\infty, \eta}^b$ cut out by the divisor $D_{\infty} = \bigcup_i V(T_i^b)$.*

Proof. The evaluated crystal is perfect over $W((R_0^{\square})_{\infty, \eta}^b)\langle F \rangle$. Consequently,

$$T_{\mathrm{\acute{e}t}}^{\mathrm{perf}}(\mathcal{E}(R_0^{\square}((q-1))_p^{\wedge})) \otimes_{\mathbf{Z}_p}^L \mathbf{F}_p \simeq T_{\mathrm{\acute{e}t}}^{\mathrm{perf}}((\mathcal{E}/p)(R_0^{\square}((q-1))_p^{\wedge}))[1].$$

Constructibility is detected modulo p , so assume that $\mathcal{E}$ is p -torsion and write $A = \overline{R}_0^{\square}((q-1))$. Evaluation identifies each cohomology module with a finitely generated unit Γ -module M . Let $\beta : M_0 \rightarrow \varphi^* M_0$ be its distinguished minimal coherent root. It is Γ -equivariant by the distinguished property.

Each $\mathrm{Fitt}_i(M_0)$ is $\mathbf{Z}_p(1)^n$ -stable. On a connected component of the open stratum $\mathrm{Spec} \overline{R}_0^{\square}((q-1))[1/T_1, \dots, 1/T_r]$, let ρ be the least index for which the restriction of $\mathrm{Fitt}_{\rho}(M_0)$ is the unit ideal (we see $\mathrm{Fitt}_{-1} = 0$). After restriction to this component, Lemma 4.23 gives $\mathrm{Fitt}_{\rho-1}(M_0) = 0$ and $\mathrm{Fitt}_{\rho}(M_0) = \mathcal{O}$. The Fitting ideal criterion [Sta26, Lemma 15.8.8, Tag 07ZD] makes M_0 locally free of rank ρ there. We note $\varphi^* M_0$ has the same rank.

Thus β is an injection between vector bundles of the same rank. Moreover β is Γ -equivariant, so by the same argument $\mathrm{coker} \beta$ must again be a vector bundle. This is forced to be rank zero, and we conclude that β is an isomorphism: in particular restricted to the open stratum $M = M_0$. It follows $T_{\mathrm{\acute{e}t}}^{\mathrm{perf}}(\mathcal{E}(R_0^{\square}((q-1))_p^{\wedge}))$ is lisse on the open stratum by (the dualized version of) Theorem 2.1. One can make a similar argument for any locally closed stratum, by first taking an appropriate quotient of $\overline{R}_0^{\square}((q-1))$ before inverting any of the T_i (using Lemma 4.24). $\square$

Remark 4.26. For the trivial log structure there is only the open stratum, so the same argument shows we obtain a derived lisse complex. In particular without log structures, the theory degenerates to the ordinary theory of (φ, Γ) -modules.

Let $\nu : (X_{\infty, \eta}^b)_{\mathrm{\acute{e}t}} \rightarrow \mathrm{Spec}((R_0^{\square})_{\infty, \eta}^b)_{\mathrm{\acute{e}t}}$ be the support map of [Bha25, Notation 6.2.1], and let $\lambda : (X_{\infty, \eta}^b)_{\mathrm{pro\acute{e}t}} \rightarrow (X_{\infty, \eta}^b)_{\mathrm{\acute{e}t}}$ be the natural morphism. For a constructible $\mathbf{Z}_p$ -complex K , set

$$\mathrm{an}^* K := \varprojlim_m \lambda^* \nu^* (K/p^m) \in D((X_{\infty, \eta}^b)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p).$$

For a Γ -equivariant constructible complex K , set

$$\mathrm{an}_\Gamma^* K := (\epsilon_\infty^*((\mathrm{an}^* K)^b))^{\hbar\Gamma} \in D((\mathrm{Spf} R, \mathcal{M})_{\eta, \mathrm{prok\acute{e}t}}, \mathbf{Z}_p),$$

where $(-)^b$ denotes transport across the tilting equivalence and $\hbar\Gamma$ denotes descent along $X_{\infty, \eta} \rightarrow X_\eta$.

Lemma 4.27. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine with framing $\square : (\mathrm{Spf} R, \mathcal{M}) \rightarrow \mathbb{T}_r^n$. Evaluation on $\Delta_{(R_0^\square)_\infty}$ gives a natural commutative diagram*

$$\begin{array}{ccc} D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\mathrm{op}} & \xrightarrow{T_{\acute{\mathrm{e}t}}} & D((\mathrm{Spf} R, \mathcal{M})_{\eta, \mathrm{prok\acute{e}t}}, \mathbf{Z}_p) \\ \mathcal{E} \mapsto \mathcal{E}(\Delta_{(R_0^\square)_\infty}) \downarrow & & \uparrow \mathrm{an}_\Gamma^* \\ (D_{\mathrm{fgu}}(W((R_0^\square)_{\infty, \eta}^b))^{\hbar\Gamma})^{\mathrm{op}} & \xrightarrow{T_{\acute{\mathrm{e}t}}^{\mathrm{perf}}} & D_{\mathrm{cons}}^{(b)}(\mathrm{Spec}(R_0^\square)_{\infty, \eta}^b, \mathbf{Z}_p)^{\hbar\Gamma} \end{array}$$

Proof. Put $A_\infty = W((R_0^\square)_{\infty, \eta}^b)$. Let U belong to the log affinoid perfectoid basis of $(X_{\infty, \eta}, \mathcal{M}_\infty)_{\mathrm{prok\acute{e}t}}$, with associated log perfectoid space $(\mathrm{Spa}(S, S^+), \mathcal{M}_S)$, and put $A_S = W(S^b)$. Applying Lemma 4.2 termwise to the continuous Γ cobar diagram, $\mathcal{E}(\Delta_{(R_0^\square)_\infty})$ corresponds functorially to an object $M_\infty \in D_{\mathrm{fgu}}(A_\infty)^{\hbar\Gamma}$. Since $(X_{\infty, \eta}, \mathcal{M}_\infty) \rightarrow (\mathrm{Spf} R, \mathcal{M})_\eta$ is the Γ -cover defining $\hbar\Gamma$ descent, it suffices to compare the two paths on U before descent. The definition of an^* , tilting, the slice comparison of [DLLZ23, Lemma 5.3.8], and base change compatibility give

$$\epsilon_\infty^*((\mathrm{an}^* T_{\acute{\mathrm{e}t}}^{\mathrm{perf}}(M_\infty))^b)(U) \simeq \mathrm{RHom}_{\widehat{D}^\varphi(A_S)} \left((M_\infty \otimes_{A_\infty}^L A_S)_p^\wedge, A_S \right).$$

On the other hand, Definition 4.18 gives

$$T_{\acute{\mathrm{e}t}}(\mathcal{E})(U) \simeq \mathrm{RHom}_{\widehat{D}^\varphi(A_S)} (\mathcal{E}(\Delta_{S^+}), A_S).$$

The comparison functor of Lemma 4.2 and the crystal property give

$$\left((M_\infty \otimes_{A_\infty}^L A_S)_p^\wedge \right) (*) \simeq M_\infty(*) \otimes_{A_\infty(*)}^{L, \blacksquare} A_S(*) \simeq \mathcal{E}(\Delta_{S^+}).$$

The first equivalence holds on the generator $A_\infty \langle F \rangle$, where both sides are $A_S(*) \langle F \rangle$, and hence on its thick closure. Moreover $0 \rightarrow A_S \langle F \rangle \xrightarrow{\cdot(F-1)} A_S \langle F \rangle \rightarrow A_S \rightarrow 0$ places A_S in this thick closure. Lemma 4.2 now identifies the two mapping complexes.

These equivalences are natural in U and equivariant for the log prism automorphisms defining the Γ -action. It follows on the saturated log affinoid perfectoid basis that the two paths agree; taking $\hbar\Gamma$ proves the claim. $\square$

Remark 4.28. The factorization in Lemma 4.27 shows that Definition 4.18 is symmetric monoidal. For $T_{\acute{\mathrm{e}t}}^{\mathrm{perf}}$, the mod p tensor comparison to test symmetric monoidality follows from [BL19, Theorem 8.4.1] and its identification with the Emerton-Kisin solution functor in [BL19, Theorems 12.5.4 and 12.6.1]; the integral statement follows by derived p -completeness. Analytification, tilting, and descent are symmetric monoidal.

The same factorization controls the essential image of $T_{\text{ét}}$.

Proposition 4.29. *Every object in the essential image of $T_{\text{ét}}$ on $D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}}$ is a p -Kummer sheaf in $D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$.*

Proof. The source satisfies strict-étale descent by Theorem 4.12, while Lemma 4.21 shows that being p -Kummer and constructible is strict-étale local. Lemma 3.4 therefore reduces the claim to a framed log affine.

Put $\mathcal{F} = T_{\text{ét}}^{\text{perf}}(\mathcal{E}(\Delta_{(R_0^{\square})_{\infty}}))$. By Proposition 4.25, it is constructible for the stratification cut out by $\bigcup_i V(T_i^b)$. Analytification and tilting give a constructible pro-étale complex: its finite-coefficient reductions come from the étale site by definition of an^* , and its restriction to every stratum cut out by $\bigcup_i V(T_i)$ is derived lisse. Lemma 4.27 gives

$$\pi^* T_{\text{ét}}(\mathcal{E}) \simeq \epsilon_{\infty}^*((\text{an}^* \mathcal{F})^b).$$

Finally, Lemma 4.22 proves the claim. $\square$

5. THE MAIN EQUIVALENCE

We now prove that

$$T_{\text{ét}} : D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}} \longrightarrow D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

is fully faithful, with essential image the p -Kummer sheaves. Theorem 4.12 and Lemmas 3.4 and 4.21 reduce the assertion to framed affines, where Theorem 4.6 describes the source by (φ, Γ) -modules.

The first step constructs crystals realizing $j_! \mathbb{L}$ for pro-étale $\mathbf{Z}_p$ -local systems on $U_{\eta} = X_{\eta} \setminus D_{\eta}$. It begins with $\mathcal{O}_{\Delta}(*D)$, whose realization is $j_! \mathbf{Z}_p$, where

$$j_! : \text{Loc}_{\mathbf{Z}_p}(U_{\eta}) \rightarrow D((X, \mathcal{M})_{\eta, \text{prokét}}, \mathbf{Z}_p).$$

Lemma 4.27 reduces full faithfulness to that of analytification, supplied by Bhatt-Lurie's prismatic Riemann-Hilbert functor [Bha25].

For essential surjectivity on the open stratum, the basic strategy is as follows. Lütkebohmert's theorem extends $\mathbb{L}$ to a Kummer-étale local system $\mathbb{L}'$ on $(X, \mathcal{M})_{\eta}$; after this a finite prime-to- p Kummer Galois cover $\pi : Y_{\eta} \rightarrow X_{\eta}$ kills its prime-to- p log inertia. This allows us to apply Theorem 7 of [IKY26], which produces a crystal $\mathcal{E}_Y$ for $\pi^* \mathbb{L}'$; tensoring with $\mathcal{O}_{\Delta}(*D_Y)$ realizes $j_{Y, \eta, !} \pi_{U_{\eta}}^* \mathbb{L}$, which we prove descends to a crystal realizing $j_! \mathbb{L}$.

We next prove a Γ -equivariant variant of Kashiwara's lemma, reducing to [EK04b]. This gives the induction on dimension, apart from full faithfulness for maps between open and closed strata. For essential surjectivity, the target is thickly generated by $j_! \mathbb{L}$ for inclusions j of locally closed strata, and the preceding constructions realize these generators.

5.1. The open stratum. We construct $\mathcal{O}_\Delta(*D)$ from the line bundle with section attached to the log structure, following the analogous log-crystalline construction of [Tsu99, §5]. Assume $(X, \mathcal{M})$ satisfies Assumption 3.1 over $W(k)[\zeta_p]$

For a sheaf of groups G , a right G -torsor P , and a left G -sheaf V , we write $P \times^G V$ for the contracted product, with $(pg, v) \sim (p, gv)$.

Definition 5.1. Let $\mathcal{M}(-D)$ be the $\mathcal{O}_X^\times$ -torsor of local sections $m \in \mathcal{M}$ such that $\alpha_X(m)\mathcal{O}_X = \mathcal{O}_X(-D)$. Let $(A, I, M_{\text{Spf } A})^a$ be a saturated log prism in $(X, \mathcal{M})_\Delta$, with structure map

$$f : (\text{Spf } A/I, M_{\text{Spf } A/I}) \rightarrow (X, \mathcal{M}).$$

Let $i : \text{Spf } A/I \rightarrow \text{Spf } A$ be the closed immersion. On $(\text{Spf } A/I)_{\text{ét}}$, set $f^*\mathcal{M}(-D) := f^{-1}\mathcal{M}(-D) \times^{f^{-1}\mathcal{O}_X^\times} \mathcal{O}_{\text{Spf } A/I}^\times$. The log structure map gives $f^*\mathcal{M}(-D) \rightarrow M_{\text{Spf } A/I}$. On $(\text{Spf } A)_{\text{ét}}$, set $\widetilde{\mathcal{M}}_A(-D) := M_{\text{Spf } A} \times_{i_*M_{\text{Spf } A/I}} i_*f^*\mathcal{M}(-D)$. Define

$$\mathcal{O}_\Delta(-D)(A, I, M_{\text{Spf } A}) := \Gamma\left(\text{Spf } A, \widetilde{\mathcal{M}}_A(-D) \times^{\mathcal{O}_{\text{Spf } A}^\times} \mathcal{O}_{\text{Spf } A}\right).$$

From this, we can construct an important finitely generated unit crystal corresponding to $j_!\mathbf{Z}_p$ under $T_{\text{ét}}$.

Definition 5.2. For $n \geq 0$, set

$$\mathcal{O}_\Delta(nD) := \underline{\text{Hom}}_{\mathcal{O}_\Delta}\left(\mathcal{O}_\Delta(-D)^{\otimes \blacksquare n}, \mathcal{O}_\Delta\right).$$

The map $\mathcal{O}_\Delta(-D) \rightarrow \mathcal{O}_\Delta$ gives transition maps $\mathcal{O}_\Delta(nD) \rightarrow \mathcal{O}_\Delta((n+1)D)$. We define

$$\mathcal{O}_\Delta(*D) := \left((\text{colim}_{n \geq 0} \mathcal{O}_\Delta(nD))_p^\wedge [1/I_\Delta]\right)_p^\wedge.$$

Strict-étale locally on a framed log affine, if D is cut out by parameters $T_1, \dots, T_r$, then $\mathcal{O}_\Delta(-D)$ is trivial and its map to $\mathcal{O}_\Delta$ is multiplication by a lift of $T_1 \cdots T_r$. Hence $\mathcal{O}_\Delta(*D)$ evaluates on the corresponding log q -de Rham prism as $R_0^\square((q-1))_p^\wedge [1/T_1, \dots, 1/T_r]_p^\wedge$. Logarithmic Frobenius carries the canonical section to its p th power up to a unit, so Frobenius on $\mathcal{O}_\Delta$ extends to $\mathcal{O}_\Delta(*D)$.

Since $\mathcal{O}_\Delta(*D)$ is a sheaf of rings, we can consider $\text{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1}$.

This category has the following local description. On a framed log affine, put $A = R_0^\square((q-1))_p^\wedge$ and $A^\circ = A[1/T_1, \dots, 1/T_r]_p^\wedge$. If $A^{(m)}$ is the degree m term of the Laurentized log q -de Rham Čech nerve, set

$$A^{\circ, (m)} = A^{(m)} \otimes_A^{\text{L}, \blacksquare} A^\circ \simeq A^{(m)}[1/T_1, \dots, 1/T_r]_p^\wedge.$$

Here the T_i denote their pullbacks to $A^{(m)}$; strictness makes the resulting localization independent of the projection to the degree zero term. These rings are the evaluations of $\mathcal{O}_\Delta(*D)$

on the Čech nerve. Lemma A.17 and the compatibility of module objects with limits give

$$D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D)) \simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A^{\circ, (m)}).$$

Here $\mathbf{Perf}$ denotes dualizable objects. Dualizability is detected termwise in a limit of symmetric monoidal categories, so this equivalence restricts to

$$\mathbf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D)) \simeq \varprojlim_{[m] \in \Delta} \mathbf{Perf}(A^{\circ, (m)}).$$

Applying $\varphi = 1$, since this is an equalizer of stable ∞ -categories, gives the analogous description.

Lemma 5.3. *We have $\mathcal{O}_{\Delta}(*D) \in D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. Moreover any crystal in $\mathbf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1}$ is also in $D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.*

Proof. This can be checked strict-étale locally by Theorem 4.12 using the log q -de Rham prism $(R_0^{\square}[[q-1]], [p]_q, M)^a$; set $A = R_0^{\square}((q-1))_p^{\wedge}$. Then we can apply the criterion in Theorem 4.6 to check the finitely generated unit condition. On a framed chart where D is defined by $T_1 \cdots T_r = 0$, the mod p coherent root is generated by $(T_1 \cdots T_r)^{-1}$, and Frobenius carries it to the root generated by $(T_1 \cdots T_r)^{-p}$. Its unitalization is precisely $A[1/T_1, \dots, 1/T_r]_p^{\wedge}$.

For the second assertion, Remark 4.4 and a Postnikov dévissage using [Hau24, Lemma 5.4.3] reduce to a finitely generated unit Frobenius module P over $A[1/T_1, \dots, 1/T_r]/p$ whose underlying module is perfect. It is therefore coherent. Put $T = T_1 \cdots T_r$, choose a coherent A/p -submodule $P_0 \subset P$ with $P_0[1/T] = P$, and let $\beta : P \rightarrow \varphi^*P$ be the inverse linearized Frobenius. For some $N \geq 0$,

$$\beta(P_0) \subseteq T^{-N} \varphi^*P_0.$$

If $(p-1)a > N$, then $T^{-a}P_0 \rightarrow \varphi^*(T^{-a}P_0)$ is a coherent root. Indeed, if $\Phi : \varphi^*P \rightarrow P$ is the linearized Frobenius and $d = (p-1)a - N > 0$, then

$$\Phi^n(\varphi^{*n}(T^{-a}P_0)) \supseteq T^{-d(1+p+\dots+p^{n-1})} T^{-a}P_0.$$

Its unitalization is therefore P . □

Theorem 5.4. *Let $(X, \mathcal{M})$ be as in Assumption 3.1 over $W(k)[\zeta_p]$. Then $T_{\text{ét}}$ fits in a commutative diagram*

$$\begin{array}{ccc} \mathbf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1, \text{op}} & \xrightarrow{\sim} & D_{\text{lis}}^{(b)}((X_{\eta} \setminus D_{\eta})_{\text{proét}}, \mathbf{Z}_p) \\ \downarrow & & \downarrow j! \\ D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}} & \xrightarrow{T_{\text{ét}}} & D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p). \end{array}$$

The top horizontal arrow is an equivalence.

Remark 5.5. Although [BS23, Corollary 3.7] describes the target using ordinary Laurent F-crystals, it is not directly suited to the functor $j_!$. The complement $X_\eta \setminus D_\eta$ need not be quasicompact; for example, $\mathbf{P}_{\mathbf{Q}_p}^1 \setminus \{\infty\} = \mathbf{A}_{\mathbf{Q}_p}^1$. Moreover, a formal model of this complement need not map to the fixed smooth proper model X with generic fiber j . Raynaud models require noncanonical admissible blowups and need not remain smooth. We fix this problem by using log formal schemes, as in Assumption 3.1.

Lemma 4.27 computes $T_{\text{ét}}(\mathcal{O}_\Delta(*D)) \simeq j_! \mathbf{Z}_p$, and Remark 4.28 identifies the bottom horizontal functor in the diagram of Theorem 5.4 with applying $j_!$ to the lisse sheaf on the open stratum.

Lemma 5.6. *Let $(X, \mathcal{M})$ be as in Assumption 3.1 over $W(k)[\zeta_p]$. The category*

$$\text{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1, \text{op}}$$

is a stack for the strict-étale topology on such $(X, \mathcal{M})$.

Proof. Let $g : (U, \mathcal{N}) \rightarrow (X, \mathcal{M})$ be strict-étale and put $D_U = D \times_X U$. There is a canonical Frobenius-equivariant equivalence

$$g^* \mathcal{O}_{\Delta, X}(*D) \simeq \mathcal{O}_{\Delta, U}(*D_U).$$

Indeed, strictness identifies $g^* \mathcal{M}(-D)$ with $\mathcal{N}(-D_U)$. In particular $\mathcal{O}_\Delta(*D)$ commutes with strict-étale pullback.

Let $U^{(\bullet)}$ be the Čech nerve of a strict-étale cover of X . Corollary A.18 gives symmetric monoidal descent for solid logarithmic Laurent crystals, under which $\mathcal{O}_{\Delta, X}(*D)$ corresponds to the algebras $\mathcal{O}_{\Delta, U^{(m)}}(*D_{U^{(m)}})$. One then also obtains descent for their module categories. Dualizability is detected termwise in a limit, and imposing the unit Frobenius structure is a limit. Hence the same descent statement holds for the category in the lemma. Taking opposites preserves limits. $\square$

Remark 5.7. Let $\pi = \zeta_p - 1$ be a uniformizer of $W(k)[\zeta_p]$. If one does not invert I_Δ and instead takes the unitalization of $\mathcal{O}_\Delta(D) \rightarrow \varphi^* \mathcal{O}_\Delta(D)$, one can calculate that the crystalline realization (i.e. the unitalization $\mathcal{E}$ of $\mathcal{O}_{\text{cris}}(D) \rightarrow \varphi^* \mathcal{O}_{\text{cris}}(D)$) satisfies

$$\text{RHom}_{D^\varphi((X, \mathcal{M})_{\pi=0}/W(k))_{\text{cris}}, \mathcal{O}_{\text{cris}}}(\mathcal{E}, \mathcal{O}_{\text{cris}}) \simeq (\text{R}\Gamma_{\text{cris}, c}((X, \mathcal{M})_{\pi=0}/W(k)))^{\varphi=1}.$$

Here compactly supported log crystalline cohomology is as defined in [Tsu99]. This shows the crystalline realization of the natural non-Laurentized variant is compatible with the expectation one would have from the open C_{st} conjecture.

Lemma 5.8. *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine. Then*

$$\text{Perf}((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1} \simeq \text{Perf}(R_0^\square((q-1)_p^\wedge[1/T_1, \dots, 1/T_r]_p^\wedge)^{\varphi=1, h\Gamma}).$$

Proof. Put $A = R_0^\square((q-1))_p^\wedge$ and $A^\circ = A[1/T_1, \dots, 1/T_r]_p^\wedge$. Evaluation is symmetric monoidal and carries $\mathcal{O}_\Delta(*D)$ to A° with its Γ -action. Lemma 5.3 embeds the source into the fgu category, and Theorem 4.6 is fully faithful there. Modules over an idempotent localization form a full subcategory, so the displayed functor is fully faithful.

Conversely, let M be a perfect unit Frobenius A° -complex with Γ action. The algebraic argument in the proof of Lemma 5.3 makes its restriction to A finitely generated unit. Theorem 4.6 gives an fgu crystal $\mathcal{E}$ with evaluation M . The localization map

$$\mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{O}_\Delta(*D)$$

is an equivalence after evaluation, hence is an equivalence by conservativity. Thus $\mathcal{E}$ is an $\mathcal{O}_\Delta(*D)$ -module. The local description $\mathrm{Perf}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D)) \simeq \varprojlim_{[m] \in \Delta} \mathrm{Perf}(A^{\circ, (m)})$ and the crystal property identify its degree m evaluation with

$$A^{\circ, (m)} \widehat{\otimes}_{A^\circ}^L M,$$

which is perfect over $A^{\circ, (m)}$. Since dualizability is detected termwise, $\mathcal{E}$ is perfect over $\mathcal{O}_\Delta(*D)$. Hence the fgu equivalence restricts to the displayed equivalence. $\square$

We now tackle full faithfulness. Our strategy will be to show this in the perfectoid setting (using the formulation of $T_{\text{ét}}$ in Lemma 4.27). However unlike Laurent F -crystals in perfect complexes, it is not possible to use only the unit Frobenius to reduce to the perfectoid setting. We explain this further in Remark 5.13.

Proposition 5.9. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine. The base change functor*

$$\mathrm{Perf}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1} \rightarrow \mathrm{Perf}(\mathbb{A}_{\text{inf}}((R_0^\square)_\infty)[1/\xi, 1/[T_1^b], \dots, 1/[T_r^b]]_p^\wedge)^{\varphi=1, h\Gamma}$$

is fully faithful.

Proof. As usual, associated to the framed log affine we have a logarithmic q -de Rham prism, so we let $A = R_0^\square((q-1))_p^\wedge$. For this particular proof we only care about a localization, so put $A^\circ := A[1/T_1, \dots, 1/T_r]_p^\wedge$ and $A_{\text{perf}}^\circ = \mathbb{A}_{\text{inf}}((R_0^\square)_\infty)[1/\xi, 1/[T_1^b], \dots, 1/[T_r^b]]_p^\wedge$. We can express the source as $\mathrm{Perf}(A^\circ)^{\varphi=1, h\Gamma}$ using Lemma 5.8; the target is already $\mathrm{Perf}(A_{\text{perf}}^\circ)^{\varphi=1, h\Gamma}$.

By the tensor-Hom adjunction, it suffices to show that

$$(4) \quad R\Gamma_{\text{cont}}(\Gamma, M)^{\varphi=1} \longrightarrow R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} A_{\text{perf}}^\circ)^{\varphi=1}$$

is an isomorphism for $M \in \mathrm{Perf}(A^\circ)^{\varphi=1, h\Gamma}$. We can also use the ordinary tensor product $M \otimes_{A^\circ} A_{\text{perf}}^\circ$ in the target because the relevant map $A^\circ \rightarrow A_{\text{perf}}^\circ$ is faithfully flat. Indeed, Kunz's theorem makes $A^+/p \rightarrow A_{\text{perf}}^+/p$ faithfully flat, so $A^+ \rightarrow A_{\text{perf}}^+$ is p -completely faithfully flat. The map $A^\circ \rightarrow A_{\text{perf}}^\circ$ is its p -completed base change after inverting $[p]_q, T_1, \dots, T_r$, so it is p -completely faithfully flat; since A° is Noetherian, it is faithfully flat.

The source and target also both have standard t -structures obtained by restricting the t -structure on $D_\bullet(A)$ and $D_\bullet(A_{\text{perf}})$. To check these are well-defined one must check truncations remain perfect, which can be done modulo p by derived p -completeness. In this setting the t -structure

exists even before applying $(-)^{h\Gamma}$, which is implicit in the proof [BS23, Proposition 3.4]: in particular they show that the top cohomology of a perfect unit Frobenius module over an $\mathbf{F}_p$ -algebra is a vector bundle. By derived Nakayama we can work modulo p . Using a Postnikov dévissage ([Hau24, Lemma 5.4.3]) we reduce to testing the claim for p -torsion M in the heart, which must lie in $\mathbf{Vect}(A^\circ/p)^{\varphi=1, h\Gamma}$.

We now prove (4) for M in $\mathbf{Vect}(A^\circ/p)^{\varphi=1, h\Gamma}$. The faithfully flat map $A^\circ/p \rightarrow A^\circ_{\text{perf}}/p$ is injective, with cokernel $(A^\circ_{\text{perf}}/A^\circ)/p$. Since M is finite projective over A°/p , tensoring this sequence with M is exact. It is therefore enough to prove $R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} (A^\circ_{\text{perf}}/A^\circ))^{\varphi=1} = 0$.

Choose $m \gg 0$ and let $(R_0^\square)_m$ be obtained from the framing by adjoining p^m th roots of the coordinates T_i . Define $\varphi^{-m}A^\circ = (R_0^\square)_m((q^{1/p^m} - 1))[1/T_1, \dots, 1/T_r]_p^\wedge$ and $\varphi^{-m}A^{\circ, \text{cyc}} = (R_0^\square)_m((q^{1/p^\infty} - 1))^\wedge[1/T_1, \dots, 1/T_r]_p^\wedge$, viewed as intermediate subrings

$$A^\circ \subset \varphi^{-m}A^\circ \subset \varphi^{-m}A^{\circ, \text{cyc}} \subset A^\circ_{\text{perf}}.$$

Since φ and Γ commute, the unit Frobenius of M gives a Γ -equivariant equivalence

$$R\Gamma_{\text{cont}}(\Gamma, M)^{\varphi=1} \simeq R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} \varphi^{-m}A^\circ)^{\varphi=1}.$$

For $m \gg 0$, the exact triangles associated to $\varphi^{-m}A^\circ/A^\circ \rightarrow A^\circ_{\text{perf}}/A^\circ \rightarrow A^\circ_{\text{perf}}/\varphi^{-m}A^\circ$ and to $\varphi^{-m}A^{\circ, \text{cyc}}/\varphi^{-m}A^\circ \rightarrow A^\circ_{\text{perf}}/\varphi^{-m}A^\circ \rightarrow A^\circ_{\text{perf}}/\varphi^{-m}A^{\circ, \text{cyc}}$ show

$$R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} (A^\circ_{\text{perf}}/A^\circ))^{\varphi=1} = 0$$

by applying Lemmas 5.11 and 5.12. The claim then follows. $\square$

We now provide the needed technical lemmas. We retain the notation of Proposition 5.9, and use the t -structure constructed in the proof.

Put $T = T_1 \cdots T_r$. Through the identification in Theorem 4.6 (using a resolution in $D_\bullet(A)^{h\Gamma}$ by $A\langle F \rangle$ induced by a root), the ring

$$\overline{A}^\circ := A^\circ/p = \bigcup_{i \geq 0} T^{-i} \overline{R}^\square((q-1))$$

is equipped with the inductive limit topology, where $T^{-i} \overline{R}^\square[[q-1]]$ is an open lattice in $T^{-i} \overline{R}^\square((q-1))$ with its $(q-1)$ -adic topology. This is a strict inductive limit of Banach modules with closed transition maps, and every compact subset is contained in $T^{-i} \overline{R}^\square((q-1))$ for some i . It is worth noting that this topology agrees with the weak topology induced on a finitely generated unit module, by using the resolution by $A\langle F \rangle$ induced by the root and showing this weak topology, modulo p , agrees with the natural topology on the colimit presentation by the root (which is what we use here).

Choose a finite $\overline{R}^\square[[q-1]]$ -submodule $M_0 \subset M$ with $M_0[1/(q-1), 1/T_1, \dots, 1/T_r] = M$. The orbit under the compact group Γ of a finite generating set of M_0 is contained in $(q-1)^{-c} T^{-i} M_0$ for some $c, i \geq 0$. Its $\overline{R}^\square[[q-1]]$ -span is therefore a finite Γ -stable lattice $M^+ \subset M$ with $M^+[1/(q-1), 1/T_1, \dots, 1/T_r] = M$.

Let $H = (1 + p^m \mathbf{Z}_p) \ltimes (p^{m-1} \mathbf{Z}_p(1))^n \subset \Gamma$. Choose a topological generator γ_0 of $1 + p^m \mathbf{Z}_p$ and generators $\gamma_1, \dots, \gamma_n$ of $(p^{m-1} \mathbf{Z}_p(1))^n$ such that $\gamma_i(T_i) = q^{p^m} T_i$. For $N \geq 0$, put $M^{(N)} = T^{-N} M^+ [1/(q-1)]$ and $\text{Fil}^a M^{(N)} = T^{-N} (q-1)^a M^+$. Then $M = \text{colim}_N M^{(N)}$ with its strict inductive limit topology. Increasing m if necessary, H is normal and uniform, and

$$(5) \quad (\gamma_i - 1)(\text{Fil}^a M^{(N)}) \subset \text{Fil}^{a+1} M^{(N)} \text{ for } 0 \leq i \leq n.$$

We use the following elementary lemma, which is similar to [KL16, Lemmas 5.6.3, 5.6.4]. For these mod p calculations, we work with a more restrictive variant of Ind-Banach $\mathbf{F}_p$ -modules. We will abbreviate this just to “LB” modules and rings, but our convention is the following. An LB module is a presentation $M = \text{colim}_{N \in \mathbf{N}} M^{(N)}$ by Banach modules with closed contractive embeddings. LB rings are algebra objects with LB underlying modules, and group actions satisfy $h \cdot M^{(N)} = M^{(N)}$. For compatible presentations, write

$$\widehat{\bigoplus_j}^{\text{LB}} M_j := \text{colim}_N \widehat{\bigoplus_j} M_j^{(N)},$$

where the completed sum is taken for the supremum norm. In the applications, all norms are normalized by the same value of $|q-1|$.

For a closed subgroup $H \leq \Gamma$, choose topological generators $h_1, \dots, h_k$ of $H \cap \mathbf{Z}_p(1)^n \simeq \mathbf{Z}_p^k$. If the image of H in $1 + p\mathbf{Z}_p$ is nontrivial, choose $h_0 \in H$ lifting a topological generator of this image. The resolution (1) and Hochschild-Serre compute $R\Gamma_{\text{cont}}(H, M)$ as the fiber of $h_0 - 1$ on

$$\text{Kos}^\bullet(M; h_1 - 1, \dots, h_k - 1),$$

where h_0 acts on the coefficients and by conjugation on the Koszul generators. If the image in $1 + p\mathbf{Z}_p$ is trivial, the Koszul complex itself computes cohomology.

This cochain model of cohomology allows us to make the following definitions, subject to the choices of generators above. A complex is *strictly contractible* if it admits a continuous contracting homotopy. For a family of complexes $C_j^\bullet$, such homotopies h_j^i are *uniformly bounded* if they preserve the Banach stages $C_j^{i,(N)}$ and

$$\sup_{j,i} \|h_j^i : C_j^{i,(N)} \longrightarrow C_j^{i-1,(N)}\| < \infty$$

for every N .

Lemma 5.10. *Let $H \leq \Gamma$ be a closed subgroup and let $B_0 \subseteq B_1 \subseteq \dots \subseteq B$ be an increasing sequence of LB rings with continuous H -actions. Suppose there are H -equivariant B_j -linear retractions $\psi_j : B_{j+1} \longrightarrow B_j$ of LB modules, and that the natural map is an isomorphism of LB modules*

$$\widehat{\bigoplus_{j \geq 0}}^{\text{LB}} \ker(\psi_j) \xrightarrow{\sim} B/B_0.$$

Let $M \in \mathbf{Vect}(B_0)$ have continuous semilinear H -action, and write $M_j = B_j \widehat{\otimes}_{B_0} M$ and $M_B = B \widehat{\otimes}_{B_0} M$. If the complexes computing continuous H -cohomology of $M_j \widehat{\otimes}_{B_j} \ker(\psi_j)$ are strictly contractible by compatible uniformly bounded homotopies, then

$$R\Gamma_{\text{cont}}(H, M_B/M) = 0.$$

Proof. Using finite projectivity of M , we obtain an isomorphism of LB modules

$$M_B/M \simeq \widehat{\bigoplus_{j \geq 0}}^{\text{LB}} (M_j \widehat{\otimes}_{B_j} \ker(\psi_j)).$$

Applying the previously constructed Koszul complex model for H -cohomology (after a choice of generators), this is carried to the corresponding completed sum of complexes computing H -cohomology. The uniform bounds extend the contracting homotopies to each Banach completed sum in the definition of $\widehat{\bigoplus}^{\text{LB}}$, and compatibility gives a contracting homotopy on the colimit. $\square$

All LB structures below use the increasing localization filtration by T^{-N} , with the $(q-1)$ -adic norm on the N th term in the LB colimit presentation. Put $M_j = B_j \widehat{\otimes}_{\bar{A}^\circ} M$ and write $\|\cdot\|_N$ for operator norms on the N th stage, using supremum norms on monomial decompositions and Koszul complexes.

Lemma 5.11. *Let M be p -torsion and in the heart of $\mathbf{Perf}(A^\circ)^{\varphi=1, h\Gamma}$. For $m \gg 0$,*

$$R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} (A_{\text{perf}}^\circ / \varphi^{-m} A^{\circ, \text{cyc}})) = 0.$$

Proof. Apply Lemma 5.10 with $B_j = \varphi^{-(m+j)} A^{\circ, \text{cyc}} / p$, $B = A_{\text{perf}}^\circ / p$, $H = (p^{m-1} \mathbf{Z}_p(1))^n$, and $B_0 \widehat{\otimes}_{\bar{A}^\circ} M$, which is finite projective by the t -structure reduction in the proof of Proposition 5.9. For $\alpha = (a_1, \dots, a_n)$, the decomposition

$$B_{j+1} = \bigoplus_{\alpha \in \{0, \dots, p-1\}^n} B_j \prod_{\nu=1}^n T_\nu^{a_\nu / p^{m+j+1}}$$

defines Γ -equivariant B_j -linear retractions ψ_j by projection onto $\alpha = 0$. With the prescribed LB structures, these give

$$\widehat{\bigoplus_{j \geq 0}}^{\text{LB}} \ker(\psi_j) \simeq B/B_0.$$

For $\alpha \neq 0$, choose the least i with $a_i \neq 0$. On $M_j \prod_\nu T_\nu^{a_\nu / p^{m+j+1}}$, the action on the coordinates and (5) give

$$\left\| (q^{a_i / p^{j+1}} - 1)^{-1} (\gamma_i - q^{a_i / p^{j+1}}) \right\|_N \leq |q-1|^{(p-1)p^{-j-1}} < 1.$$

The geometric series gives $(\gamma_i - 1)^{-1}$ with norm at most $|q-1|^{-1}$. Thus $h_{j,\alpha} = (\gamma_i - 1)^{-1} \iota_i$ contracts the Koszul complex of this module, where ι_i is exterior contraction with $\iota_i(e_\nu) = \delta_{i\nu}$ for the basis $e_1, \dots, e_n$. These contractions preserve the stages, commute with their inclusions,

and satisfy $\|h_{j,\alpha}\|_N \leq |q-1|^{-1}$. Lemma 5.10 gives the vanishing for H , and Hochschild-Serre gives it for Γ . $\square$

Lemma 5.12. *Let M be p -torsion and in the heart of $\mathbf{Perf}(A^\circ)^{\varphi=1, h\Gamma}$. For $m \gg 0$,*

$$R\Gamma_{\text{cont}}(\Gamma, M \otimes_{A^\circ} (\varphi^{-m} A^{\circ, \text{cyc}} / \varphi^{-m} A^\circ)) = 0.$$

Proof. Apply Lemma 5.10 with $B = \varphi^{-m} A^{\circ, \text{cyc}} / p$,

$$B_j = (\overline{R}_0^\square)_m((q^{1/p^{m+j}} - 1))[1/T_1, \dots, 1/T_r],$$

with $B_0 \widehat{\otimes}_{A^\circ} M$, and $H = (1 + p^m \mathbf{Z}_p) \ltimes N$, where $N = (p^{m-1} \mathbf{Z}_p(1))^n$. The decomposition

$$B_{j+1} = \bigoplus_{a=0}^{p-1} B_j q^{a/p^{m+j+1}}$$

defines Γ -equivariant B_j -linear retractions ψ_j by projection onto $a = 0$. As in the preceding proof, $\widehat{\bigoplus}_j^{\text{LB}} \ker(\psi_j) \simeq B/B_0$.

Choose $\gamma_0 = 1 + p^m u$ generating $1 + p^m \mathbf{Z}_p$, with $u \in \mathbf{Z}_p^\times$. For $1 \leq a < p$, put $C_{j,a}^\bullet = \text{Kos}_N^\bullet(M_j q^{a/p^{m+j+1}})$. For $I \subseteq \{1, \dots, n\}$, write $e_I = \bigwedge_{i \in I} e_i$, with indices in increasing order. The relation $\gamma_0 \gamma_i \gamma_0^{-1} = \gamma_i^{1+p^m u}$ gives the action

$$\gamma_0(y e_I) = \left(\prod_{i \in I} \frac{\gamma_i^{1+p^m u} - 1}{\gamma_i - 1} \right)^{-1} \gamma_0(y) e_I.$$

For m sufficiently large, this action satisfies

$$\left\| (q^{au/p^{j+1}} - 1)^{-1} (\gamma_0 - q^{au/p^{j+1}}) \right\|_N \leq |q-1|^{p^{-j}(1-p^{-m}-p^{-1})} < 1$$

on $C_{j,a}^\bullet$. The geometric series gives an inverse chain map $(\gamma_0 - 1)^{-1}$ of norm at most $|q-1|^{-1}$. It supplies compatible uniformly bounded contractions of $\text{fib}(\gamma_0 - 1 : C_{j,a}^\bullet \rightarrow C_{j,a}^\bullet)$. The sum of these fibers over $1 \leq a < p$ computes H -cohomology of $M_j \widehat{\otimes}_{B_j} \ker(\psi_j)$. Lemma 5.10 gives the vanishing for H , and Hochschild-Serre gives it for Γ . $\square$

Remark 5.13. It is tempting to try to argue the previous result would only need to use φ , as this is the case for ordinary (φ, Γ) modules using invariance of étale φ -modules under completed perfection. Consider the map

$$A = k((q-1))\langle T \rangle[1/T] \rightarrow B = k((q^{1/p^\infty} - 1))^\wedge \langle (T^\flat)^{1/p^\infty} \rangle[1/T^\flat],$$

with $T \mapsto T^\flat$. If full faithfulness followed just from invariance under this completed perfection, then $\mathbf{Perf}(A)^{\varphi=1} \rightarrow \mathbf{Perf}(B)^{\varphi=1}$ would be fully faithful. By Katz's theorem this would force $H_{\text{ét}}^1(A, \mathbf{F}_p) \rightarrow H_{\text{ét}}^1(B, \mathbf{F}_p)$ to be an isomorphism. One can check that the Artin-Schreier class on B defined by

$$f = \sum_{n \geq 1} (q-1)^n (T^\flat)^{-1+1/p^n}$$

is not pulled back from A , so the étale cohomology groups are not isomorphic via the natural base change map. The issue is passage to B mixes completed and decompleted perfection.

Proposition 5.14. *The top horizontal functor in Theorem 5.4 is fully faithful.*

Proof. Lemma 5.6 and strict-étale descent for $D_{\text{lis}}^{(b)}((X_\eta \setminus D_\eta)_{\text{proét}}, \mathbf{Z}_p)$, together with Lemma 3.4, reduce the claim to a framed log affine. Lemma 5.8 identifies the source with $\text{Perf}(R_0^\square((q-1)_p^\wedge[1/T_1, \dots, 1/T_r]_p^\wedge)^{\varphi=1, h\Gamma})$, and Proposition 5.9 makes base change to $\mathbb{A}_{\text{inf}}((R_0^\square)_\infty)[1/\xi, 1/[T_1^\flat], \dots, 1/[T_r^\flat]_p^\wedge]$ fully faithful. Lemma 4.27 then identifies $T_{\text{ét}}$ with tilting, analytification, and continuous descent along the perfectoid pro-Kummer Γ -torsor. Tilting and descent are equivalences.

Thus we must investigate full faithfulness of analytification. Put

$$U = \text{Spec}(R_0^\square{}_\infty{}_\eta^\flat[1/T_1^\flat, \dots, 1/T_r^\flat]).$$

For $\mathbb{L}, \mathbb{L}' \in D_{\text{lis}}^{(b)}(U_{\text{ét}}, \mathbf{F}_p)$, the monodromy of each $H^i \underline{\text{RHom}}(\mathbb{L}, \mathbb{L}')$ is contained in a finite group $\text{GL}_{m_i}(\mathbf{F}_p)$. Boundedness gives one finite étale cover $U' \rightarrow U$ on which $H^i \underline{\text{RHom}}(\mathbb{L}, \mathbb{L}') \simeq \mathbf{F}_p^{m_i}$ for every i . On U' , [Bha25, Remark 6.2.7] gives

$$\text{R}\Gamma(U', \mathbf{F}_p) \xrightarrow{\sim} \text{R}\Gamma(U'^{\text{an}}, \mathbf{F}_p).$$

Postnikov dévissage and finite étale descent give full faithfulness modulo p ; derived Nakayama gives the assertion. $\square$

We next prove essential surjectivity after passing, on a framed strict-étale chart, to a finite prime-to- p Kummer cover. Proposition 5.15 descends the object constructed on the cover, using the full faithfulness already proved.

Proposition 5.15. *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine with boundary coordinates $T_1, \dots, T_r$, and let N be prime to p . Put*

$$R_N = (R[S_1, \dots, S_r]/(S_1^N - T_1, \dots, S_r^N - T_r))_p^\wedge,$$

and equip $\text{Spf } R_N$ with its induced log structure $\mathcal{M}_N$. Then we have

$$(\text{Perf}((\text{Spf } R_N, \mathcal{M}_N)_\Delta, \mathcal{O}_\Delta(*D_N))^{\varphi=1, \text{op}})^{h\mu_N^r} \simeq \text{Perf}((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1, \text{op}}.$$

Proof. Let $f_N : (\text{Spf } R_N, \mathcal{M}_N) \rightarrow (\text{Spf } R, \mathcal{M})$ be the Kummer map. Since $f_N^* D = N D_N$, there is a canonical equivalence

$$f_N^* \mathcal{O}_\Delta(*D) \simeq \mathcal{O}_\Delta(* (N D_N)) \simeq \mathcal{O}_\Delta(*D_N).$$

Thus pullback induces a functor on the corresponding module categories

$$\text{Perf}((\text{Spf } R, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1, \text{op}} \rightarrow \text{Perf}((\text{Spf } R_N, \mathcal{M}_N)_\Delta, \mathcal{O}_\Delta(*D_N))^{\varphi=1, \text{op}}$$

we will use to induce the desired equivalence.

Base change of the framing along $T_i \mapsto S_i^N$ gives R_N a framing with boundary coordinates $S_1, \dots, S_r$, so it also has an associated q -de Rham prism. On the degree zero log q -de Rham term, the pullback is base change along

$$A := R_0^\square((q-1))_p^\wedge[1/T_1, \dots, 1/T_r]_p^\wedge \longrightarrow B := A[S_1, \dots, S_r]/(S_1^N - T_1, \dots, S_r^N - T_r).$$

Because $N \in \mathbf{Z}_p^\times$ and the T_i are units in A , this is a finite étale μ_N^r -torsor.

After rescaling the first r coordinates by N^{-1} , the two actions of Γ make $A \rightarrow B$ a Γ -equivariant μ_N^r -torsor. It is also Frobenius-equivariant by functoriality of the q -de Rham prisms. Lemma 5.8 identifies pullback with extension of scalars from A to B , so the proposition therefore reduces to finite étale descent in (φ, Γ) -equivariant perfect complexes:

$$\mathrm{Perf}(A)^{\varphi=1, h\Gamma} \simeq (\mathrm{Perf}(B)^{\varphi=1, h\Gamma})^{h\mu_N^r}.$$

Taking opposites proves the assertion.

Pullback compatibility is clear. □

Remark 5.16. On a framed log affine, $[(X_N, \mathcal{M}_N)/\mu_N^r]$ is the N th root stack of $(X, \mathcal{M})$. Thus the proposition is the local form of finite étale descent from this root stack; the local form suffices because the equivalence can be checked strict-étale locally.

Corollary 5.17. *The top horizontal functor in Theorem 5.4 is an equivalence.*

Proof. By Proposition 5.14, it remains to prove essential surjectivity. Let $\mathbb{L} \in D_{\mathrm{lis}}^{(b)}((X_\eta \setminus D_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$. Since X_η is quasicompact and smooth, it has finitely many connected components, each irreducible; their nonempty intersections with $X_\eta \setminus D_\eta$ remain irreducible. The amplitude of a derived lisse sheaf is locally constant, so $\mathbb{L}$ is globally bounded. Bounded objects are generated under shifts and fibers by their hearts, so without loss of generality we may assume $\mathbb{L} \in D_{\mathrm{lis}}^{(b)}((X_\eta \setminus D_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)^\heartsuit$.

This assertion is strict-étale local on X . Indeed, if a local system $\mathbb{L}$ is realized locally by objects $\mathcal{E}_i$, then full faithfulness gives unique identifications of the $\mathcal{E}_i$ on pairwise overlaps inducing the identity on $\mathbb{L}$. The cocycle condition holds after applying the fully faithful realization functor, and Lemma 5.6 glues the $\mathcal{E}_i$. Thus we may assume that $X = \mathrm{Spf} R$ is a framed log affine.

Let $\mathbb{L}$ be such a local system. By [DLLZ23, Corollary 6.3.4], $\mathbb{L}^{\mathrm{ext}} := j_{\mathrm{k\acute{e}t},*} \mathbb{L}$ is a Kummer-étale $\mathbf{Z}_p$ -local system on the log generic fiber. We want to argue there is a single prime-to- p Kummer cover $f_N : (X_N, \mathcal{M}_N) \rightarrow (X, \mathcal{M})$ as in Proposition 5.15 such that $f_N^* \mathbb{L}^{\mathrm{ext}}$ is p -Kummer. Without loss of generality we assume $(\mathrm{Spf} R, \mathcal{M})_\eta$ is connected, as we can take the least common multiple of such N on each component. By [IKY26, Definition 4] we want to show that if $\bar{x}$ is a log geometric point of $(\mathrm{Spf} R, \mathcal{M})_\eta$ and $(\mathrm{Spf} R, \mathcal{M})_{\eta, \bar{x}}$ denotes strict localization (at the underlying geometric point) with the induced log structure then

$$\pi_1^{\mathrm{k\acute{e}t}}((\mathrm{Spf} R, \mathcal{M})_{\eta, \bar{x}}, \bar{x}) \rightarrow \mathrm{Aut}_{\mathbf{Z}_p}(\mathbb{L}_{\bar{x}}^{\mathrm{ext}})$$

has pro- p image after pullback along f_N for some N . Here ρ is the associated representation to $\mathbb{L}^{\mathrm{ext}}$. The key observation is that $\mathrm{Aut}_{\mathbf{Z}_p}(\mathbb{L}_{\bar{x}}^{\mathrm{ext}}) \rightarrow \mathrm{GL}_n(\mathbf{F}_p)$ has pro- p kernel, where n is

the rank of the mod p reduction of $\mathbb{L}$ (which makes sense by connectedness). Thus to obtain a pro- p image we can take N to be the prime-to- p part of $|\mathrm{GL}_n(\mathbf{F}_p)|$, as the effect of pullback along f_N takes N th powers in the image (see the proof of Proposition 5 in [IKY26]).

By the (dualized) Koshikawa-Yao correspondence [IKY26, Theorem 7]

$$\mathrm{Perf}((X_N, \mathcal{M}_N)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1, \mathrm{op}} \simeq D_{\mathrm{lis}}^{(b), p\mathrm{Kum}}((X_N, \mathcal{M}_N)_\eta, \mathbf{Z}_p),$$

this p -Kummer local system is realized by a Laurent F -crystal $\mathcal{E}$ on $(X_N, \mathcal{M}_N)_\Delta$, so $T_{\mathrm{ét}}(\mathcal{E}) \simeq f_N^* \mathbb{L}^{\mathrm{ext}}$. Let $j_N : X_{N, \eta} \setminus D_{N, \eta} \hookrightarrow (X_N, \mathcal{M}_N)_\eta$ denote the open-stratum inclusion. Since $T_{\mathrm{ét}}(\mathcal{O}_\Delta(*D_N)) \simeq j_{N,!} \mathbf{Z}_p$ by the compatibility of the vertical arrows before Theorem 5.4, the symmetric monoidality of $T_{\mathrm{ét}}$ gives

$$T_{\mathrm{ét}}(\mathcal{E} \otimes \mathcal{O}_\Delta(*D_N)) \simeq j_{N,!} f_N^* \mathbb{L}.$$

The sheaf $j_{N,!} f_N^* \mathbb{L}$ has its canonical descent datum along the finite étale torsor on the open stratum. The Čech nerve of the localized map in Proposition 5.15 is finite étale in every degree. Proposition 5.14 therefore lifts the descent datum uniquely to $\mathcal{E} \otimes \mathcal{O}_\Delta(*D_N)$, and full faithfulness gives the cocycle condition. Proposition 5.15 now descends this object to $\mathrm{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1, \mathrm{op}}$. Compatibility with realization shows that it realizes $j_! \mathbb{L}$. $\square$

5.2. Proof of the main theorem. We will show that

$$T_{\mathrm{ét}} : D_{\mathrm{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\mathrm{op}} \longrightarrow D_{\mathrm{cons}, D}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence onto the full subcategory of p -Kummer sheaves $D_{\mathrm{cons}, D}^{(b), p\mathrm{Kum}}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$. Strict-étale descent on the source and target, established in Theorem 4.12 and Lemma 4.21, permits us to work on framed log affines for the proof.

The remainder of the proof beyond the already established equivalence for the open stratum is an induction on dimension using a variant of Kashiwara's lemma in Lemma 5.18 for closed strata, together with full faithfulness for maps between open and closed strata. The following toy model summarizes the argument.

A toy model. Consider the disk $\mathbb{D}_{\mathbf{Q}_p(\zeta_p)}$ punctured at the origin; we use the coordinate T so $\mathbb{D} = \mathrm{Spa} \mathbf{Q}_p(\zeta_p)\langle T \rangle$, equipped with the log structure $\mathcal{M}_{V(T)}$. We use the usual formal model $(\mathrm{Spf} \mathbf{Z}_p[\zeta_p]\langle T \rangle, \mathcal{M}_{V(T)})$.

Let $j : \mathbb{D}^\times \hookrightarrow \mathbb{D}$ and $i : V(T) \hookrightarrow \mathbb{D}$ denote the two strata. If $A = \mathbf{Z}_p\langle T \rangle((q-1))_p^\wedge$ and $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)$, the setup of the argument is summarized by the diagram

$$\begin{array}{ccccc} \left\{ \begin{array}{c} \text{open stratum unit} \\ \text{Laurent crystals} \end{array} \right\}^{\mathrm{op}} & \xrightarrow{\quad} & \left\{ \begin{array}{c} \text{fgu Laurent crystals} \\ \text{on } (\mathrm{Spf} \mathbf{Z}_p[\zeta_p]\langle T \rangle, \mathcal{M}_{V(T)})_\Delta \end{array} \right\}^{\mathrm{op}} & \xrightarrow{\quad (i_+(-))[-1] \quad} & (D_{\mathrm{fgu}}(A/(T))^{h\Gamma})^{\mathrm{op}} \\ \downarrow T_{\mathrm{ét}} & & \downarrow T_{\mathrm{ét}} & & \downarrow T_{\mathrm{ét}} \\ \left\{ \begin{array}{c} p\text{-Kummer local systems} \\ \text{on } \mathbb{D}^\times \end{array} \right\} & \xrightarrow{j_!} & \left\{ \begin{array}{c} p\text{-Kummer constructible sheaves} \\ \text{on } (\mathbb{D}, \mathcal{M}_{V(T)}) \end{array} \right\} & \xleftarrow{i_*} & \left\{ \begin{array}{c} p\text{-Kummer local systems} \\ \text{on } (V(T), i^* \mathcal{M}_{V(T)}) \end{array} \right\}. \end{array}$$

The left vertical arrow is an equivalence by Theorem 5.4, and tensoring with $\mathcal{O}_\Delta(*V(T))$ identifies with $j_!$ under $T_{\text{ét}}$. The factor of Γ corresponding to T acts trivially on $A/(T)$, but its action on objects records the normal Kummer inertia on the logarithmic point. The dashed arrow is supplied by the local Kashiwara equivalence below: since $i^! = Li^*[-1]$ and $T_{\text{ét}}$ is contravariant, $(i_+(-))[-1]$ identifies with i_* under realization.

With Kashiwara's lemma what we have so far (generalized to a hollow log point to deal with the origin) is enough for essential surjectivity once we have derived full faithfulness, because étale sheaves of the form $j_!\mathbb{L}$ and $i_*\mathbb{L}'$ generate $D_{\text{cons}, V(T)}^{(b)}((\mathbb{D}, \mathcal{M}_{V(T)}), \mathbf{Z}_p)$ under shifts and fibers; the same holds after imposing the p -Kummer condition. The functor $T_{\text{ét}}$ lands in the full subcategory of p -Kummer constructible sheaves along the stratification $(\mathbb{D}^\times, V(T))$ of the log adic generic fiber and is compatible with shifts and fibers. One need only note that $D_{\text{fgu}}((\text{Spf } \mathbf{Z}_p[\zeta_p]\langle T \rangle, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ is closed under shifts and fibers to conclude the argument.

For full faithfulness, it is not quite enough. The strategy will be to place any $\mathcal{E} \in D_{\text{fgu}}((\text{Spf } \mathbf{Z}_p[\zeta_p]\langle T \rangle, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ into a fiber sequence with an “open” and “closed” part. We will use the corresponding stratum on the adic generic fiber as a subscript to indicate the support of a crystal. When comparing $\text{RHom}(\mathcal{E}, \mathcal{E}')$ with $\text{RHom}(T_{\text{ét}}(\mathcal{E}'), T_{\text{ét}}(\mathcal{E}))$ to deduce the comparison one must ultimately show full faithfulness in the form

$$\text{RHom}(\mathcal{E}_{\mathbb{D}^\times}, \mathcal{E}'_{V(T)}) \simeq \text{RHom}(T_{\text{ét}}(\mathcal{E}'_{V(T)}), T_{\text{ét}}(\mathcal{E}_{\mathbb{D}^\times})).$$

Here the corresponding étale realizations are supported on the logarithmic point at the origin and on the punctured disk $\mathbb{D}^\times$. The trick is similar to what was already done in the proof of essential surjectivity of Theorem 5.4, where we reduce the statement to a prime-to- p Kummer cover $\mathbb{D}_N$. Unlike that case we do not work on the open stratum, so this does not become an étale cover. However as we are only interested in mapping complexes, we can show that one can still get away with taking μ_N^r -invariants to obtain the claim for $\mathbb{D}$.

Put $\mathbb{L} = T_{\text{ét}}(\mathcal{E}_{\mathbb{D}_N^\times})$ and $i_*\mathbb{L}' = T_{\text{ét}}(\mathcal{E}'_{V(T)})$. We may assume that $\mathbb{L}$ extends across the origin to the p -Kummer derived local system $Rj_*\mathbb{L}$. Thus there is

$$\mathcal{E}_{\mathbb{D}_N} \in \text{Perf}((\text{Spf } \mathbf{Z}_p[\zeta_p]\langle T^{1/N} \rangle, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1}$$

such that $T_{\text{ét}}(\mathcal{E}_{\mathbb{D}_N}) \simeq Rj_*\mathbb{L}$ and $\mathcal{E}_{\mathbb{D}_N^\times} \simeq \mathcal{E}_{\mathbb{D}_N}(*V(T))$. On the Kummer-étale side, one has

$$\text{RHom}(i_*\mathbb{L}', Rj_*\mathbb{L}) = 0.$$

Indeed, the adjunction between j^* and Rj_* identifies this with $\text{RHom}(j^*i_*\mathbb{L}', \mathbb{L}) = 0$. Applying $\text{RHom}(i_*\mathbb{L}', -)$ to the triangle

$$j_!\mathbb{L} \longrightarrow Rj_*\mathbb{L} \longrightarrow i_*i^*Rj_*\mathbb{L}$$

and using the adjunction between i^* and i_* gives $\text{RHom}(i_*\mathbb{L}', j_!\mathbb{L}) \simeq \text{RHom}(\mathbb{L}', i^*Rj_*\mathbb{L})[-1]$. This now consists of sheaves supported at the origin $V(T) \subset \mathbb{D}_N$, so producing an analogous isomorphism on the crystal side would reduce the needed result down one dimension to a point where the claim is known (we will use an induction on dimension). For this to be plausible one

needs $Rj_*\mathbb{L}$ to be p -Kummer, so there is a corresponding crystal; this is the point of passing to the cover. Using exactly this strategy, one can show

$$\mathrm{RHom}(\mathcal{E}_{\mathbb{D}_N^\times}, \mathcal{E}'_{V(T)}) \simeq \mathrm{RHom}(\mathrm{fib}(\mathcal{E}_{\mathbb{D}_N} \rightarrow \mathcal{E}_{\mathbb{D}_N}(*V(T))), \mathcal{E}'_{V(T)})[-1].$$

The remaining closed term is computed over the quotient by T , so it is indeed a dimension lower. Thus one can reduce to a hollow point where the claim can be checked directly.

Notation for this section. Consider a framed log affine $(\mathrm{Spf} R, \mathcal{M})$ with framing $\square$ as in Definition 3.2, and put $A = R_0^\square((q-1))_p^\wedge$. For $\Lambda \subseteq \{1, \dots, r\}$, put

$$J_\Lambda = (T_i \mid i \in \Lambda).$$

For $Z_\Lambda = V(J_\Lambda)$, put $D|_{Z_\Lambda} = \bigcup_{i \notin \Lambda} (D_i \cap Z_\Lambda)$ and consider $D_{\mathrm{fgu}}(A/J_\Lambda)^{h\Gamma}$. The group $\Gamma_\Lambda = \prod_{i \in \Lambda} \mathbf{Z}_p(1)_i$ fixes A/J_Λ . Give Z_Λ the pullback log structure, which now has a hollow component. Since we continue using the full group Γ , we see $T_{\mathrm{ét}}$ factors as

$$T_{\mathrm{ét}} : (D_{\mathrm{fgu}}(A/J_\Lambda)^{h\Gamma})^{\mathrm{op}} \longrightarrow D_{\mathrm{cons}, D|_{Z_\Lambda}}^{(b), \mathrm{pKum}}((Z_\Lambda, \mathcal{M}|_{Z_\Lambda})_\eta, \mathbf{Z}_p)$$

followed by extension by zero. One can see this using Lemma 4.27. The target is constructible for the remaining coordinate stratification.

We now prove a local version of Kashiwara's lemma. We write $D_{\mathrm{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)_Z$ for the full subcategory of finitely generated unit Laurent crystals supported on Z . Let us explain the general globalized notion of this when $D = \bigcup_i D_i$ is a simple normal crossings divisor with irreducible components D_i . For any nonempty intersection $Z = D_1 \cap \dots \cap D_c$ of distinct irreducible components, define

$$D_{\mathrm{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)_Z := \bigcap_{a=1}^c \ker(- \otimes^{L, \square} \mathcal{O}_\Delta(*D_a)).$$

Similarly one may make a definition for connected components. This covers all Z which are smooth closed log strata of the stratification induced by D ; connected components are necessarily of pure codimension. On a compatible framing, Theorem 4.6 identifies this supported ambient category with the source of the following lemma.

Lemma 5.18 (Kashiwara's lemma). *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine, put $A = R_0^\square((q-1))_p^\wedge$ and $J = J_\Lambda = (T_\lambda \mid \lambda \in \Lambda)$, and let $i : Z = V(J) \hookrightarrow \mathrm{Spf} R$ so that under Theorem 4.6 one has*

$$D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)_Z \simeq D_{\mathrm{fgu}}(A)_J^{h\Gamma},$$

the analogous category of finitely generated unit modules supported on Z .

Then one has an equivalence

$$i^! := \left((A/J) \otimes_A^{L, \square} - \right) [-|\Lambda|] : D_{\mathrm{fgu}}(A)_J^{h\Gamma} \xrightarrow{\sim} D_{\mathrm{fgu}}(A/J)^{h\Gamma}$$

We write i_+ for the inverse equivalence.

Proof. Put $\overline{A} = A/p$ and $\overline{J} = J\overline{A}$. The categories and functor above satisfy the hypotheses of Lemma 2.3; the support condition is detected modulo p by derived Nakayama. It is therefore enough to prove that

$$i^! := \left((\overline{A}/\overline{J}) \otimes_{\overline{A}}^{\mathbf{L}, \blacksquare} - \right) [-|\Lambda|] : D_{\text{fgu}}(\overline{A})_{\overline{J}}^{h\Gamma} \longrightarrow D_{\text{fgu}}(\overline{A}/\overline{J})^{h\Gamma}$$

is an equivalence. The functor is well-defined: one can use Lemma 4.24, and in particular $i^!$ is Γ -equivariant.

In this case, the functor $i^!$ is an equivalence *before* we apply $(-)^{h\Gamma}$. One may use the proof of [EK04b, Corollary 5.11.3] to the present regular F -finite situation (noting we have access to important ingredients like roots by [Bli08]). Thus one obtains an adjoint equivalence

$$i_+ : D_{\text{fgu}}(\overline{A}/\overline{J}) \rightleftarrows D_{\text{fgu}}(\overline{A})_{\overline{J}} : i^!.$$

This survives after applying a bar construction $(-)^{h\Gamma}$ to both sides, because evaluation on the degree zero term is conservative. Thus one may test $i^!i_+ \simeq \text{id}$ and $i_+i^! \simeq \text{id}$ after evaluation on the degree zero term, at which point it reduces to the ordinary Kashiwara lemma for finitely generated unit modules. $\square$

For $i : Z_{\Lambda} \hookrightarrow \text{Spf } R$ and $M \in D_{\text{fgu}}(A/J_{\Lambda})^{h\Gamma}$, the equivalence above in Lemma 5.18 gives

$$T_{\text{ét}}(i_+M) \simeq i_*T_{\text{ét}}(M)[-|\Lambda|].$$

It is also easy to give a formula for $i_+i^!(-)$ even when the input is not supported on Z . On the same chart, we also remark that local cohomology along Z is computed by

$$R\Gamma_Z(M) \simeq M \otimes_A^{\mathbf{L}, \blacksquare} \bigotimes_{i \in \Lambda}^{\mathbf{L}, \blacksquare} [A \longrightarrow (A[1/T_i])_p^{\wedge}].$$

Completed base change to A/J kills every localized term, so $i^!R\Gamma_Z(M) \simeq i^!M$. The complex $R\Gamma_Z(M)$ is canonically Γ -equivariant, since each $\gamma \in \Gamma$ carries every T_i to a unit multiple of itself. Checking $R\Gamma_Z(M)$ is finitely generated unit is straightforward, so Kashiwara's lemma therefore gives

$$R\Gamma_Z(M) \simeq i_+i^!M.$$

In the toy model at the start of the subsection, the following is the Laurent crystal analogue of $\text{RHom}(i_*\mathbb{L}', Rj_*\mathbb{L}) = 0$ (any Kummer-étale local system in the second argument can be written this way using [DLLZ23]).

Lemma 5.19. *Let $(\text{Spf } R, \mathcal{M})$ be a framed log affine, put $A = R_0^{\square}((q-1))_p^{\wedge}$, and fix one boundary component $Z = D_i = V(T_i)$. If*

$$E \in \text{Perf}(A)^{\varphi=1, h\Gamma}$$

and $Q \in D_{\text{fgu}}(A/(T_i))^{h\Gamma}$, then

$$\text{RHom}(E, i_+Q) = 0.$$

Proof. Put $\bar{A} = A/p$ and $T = T_i$. Derived Nakayama reduces the assertion modulo p .

Using Lemma 4.2 and the Emerton-Kisin correspondence in the regular case, established in [Mat23, Theorem 6.20], before taking Γ -invariants we identify the mapping complex $\mathrm{RHom}(E, i_+Q)$ of finitely generated unit Frobenius modules over $\bar{A}$ with $\mathrm{RHom}(i_*\mathcal{F}, L)$ on $X = \mathrm{Spec} \bar{A}$. Here $\mathcal{F}$ is a bounded constructible complex of étale $\mathbf{F}_p$ -sheaves on $Z = \mathrm{Spec}(\bar{A}/(T))$ and L is a bounded complex with lisse cohomology on X ; we absorb the shift in Kashiwara's lemma into $\mathcal{F}$.

There is a subtlety in translating continuity to the étale side, but we can avoid it here: it is enough to show that $\gamma_i - 1$ acts invertibly on $\mathrm{RHom}(i_*\mathcal{F}, L)$, where γ_i is the chosen topological generator of $\mathbf{Z}_p(1)_i$. Naturality transfers this operator through the Emerton-Kisin correspondence. The Koszul resolution (1) for the toric subgroup $\mathbf{Z}_p(1)^n$, followed by Hochschild-Serre, then gives the desired vanishing.

By adjunction and the fact that L is derived lisse, we have

$$\mathrm{RHom}_X(i_*\mathcal{F}, L) \simeq i_*\mathrm{RHom}_Z(\mathcal{F}, i^!L) \simeq i_*\mathrm{RHom}_Z(\mathcal{F}, i^*L \otimes_{\mathbf{F}_p}^L i^!\mathbf{F}_p).$$

A Postnikov dévissage in $\mathcal{F}$ and L reduces the claim to $\mathcal{F}$ constructible and L lisse, both concentrated in degree zero. The group $\mathbf{Z}_p(1)_i$ fixes Z , so its actions on $\mathcal{F}$ and i^*L factor through finite automorphism groups: a constructible $\mathbf{F}_p$ -sheaf on a noetherian scheme has finite dimensional $\mathbf{F}_p$ -vector space of endomorphisms, as one sees from a finite stratification and one geometric stalk on each connected stratum. Thus some $H = p^\ell \mathbf{Z}_p(1)_i$ acts trivially on both. Put $\gamma = \gamma_i^{p^\ell}$. It remains to show that $\gamma - 1$ acts invertibly on $i^!\mathbf{F}_p$.

For a geometric point $\bar{z}$ of Z , put $B = \mathcal{O}_{X, \bar{z}}^{\mathrm{sh}}$ and $K = (B[1/T]/B)[-1]$. The calculation following Lemma 5.18 and the Artin-Schreier sequence give

$$(i^!\mathbf{F}_p)_{\bar{z}} \simeq \mathrm{fib}(\varphi - 1 : K \longrightarrow K).$$

Since γ fixes $B/(T)$ and sends T to $q^{p^{\ell+1}}T$, the operator $\gamma - 1$ acts by the units $q^{-mp^{\ell+1}} - 1$ on the graded pieces of the filtration $T^{-m}B/B$ on $B[1/T]/B$. It is therefore invertible on K and, as it commutes with Frobenius, on $(i^!\mathbf{F}_p)_{\bar{z}}$. This proves the claim on stalks.

Thus $\gamma - 1$ acts invertibly on $\mathrm{RHom}_X(i_*\mathcal{F}, L)$. Finally, $\gamma - 1 = (\gamma_i - 1)^{p^\ell}$ in characteristic p , so $\gamma_i - 1$ is also invertible. Taking global sections and applying the preceding group cohomology calculation proves the lemma. $\square$

Remark 5.20. As pointed out in the toy model, this sort of vanishing is entirely false in the non-logarithmic setting: there usually are nonzero maps in $\mathrm{RHom}(i_*\mathcal{F}, L)$. The key point is that in the log setting, due to [DLLZ23, Corollary 6.3.4] we may write $L = j_*L'$. But then by adjunction the RHom is zero. It is also possible to prove the previous result working entirely in $D_{\mathrm{fgu}}(A)^{h\Gamma}$ but (in the author's opinion) the previous proof makes what is going on slightly more transparent.

We now introduce an assumption which will be our induction step for the main theorem. To be clear, the structure of the induction will be to induct over all framed log affines on the size of $|\Lambda|$.

Assumption 5.21. Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine. Assume for all i that the restriction of $T_{\text{ét}}$ to $D_{\text{fgu}}(A/J_{\{i\}})^{h\Gamma}$ is an equivalence onto the full subcategory of $D_{\text{cons}, D}^{(b), \text{pKum}}((\mathrm{Spf} R, \mathcal{M})_\eta, \mathbf{Z}_p)$ consisting of sheaves supported on $V(T_i)$.

As in the toy model, what we will try to do is reduce the general claim to a claim which is one dimension lower, which is where the assumption appears.

Lemma 5.22. Suppose Assumption 5.21 holds. If $\mathcal{E}, \mathcal{E}' \in D_{\text{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ satisfy

$$\mathcal{E}(*D) = \mathcal{E}'(*D) = 0,$$

then the natural map

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \longrightarrow \mathrm{RHom}(T_{\text{ét}}(\mathcal{E}'), T_{\text{ét}}(\mathcal{E}))$$

is an equivalence.

Moreover, if $\mathcal{E} \in \mathrm{Perf}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1}$ and $\mathcal{E}' \in D_{\text{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ satisfies $\mathcal{E}'(*D) = 0$, then $\mathrm{RHom}(\mathcal{E}, \mathcal{E}') = 0$.

Proof. As we work on a framed log affine, we may write $D = \bigcup_{i=1}^r D_i$, where $D_i = V(T_i)$. It is tempting to try to deduce the first assertion directly from Kashiwara's equivalence in Lemma 5.18, but one must be careful because D itself need not be smooth. All calculations can be done with (φ, Γ) -modules in $D_{\text{fgu}}(A)^{h\Gamma}$ by Theorem 4.6.

The induction hypothesis provides a natural strategy, which is to place $\mathrm{RHom}(\mathcal{E}, \mathcal{E}')$ in a triangle where we already know $T_{\text{ét}}$ is fully faithful on two of the terms. We start with the triangle

$$\mathrm{R}\Gamma_{D_1}(\mathcal{E}) \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}(*D_1)$$

As remarked after Lemma 5.18 the first term corresponds to $i_+ i^! \mathcal{E}$ and is supported on D_1 . After applying $T_{\text{ét}}$ we get

$$j_! j^* T_{\text{ét}}(\mathcal{E}) \longrightarrow T_{\text{ét}}(\mathcal{E}) \longrightarrow i_* i^* T_{\text{ét}}(\mathcal{E}).$$

Iteratively taking $\mathcal{E}(*D_1 \cup D_2)$ up to $\mathcal{E}(*D) = 0$, we get similar triangles where the first terms are supported on D_i .

This gives a finite filtration on $\mathcal{E}$ in $D_{\text{fgu}}(A)^{h\Gamma}$. For $\mathcal{E}$ the first term is $\mathrm{R}\Gamma_{D_1}(\mathcal{E})$, and then on the cofiber $\mathcal{E}(*D_1)$ we apply $\mathrm{R}\Gamma_{D_2}(\mathcal{E}(*D_1))$, and we continue until the cofiber is zero when we have exhausted the irreducible components. Considering the map

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \longrightarrow \mathrm{RHom}(T_{\text{ét}}(\mathcal{E}'), T_{\text{ét}}(\mathcal{E}))$$

by finiteness of the filtration we reduce to the case where $\mathcal{E}$ is supported on a single D_i . Then using the triangle $\mathrm{R}\Gamma_{D_i}(\mathcal{E}') \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E}'(*D_i)$, $\mathrm{RHom}(\mathcal{E}, -)$ vanishes on the last term (as does the corresponding RHom after applying $T_{\text{ét}}$); thus the claim reduces to the comparison

map being an isomorphism in the case both are supported on D_i , which we know by the induction hypothesis.

The localization dévissage above shows that every $\mathcal{E}'$ with $\mathcal{E}'(*D) = 0$ lies in the thick subcategory generated by bounded finitely generated unit crystals supported on the smooth divisors D_i . By Lemma 5.18, each such crystal is of the form $i_+ M$ for $M \in D_{\text{fgu}}(A/J_{\{i\}})^{h\Gamma}$; hence Lemma 5.19 and exactness of $\text{RHom}(\mathcal{E}, -)$ gives the last assertion. $\square$

We next treat maps between objects supported on the open stratum and its complement.

Lemma 5.23. *Under Assumption 5.21, let*

$$P \in \text{Perf}((\text{Spf } R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1}$$

*and let $Q \in D_{\text{fgu}}((\text{Spf } R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ satisfy $Q(*D) = 0$. Then the natural map*

$$\text{RHom}(P, Q) \longrightarrow \text{RHom}(T_{\text{ét}}(Q), T_{\text{ét}}(P))$$

is an equivalence.

Proof. Let $j : U_{\eta} \hookrightarrow (\text{Spf } R, \mathcal{M})_{\eta}$ be the complement of D . Theorem 5.4 identifies $T_{\text{ét}}(P)$ with $j_! \mathbb{L}$ for a p -Kummer derived local system $\mathbb{L}$ on the open stratum.

By [DLLZ23, Corollary 6.3.4], the cohomology sheaves of $\mathbb{L}$ extend across the boundary as Kummer-étale local systems. The vanishing in [DLLZ23, Theorem 4.6.1], applied modulo p^n and passed to the inverse limit, makes the spectral sequence

$$E_2^{a,b} = R^a j_* H^b(\mathbb{L}) \implies H^{a+b}(Rj_* \mathbb{L})$$

degenerate. Thus $Rj_* \mathbb{L}$ is derived lisse, and, as in the essential surjectivity component of Theorem 5.4, for every log geometric point $\bar{x}$ and $b \in \mathbf{Z}$,

$$\text{im} \left(I_{\bar{x}}^{\log, (p')} \longrightarrow \text{Aut}_{\mathbf{Z}_p} (H^b(Rj_* \mathbb{L})_{\bar{x}}) \right)$$

is finite. Thus for each cohomology sheaf there is some prime-to- p Kummer cover making it p -Kummer by [IKY26, Proposition 5]. A Postnikov dévissage as in [Hau24, Lemma 5.4.3] and boundedness give a common exponent for these finite images, so a single prime-to- p root cover $\text{Spf } R_N$ makes $Rj_* \mathbb{L}$ p -Kummer.

Let $f_N : (\text{Spf } R_N, \mathcal{M}_N) \rightarrow (\text{Spf } R, \mathcal{M})$ be the corresponding map. We claim that there is a commutative diagram

$$\begin{array}{ccc} \text{RHom}(f_N^* P, f_N^* Q)^{h\mu_N^r} & \xrightarrow{T_{\text{ét}}} & \text{RHom}(T_{\text{ét}}(f_N^* Q), T_{\text{ét}}(f_N^* P))^{h\mu_N^r} \\ \sim \uparrow & & \sim \uparrow \\ \text{RHom}(P, Q) & \xrightarrow{T_{\text{ét}}} & \text{RHom}(T_{\text{ét}}(Q), T_{\text{ét}}(P)) \end{array}$$

which would then reduce the claim to showing $\mathrm{RHom}(f_N^*P, f_N^*Q) \simeq \mathrm{RHom}(T_{\text{ét}}(f_N^*Q), T_{\text{ét}}(f_N^*P))$ via the canonical map induced by $T_{\text{ét}}$. We first must explain the natural actions of μ_N^r . For the Kummer-étale side, we note $T_{\text{ét}}$ is compatible with pullbacks and $f_{N,\eta}$ is a μ_N^r -torsor for the Kummer-étale topology; this gives the desired action, and shows the vertical arrow on the right is an equivalence.

For the finitely generated unit (φ, Γ) module or Laurent crystal side it is slightly more subtle. As usual let $A = R_0^\square((q-1))_p^\wedge$ be the base so that $D_{\text{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \simeq D_{\text{fgu}}(A)^{h\Gamma}$ under Theorem 4.6, and let A_N be the corresponding base for the prime-to- p Kummer cover $(\mathrm{Spf} R_N, \mathcal{M}_N)$. Similarly to Proposition 5.15, $A_N = \bigoplus_{\alpha \in (\mathbf{Z}/N\mathbf{Z})^r} A \cdot S^\alpha$ with a natural μ_N^r -action, which gives $\mathrm{RHom}(f_N^*P, f_N^*Q)$ a μ_N^r -action.

Since N is invertible, for any $M \in D_{\text{fgu}}(A)^{h\Gamma}$ the homotopy fixed points of its pullback are computed by taking the degree zero summand in the decomposition $A_N = \bigoplus_{\alpha \in (\mathbf{Z}/N\mathbf{Z})^r} A \cdot S^\alpha$. The projection onto this summand is Γ -equivariant. It is also Frobenius equivariant, since Frobenius permutes the summands by $\alpha \mapsto p\alpha$ and $(p, N) = 1$. The extension/restriction adjunction (in Frobenius modules) and preservation of limits by $\mathrm{RHom}(P, -)$ (to move $h\mu_N^r$ inside) therefore give

$$\begin{aligned} \mathrm{RHom}(f_N^*P, f_N^*Q)^{h\mu_N^r} &\simeq \mathrm{RHom}\left(P, (A_N \widehat{\otimes}_A^L Q)^{h\mu_N^r}\right) \\ &\simeq \mathrm{RHom}(P, Q). \end{aligned}$$

Pullback compatibility of $T_{\text{ét}}$ and naturality makes the comparison on the cover μ_N^r -equivariant, proving commutativity of the diagram and reducing the claim to the prime-to- p Kummer cover.

On the cover $\mathrm{Spf} R_N$, $Rj_*\mathbb{L}$ is p -Kummer. The contravariant version of the Koshikawa-Yao correspondence [IKY26, Theorem 7] therefore gives a perfect unit Laurent crystal $\mathcal{E}_0$ realizing it. Since $\mathcal{E}_0(*D_N)$ and f_N^*P both realize $j_!f_N^*\mathbb{L}$, open stratum full faithfulness in Theorem 5.4 gives

$$f_N^*P \simeq \mathcal{E}_0(*D_N)$$

with $\mathcal{E}_0 \in \mathrm{Perf}((X_N, \mathcal{M}_N)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\varphi=1}$. We now work on the root cover and omit the pullbacks and subscripts, as we have already made the reduction. Put $\mathcal{E}_Z = R\Gamma_D(\mathcal{E}_0)$. The localization triangle is

$$\mathcal{E}_Z \longrightarrow \mathcal{E}_0 \longrightarrow P.$$

The proofs of [EK04b, Propositions 5.11.5 and 6.8.1] show that $\mathcal{E}_Z$ is bounded finitely generated unit and satisfies $\mathcal{E}_Z(*D) = 0$. The last assertion of Lemma 5.22 gives $\mathrm{RHom}(\mathcal{E}_0, Q) = 0$.

By construction, $T_{\text{ét}}(\mathcal{E}_0) \simeq Rj_*\mathbb{L}$. Since $Q(*D) = 0$, symmetric monoidality and the calculation preceding Theorem 5.4 give $j^*T_{\text{ét}}(Q) = 0$. The adjunction between j^* and Rj_* therefore gives

$$\mathrm{RHom}(T_{\text{ét}}(Q), T_{\text{ét}}(\mathcal{E}_0)) \simeq \mathrm{RHom}(j^*T_{\text{ét}}(Q), \mathbb{L}) = 0.$$

Applying $\mathrm{RHom}(-, Q)$ to the localization triangle and $\mathrm{RHom}(T_{\text{ét}}(Q), -)$ to its realization gives a map of fiber sequences

$$\begin{array}{ccccc}
\mathrm{RHom}(P, Q) & \longrightarrow & \mathrm{RHom}(\mathcal{E}_0, Q) & \longrightarrow & \mathrm{RHom}(\mathcal{E}_Z, Q) \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{RHom}(T_{\acute{\mathrm{e}}\mathrm{t}}(Q), T_{\acute{\mathrm{e}}\mathrm{t}}(P)) & \longrightarrow & \mathrm{RHom}(T_{\acute{\mathrm{e}}\mathrm{t}}(Q), T_{\acute{\mathrm{e}}\mathrm{t}}(\mathcal{E}_0)) & \longrightarrow & \mathrm{RHom}(T_{\acute{\mathrm{e}}\mathrm{t}}(Q), T_{\acute{\mathrm{e}}\mathrm{t}}(\mathcal{E}_Z)).
\end{array}$$

The two middle terms vanish by the preceding two paragraphs, while the right vertical map is an equivalence by the first assertion of Lemma 5.22. The left vertical map is therefore an equivalence. This proves the comparison on the root cover, so by the earlier reduction we are done. $\square$

Proposition 5.24. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine, and suppose Assumption 5.21 holds. Then the étale realization*

$$T_{\acute{\mathrm{e}}\mathrm{t}} : D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\mathrm{op}} \longrightarrow D_{\mathrm{cons}, D}^{(b), \mathrm{pKum}}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

is an equivalence.

Proof. The hypothesis gives us full faithfulness on full subcategories of $D_{\mathrm{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$ that have the form

$$D_{\mathrm{fgu}}(A/J_{\Lambda})^{h\Gamma}$$

with $|\Lambda| = 1$ under Theorem 4.6 (here as usual $A = R_0^{\square}((q-1))_p^{\wedge}$ with its Γ action).

The desired claim is for $\Lambda = \emptyset$, so we will show full faithfulness $|\Lambda| = 1$ implies the claim for $\Lambda = \emptyset$. We reduce full faithfulness to maps between two objects supported on D , between two objects supported on the open stratum, and from an object supported on the open stratum to an object supported on D . These cases are Lemma 5.22, Theorem 5.4, and Lemma 5.23, respectively. The middle case does not use the induction hypothesis, but the others do.

Let $\mathcal{E}, \mathcal{E}' \in D_{\mathrm{fgu}}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. Put $\mathcal{E}_Z = R\Gamma_D(\mathcal{E})$ and $\mathcal{E}_U = \mathcal{E}(*D)$, and use primes for the variants applied to $\mathcal{E}'$. It is straightforward to check that $\mathcal{E}_Z$ and $\mathcal{E}_U$ remain finitely generated unit, and similarly for $\mathcal{E}'_Z$ and $\mathcal{E}'_U$. We will later check the open terms are actually in $\mathrm{Perf}((\mathrm{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1}$.

The associated localization triangles give a square of fiber sequences

$$\begin{array}{ccccc}
\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_Z) & \longrightarrow & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_Z) & \longrightarrow & \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z) \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}') & \longrightarrow & \mathrm{RHom}(\mathcal{E}, \mathcal{E}') & \longrightarrow & \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}') \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U) & \longrightarrow & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_U) & \longrightarrow & \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_U).
\end{array}$$

In fact every row and column is a fiber sequence. Since $\mathcal{E}_Z(*D) = 0$ and $\mathcal{E}'_U \simeq \mathcal{E}'_U(*D)$, the lower right term $\mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_U)$ vanishes. After applying $T_{\text{ét}}$, let $j : U_\eta \hookrightarrow (X, \mathcal{M})_\eta$ denote the complement of D . The adjunction between $j_!$ and j^* identifies the analogous term with

$$\mathrm{RHom}(j^*T_{\text{ét}}(\mathcal{E}'_U), j^*T_{\text{ét}}(\mathcal{E}_Z)) = 0,$$

since $T_{\text{ét}}(\mathcal{E}_Z)$ is supported on the complement of U . The same vanishing gives $\mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}') \simeq \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z)$ and $\mathrm{RHom}(\mathcal{E}, \mathcal{E}'_U) \simeq \mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U)$. Taking the total fiber of the bottom right square by columns gives $\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_Z)$; by rows, it is the fiber of

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \rightarrow \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z) \oplus \mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U)$$

and hence

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \longrightarrow \mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z) \oplus \mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U) \longrightarrow \mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_Z[1]).$$

after rotating the fiber sequence. Exactness of the étale realization gives the corresponding fiber sequence after applying $T_{\text{ét}}$, and naturality gives a map between them. To deduce that the map induced by $T_{\text{ét}}$ on the first term is an equivalence, since it is a fiber sequence we need only check the induced maps on the second and third terms are isomorphisms.

We verify that the open terms satisfy the hypotheses of Theorem 5.4 and Lemma 5.23. Put $A^\circ = A[1/T_1, \dots, 1/T_r]_p^\wedge$. Theorem 4.6 identifies $\mathcal{E}_U$ with an object $M \in D_{\text{fgu}}(A)^{h\Gamma}$ satisfying $M \simeq A^\circ \otimes_A^{L, \square} M$. Modulo p , the Fitting ideal argument in the proof of Proposition 4.25 makes the minimal roots of the cohomology modules of M vector bundles over A°/p and their root maps isomorphisms. Hence $M//p$ is perfect over A°/p , and therefore M is perfect over A° . Lemma 5.8 then gives

$$\mathcal{E}_U \in \mathrm{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta(*D))^{\varphi=1}.$$

The same argument of course applies to $\mathcal{E}'_U$.

Lemma 5.22 treats $\mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z)$ using the induction hypothesis. Theorem 5.4 treats $\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U)$. Lemma 5.23 treats $\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_Z[1])$ again under the induction hypothesis. Hence the comparison for $\mathcal{E}$ and $\mathcal{E}'$ is an equivalence.

Finally, we address essential surjectivity. Since the source is idempotent complete and $T_{\text{ét}}$ is exact and fully faithful, its essential image is thick. The target is thickly generated by extensions by zero from the open stratum and by its full subcategories supported on the divisors $V(T_i)$. The former lie in the essential image by Theorem 5.4, and the latter do by Assumption 5.21 and Lemma 5.18. $\square$

Corollary 5.25. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine. Then*

$$T_{\text{ét}} : D_{\text{fgu}}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\text{op}} \longrightarrow D_{\text{cons}, D}^{(b), \text{pKum}}((\mathrm{Spf} R, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence.

Proof. We run a descending induction on $|\Lambda|$ for the categories $(D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma})^{\text{op}}$, in the notation **notation** fixed at the start of this subsection. We claim these are equivalent to the

full subcategories of $D_{\text{cons}, D}^{(b), \text{pKum}}((\text{Spf } R, \mathcal{M})_\eta, \mathbf{Z}_p)$ supported on $(\bigcap_{i \in \Lambda} D_i)_\eta$ (with its pullback log structure). Equivalently, they should be equivalent to p -Kummer constructible sheaves on $(\bigcap_{i \in \Lambda} D_i)_\eta$ with its induced stratification by restricting the divisor.

In the base case $|\Lambda| = r$, the equivalence can be checked directly. Here this is because the corresponding stratum on the log adic generic fiber is an ordinary framed affine $Z = \bigcap_{i=1}^r D_i$ over a rank $n - r$ torus with a rank r hollow log structure. Equipping the target torus with the hollow chart $\mathbf{N}^r \rightarrow 0$, we get an induced strict-étale framing

$$\square : (Z, \mathcal{M}|_Z) \longrightarrow (\text{Spf } W(k)[\zeta_p] \langle T_{r+1}^\pm, \dots, T_n^\pm \rangle, (\mathbf{N}^r \rightarrow 0)^a).$$

The torus has a natural perfectoid cover $\text{Spf } W(k)^{\text{cyc}} \langle T_{r+1}^{\pm 1/p^\infty}, \dots, T_n^{\pm 1/p^\infty} \rangle$ with hollow log structure $\mathbf{N}[1/p]^r \rightarrow 0$, so using $\square$ we obtain an associated perfectoid cover $(R_0^\square)_{\infty, Z}[1/p]$ again with the chart $\mathbf{N}[1/p]^r \rightarrow 0$; we let $(A/J_\Lambda)_{\text{perf}}$ denote $\mathbb{A}_{\text{inf}}((R_0^\square)_{\infty, Z}[1/p]^b)$. Katz's theorem, tilting, and pro-Kummer-étale descent for the same full group Γ give

$$(\text{Perf}((A/J_\Lambda)_{\text{perf}})^{\varphi=1, h\Gamma})^{\text{op}} \xrightarrow{\sim} D_{\text{lisce}}^{(b), \text{pKum}}((Z, \mathbf{N}^r)_\eta, \mathbf{Z}_p) \xrightarrow{i_*} D_{\text{cons}, D}^{(b)}((\text{Spf } R, \mathcal{M})_\eta, \mathbf{Z}_p).$$

In this case the finitely generated unit condition is equivalent to being perfect with unit Frobenius by the Fitting ideal argument of Proposition 4.25. Moreover,

$$D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma} \simeq \text{Perf}((A/J_\Lambda)_{\text{perf}})^{\varphi=1, h\Gamma}$$

by [BS23, Proposition 3.6] and Lemma 2.3; thus the desired claim follows. We note Lemma 4.27 identifies construction with $T_{\text{ét}}$.

The proof of Proposition 5.24 applied to $D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma}$ supplies the induction step, deducing the claim for Λ from the claims for $\Lambda \cup \{i\}$ for all $i \notin \Lambda$. Strictly speaking, as stated the proposition would only apply with the group Γ/Γ_Λ , where $\Gamma_\Lambda = \prod_{i \in \Lambda} \mathbf{Z}_p(1)_i \simeq \mathbf{Z}_p(1)^{|\Lambda|}$. We actually need to deduce the equivalence

$$D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma, \text{op}} \simeq D_{\text{cons}, D|_{Z_\Lambda}}^{(b), \text{pKum}}((Z_\Lambda, \mathcal{M}|_{Z_\Lambda})_\eta, \mathbf{Z}_p)$$

for the induction step; here the target has a hollow component to its log structure, and is therefore not covered by Assumption 3.1. Fortunately, the proof of the open stratum comparison (Theorem 5.4) still applies with the full group Γ , or equivalently where a rank $|\Lambda|$ hollow log structure is allowed on the open stratum. The argument is identical once we can pass fully faithfully to the perfectoid level; thus since we only need a local comparison, the only modification needed is the decompletion result in Proposition 5.9. The reduction to cohomology lemmas holds verbatim, so one simply observes that the helper lemmas (applying Lemma 5.10) work with the larger group. For essential surjectivity one must also check the equivalence in [DLLZ23, Corollary 6.3.4] works in the hollow setting, but this is immediate by keeping track of hollow inertia on both sides. $\square$

The main theorem immediately follows.

Theorem 5.26. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $W(k)[\zeta_p]$. Let $D_{\text{cons}, D}^{(b), \text{pKum}}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$ denote the full subcategory of p -Kummer constructible sheaves in $D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_\eta, \mathbf{Z}_p)$. Then*

$$T_{\text{ét}} : D_{\text{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\text{op}} \xrightarrow{\sim} D_{\text{cons}, D}^{(b), \text{pKum}}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

is a symmetric monoidal equivalence.

Proof. By Lemma 3.4 and strict-étale descent (for the source Theorem 4.12 and for the target Lemma 4.21), we may first work on a framed log affine $(\text{Spf } R, \mathcal{M})$. The claim is then Corollary 5.25.

The fact that the equivalence is symmetric monoidal can also be tested locally, where via Lemma 4.27 it reduces to symmetric monoidality in Theorem 2.6. $\square$

We must also understand perverse t -exactness. We use a definition of the perverse t -structure on the Kummer-étale side of the correspondence which is similar to [BH22].

Definition 5.27. For each connected logarithmic stratum $j_U : U_{\eta} \hookrightarrow (X, \mathcal{M})_{\eta}$, put $d_U = \dim(U_{\eta})$. The p -perverse t -structure of [Gab04] on the full subcategory of p -Kummer sheaves in $D_{\text{cons}, D}^{(b)}((X, \mathcal{M})_{\eta}, \mathbf{Z}_p)$ is defined by

$${}^p D^{\leq 0} = \{K \mid j_U^* K \in D^{\leq -d_U}(U_{\eta}, \mathbf{Z}_p) \text{ for every } U_{\eta}\},$$

$${}^p D^{\geq 0} = \{K \mid j_U^! K \in D^{\geq -d_U}(U_{\eta}, \mathbf{Z}_p) \text{ for every } U_{\eta}\}.$$

Write ${}^p H^i$ for its cohomology functors. The p^+ -perverse t -structure is

$${}^{p^+} D^{\leq 0} = \{K \in {}^p D^{\leq 1} \mid {}^p H^1(K) \text{ is locally bounded } p\text{-power torsion}\},$$

$${}^{p^+} D^{\geq 0} = \{K \mid K//p \in {}^p D^{\geq 0}((X, \mathcal{M})_{\eta}, \mathbf{F}_p)\}.$$

One may check these t -structures restrict to the p -Kummer subcategory. The argument of [BH22, Construction 4.3 and Proposition 4.6], applied to the defining bounds after j_U^* and $j_U^!$, gives the following reduction criteria. For every $n \geq 1$, derived reduction modulo p^n is right t -exact for p -perversity and left t -exact for p^+ -perversity. Moreover, $K \in {}^p D^{\leq 0}$ if and only if $K//p \in {}^p D^{\leq 0}((X, \mathcal{M})_{\eta}, \mathbf{F}_p)$, while $K \in {}^{p^+} D^{\geq 0}$ if and only if $K//p \in {}^p D^{\geq 0}((X, \mathcal{M})_{\eta}, \mathbf{F}_p)$.

Using the equivalence with the essential image, the main content of the following proposition can be reduced to the analogous perverse t -exactness of the Emerton-Kisin correspondence.

Proposition 5.28. *Assume that $(X, \mathcal{M})_{\eta}$ is pure of dimension d . The equivalence of Theorem 5.26 with its essential image, shifted by $[d]$, is t -exact for the standard t -structure on the source and the p^+ -perverse t -structure of Definition 5.27 on the target.*

Proof. The source t -structure is strict-étale local by Corollary A.19, and the same holds for the bounds in Definition 5.27. We may therefore work on a framed log affine.

We first prove the claim modulo p and therefore use the base $\overline{A} = \overline{R}^{\square}((q-1))$. We may consider $M \in D_{\text{fgu}}(\overline{A})^{h\Gamma}$, using Theorem 4.6.

For $\Lambda \subseteq \{1, \dots, r\}$, put

$$\overline{A}_\Lambda = (\overline{A}/(T_i \mid i \in \Lambda)) [1/T_j \mid j \notin \Lambda, j \leq r]$$

and write M_Λ for the derived base change $M \otimes_{\overline{A}}^L \overline{A}_\Lambda$.

Let $j_\Lambda : X_{\eta, \Lambda} \hookrightarrow (X, \mathcal{M})_\eta$ be the corresponding stratum, of dimension $d - |\Lambda|$. Lemma 4.24 and the Fitting ideal argument of Proposition 4.25 show that each $H^i(M_\Lambda)$ is finite projective over $\overline{A}_\Lambda$. Compatibility with restriction, Lemma 4.27, and the contravariant Katz correspondence therefore yield

$$j_\Lambda^* T_{\text{ét}, \mathbf{F}_p}(M)[d] \in D^{\leq -(d-|\Lambda|)} \iff M_\Lambda \in D^{\geq -|\Lambda|}.$$

We claim that $M \in D^{\geq 0}$ if and only if $M_\Lambda \in D^{\geq -|\Lambda|}$ for every Λ , where we use the ordinary t -structure on $D_{\text{fgu}}(\overline{A})^{h\Gamma}$. Applying the coconnectivity criterion of [ATJLS10, Lemma 3.1] along the coordinate stratification, using $i_\Lambda^! M \simeq M_\Lambda[-|\Lambda|]$, gives the claim.

Definition 5.27 now gives

$$T_{\text{ét}, \mathbf{F}_p}(M)[d] \in {}^p D^{\leq 0} \iff M \in D^{\geq 0}.$$

Thus the contravariant equivalence exchanges coconnectivity on the source with perverse connectivity on the target. By orthogonality, it also exchanges connectivity with perverse coconnectivity, proving mod p perverse t -exactness.

To deduce the general claim, we see $M \in D^{\leq 0}$ if and only if $M//p \in D^{\leq 0}$ by derived Nakayama. On the kummer-étale side we have analogous reduction criteria explained in Definition 5.27, so for the ≤ 0 part we may reduce to the mod p assertion. Orthogonality then proves the general claim. $\square$

It is also possible to extract ordinary perverse sheaves from the correspondence. We warn that extracting ordinary constructible sheaves in the derived sense cannot work without altering the crystal side of the correspondence, as these are not a full subcategory of the derived category of Kummer-étale sheaves (simply because Kummer-étale cohomology and étale cohomology differ).

Remark 5.29. Following [KL16, Theorem 7.5.6], call $\mathcal{E} \in D_{\text{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ corresponding to an object in the ordinary heart of constructible sheaves *effective* if, on every framed strict-étale chart, its associated (φ, Γ) -module M satisfies the following condition for every boundary coordinate T_i : the i th $\mathbf{Z}_p(1)$ -factor acts trivially on

$$(A/(T_i)) \otimes_A^{L, \blacksquare} M$$

in the derived category. This condition is strict-étale local and independent of the framing. Theorem 5.26 gives an equivalence

$$T_{\text{ét}} : D_{\text{fgu}}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\text{eff, op}} \xrightarrow{\sim} D_{\text{cons}, D}^\heartsuit(X_{\eta, \text{proét}}, \mathbf{Z}_p).$$

where on the target the ordinary t -structure is used.

In the derived setting, it seems plausible that one could take the “effective part” of an RHom to obtain the derived category of constructible sheaves along D .

We remark that trivial modifications of the previous arguments allow one to deduce the theorem over a slightly more general base. In particular our restriction to $W(k)[\zeta_p]$ up until this point is mainly for expository reasons to avoid tracking additional variables and cases throughout.

Corollary 5.30. *Theorem 5.26 and all preceding results stated over $W(k)[\zeta_p]$ admit corresponding variants over $W(k)[\zeta_{p^\ell}]$ for every $\ell \geq 0$.*

Proof. Corollary 3.19 is the only nontrivial change needed in the argument. With this, one can deduce Theorem 4.12 verbatim (as it uses nothing about Γ). The following arguments are insensitive to the change in prismatic ideal or Γ , except for Proposition 5.9; however there the reduction is to vanishing of a cohomology group, so changing Γ from $(1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ to $(\mu_{p-1} \times (1 + p\mathbf{Z}_p)) \ltimes \mathbf{Z}_p(1)^n$ does not change the result by using Hochschild-Serre. $\square$

Remark 5.31. The entire local theory seems to have a variant which will work over $\mathbf{A}_K$, but unfortunately the globalization step does not appear to work. One would want to have a prism $(\mathbf{A}_K^+, \varphi^n([p]_q))$ which satisfies

$$\mathbf{A}_K^+[1/\varphi^n([p]_q)]_p^\wedge \simeq \mathbf{A}_K.$$

However it seems such a prism cannot exist in general without dropping some desirable properties. See [Wat26, Appendix 2, Proposition 23] for more details. The argument does use that the mod p reduction should be $\mathbf{E}_K^+$, the ring of integers of the field of norms. It is still possible that $\mathbf{A}_K^+$ exists by dropping this condition, but this seems unnatural to do.

The following explicit (φ, Γ) -module gives a constructible sheaf with nontrivial gluing between the two strata.

Example 5.32. Let $(X, \mathcal{M}) = (\mathrm{Spf} \mathbf{Z}_p\langle T \rangle, \mathbf{N})$ with log structure induced by $D = \{T = 0\}$, and let $j : U_\eta = X_\eta \setminus D_\eta \hookrightarrow (X_\eta, \mathbf{N})$. Put $\overline{\mathbf{R}} = \mathbf{F}_p\langle T \rangle$ and $A = \overline{\mathbf{R}}^\square((q-1))$, using the framing given by T .

Let $f : \mathcal{E} \rightarrow \mathbb{D} = X_\eta$ be the projective completion of the Tate family with parameter $x = p^p T$, given in affine coordinates by

$$v^2 + uv = u^3 + a_4(x)u + a_6(x),$$

where a_4 and a_6 are the standard coefficients for the Tate curve.

It is smooth away from $T = 0$, where it becomes a split nodal cubic. The ordinary Zariski constructible sheaf

$$\mathcal{F} = R^1 f_* \mathbf{F}_p$$

has rank two on U_η and rank one at the origin. One may describe $\mathcal{E}[p]$ on U_η as an extension of $\mathbf{F}_p$ by μ_p with Kummer class x (see [BC09, Example 9.2.1]). Since $x^{1/p} = pT^{1/p}$, this is the nonzero Kummer class of T . The sheaf $\mathcal{F}$ is a nonsplit extension

$$0 \longrightarrow \mathbf{F}_p \longrightarrow \mathcal{F} \longrightarrow j_! \mathbf{F}_p(-1) \longrightarrow 0.$$

The corresponding (φ, Γ) -module has underlying module

$$M = A[1/T]e_1 \oplus Ae_2.$$

The submodule $A[1/T]e_1$ and quotient Ae_2 correspond respectively to $j_! \mathbf{F}_p(-1)$ and $\mathbf{F}_p$, in accordance with the contravariance of $T_{\text{ét}}$. Frobenius fixes both basis vectors. For $\gamma = (a_\gamma, b_\gamma) \in \Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)$, we set $\gamma(e_1) = \bar{a}_\gamma e_1$ and

$$\gamma(e_2) = e_2 + \bar{b}_\gamma e_1,$$

where the bars denote reduction modulo p . This realizes $\mathcal{F}$ (applying $T_{\text{ét}, \mathbf{F}_p}$).

We note that Theorem 6.10 does not apply directly in this setting, since the example is not strict log smooth.

6. PUSHFORWARD COMPATIBILITY

6.1. Six functors and Laurent stacks. The category $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta)$ is large enough that the six-functor results imply their Laurent analogues. For ordinary formal schemes, this is carried out in [Kip26b, Kip26a] by constructing an analytic stack $X^{\Delta, \bullet}$. We construct the logarithmic analogue and compare it with our site-theoretic category, and deduce several useful six functors inputs that we will use.

Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$, where K is a complete discretely valued field of mixed characteristic $(0, p)$ with perfect residue field k . As explained in more detail in Definition A.1, the ordinary log prismatic stack is defined by the fiber product

$$\begin{array}{ccc} (X, \mathcal{M})^\Delta & \longrightarrow & X^\Delta \\ \downarrow & & \downarrow \\ \mathbf{Z}_p^\Delta \times \text{Log} & \longrightarrow & \text{Log}^\Delta \end{array}$$

where Log is the formal p -completion of Olsson's log stack [Ols03]. This is a relative prismatic stack over Log , viewed as a “ δ -stack”, so the constructions of [Kip26a] apply by base change.

In [Kip26a, Construction 2.2.7], the éid topology is defined. Roughly, one takes the topology generated by étale covers and integral descendable covers; as explained in [Kip26a, 2.2.5], integrality for formal schemes is tested on ordinary schemes, so for formal affines one checks it on rings modulo an ideal of definition. We will always use integrality in this sense.

Definition 6.1. For a formal prestack $\mathcal{Y}$ over $\mathbf{Z}_p^\Delta$, let $\text{sh}_{\text{éid}} \mathcal{Y}$ denote the éid sheafification of the associated presheaf on $\text{fSch}/\mathbf{Z}_p^{\text{aff}}$ from [Kip26a, Construction 2.2.7]. We define

$$\mathcal{Y}^\bullet := (-)_{\text{Kipp}}^\bullet(\text{sh}_{\text{éid}} \mathcal{Y}),$$

which we call informally the *analytic realization* where

$$(-)^\blacksquare_{\text{Kipp}} : (\mathbf{Z}_p^\Delta)_{\text{éid}} \longrightarrow \text{AnStack}/_{\mathbf{Z}_p^\Delta, \blacksquare}$$

is the colimit-preserving left exact functor of [Kip26a, Theorem 2.3.3(A) and §3.6.1]. We note $(-)^{\blacksquare}$ preserves finite limits, and $(\text{Spf } B)^{\blacksquare} = \text{Spa}(B, B)$ for a bounded prism B .

Strictly speaking, this construction not the same as in [Kip26a]; however this definition will be more convenient for us, as the log prismatic stack is a type of relative stack so we need a version of analytification which also works there. It turns out, however, that this is the same as Kipp's construction. In [Kip26a] the topos $\text{Sh}(\text{Spf}(\mathbf{Z}_p)_{\sqrt{\infty}, \text{ét}})$ is introduced, or sheaves for the topology generated by infinite root covers and étale covers. The functor $\text{Spf } R \mapsto \text{Spa}(\Delta_R, \Delta_R)$ sends covers in this topology to effective epimorphisms of analytic stacks. It is shown that semiperfectoids form a basis of the topology $\text{Spf}(\mathbf{Z}_p)_{\sqrt{\infty}, \text{ét}}$, and so one obtains a colimit preserving functor

$$\text{Sh}(\text{Spf}(\mathbf{Z}_p)_{\sqrt{\infty}, \text{ét}}) \rightarrow \text{AnStack}/_{\mathbf{Z}_p^\Delta, \blacksquare}$$

induced by $\text{Spf } R \mapsto \text{Spa}(\Delta_R, \Delta_R)$ on the semiperfectoid basis. This is fine in the non-logarithmic setting, but unfortunately in the logarithmic setting semiperfectoids are problematic (they may not even have maps to the log prismatic stack, as they often lack fine log structures).

Fortunately the two constructions are the same, although this is not obvious. What is required is to show that for an infinite root/étale cover $\text{Spf } R \rightarrow X$ that

$$\text{Spf } \Delta_R \rightarrow X^\Delta$$

is an effective epimorphism in the éid topos. Then we may use the corresponding colimit presentation coming from the Čech nerve to present X^Δ in the éid topos, and then since both functors are colimit preserving we get the same colimit presentation in analytic stacks.

In fact we claim more generally that for an infinite root/étale cover $\text{Spf } A \rightarrow \text{Spf } B$ that after prismatization and éid sheafification we get an effective epimorphism in the éid topos. It suffices to check this on generating covers; for étale covers it's clear, so one must show

$$\text{sh}_{\text{éid}} V_\infty^\Delta \rightarrow \text{sh}_{\text{éid}} V^\Delta$$

is an effective epimorphism where $V = \text{Spf } \mathbf{Z}_p[t_i]_{i \in I}$ and $V_\infty = \text{Spf } \mathbf{Z}_p[t_i^{1/p^\infty}]_{i \in I}$. In the affine case it is easy to see the éid sheafification is in fact redundant, because $W(-)$ is an éid sheaf in spectra by [Kip26a, 3.2.1] (in particular one may apply the argument in the easy affine case of [BL22, Lemma 7.3]). The argument of [BL22, Lemma 6.3] applies to see we get an effective epimorphism. Consider an R -point of V^Δ . One explicit cover $R \rightarrow S$ constructed there (for S -point lift of V_∞^Δ) is in fact an éid cover. Indeed unwinding their proof one first constructs, after orienting the Cartier-Witt divisor as $(d) \rightarrow W(R)$ by localizing,

$$B = W(R) \langle x_i^{1/p^\infty} \rangle_{i \in I} \left\{ \frac{x_i - \tilde{t}_i}{d} \right\}_{(p, d)}^\wedge$$

where $\tilde{t}_i$ are lifts of t_i to $W(R)$ and $\delta(x_i^{1/p^n}) = 0$. Then $S = B \otimes_{W(R)} R$ (underived by their flatness result). Descendability follows from the proof of [Kip26b, Proposition 3.2.1.8]. For integrality, Lemma A.14 shows that $\pi_0(B//\langle p, d \rangle)$ is integral over the algebra obtained from $\pi_0(W(R)//\langle p, d \rangle)$ by adjoining compatible p -power roots of the images of the $\tilde{t}_i$. This algebra is itself integral over $\pi_0(W(R)//\langle p, d \rangle)$. Since the image of $\langle p, d \rangle$ in R is nilpotent, base change and lifting across nilpotent ideals show that $R \rightarrow S$ is integral, hence an *  id* cover.

Thus, we use Definition 6.1 throughout. In particular, we may then safely define

$$\begin{array}{ccc} (X, \mathcal{M})^{\Delta, \blacksquare} & \longrightarrow & X^{\Delta, \blacksquare} \\ \downarrow & & \downarrow \\ (\mathbf{Z}_p^{\Delta} \times \text{Log})^{\blacksquare} & \longrightarrow & \text{Log}^{\Delta, \blacksquare} \end{array}$$

where the notation on the bottom is interpreted according to Definition 6.1, and thus includes *  id* sheafification. Finite limit preservation in that definition shows one may also construct $(X, \mathcal{M})^{\Delta, \blacksquare}$ by applying analytic realization to the already existing formal stack over $\mathbf{Z}_p^{\Delta}$.

The logarithmic Hodge-Tate substack is the fiber product $(X, \mathcal{M})^{\Delta, \blacksquare} \times_{\mathbf{Z}_p^{\Delta, \blacksquare}} \mathbf{Z}_p^{\text{HT}, \blacksquare}$, where $\mathbf{Z}_p^{\text{HT}, \blacksquare}$ is as in [Kip26a]. Following [Kip26a, Construction 4.3.7 and  4.3.9], we write $(X, \mathcal{M})^{\mathcal{L}, \blacksquare}$ for the algebraic nonvanishing locus of the Hodge-Tate ideal, further base changed along $\text{colim}_n \text{Spa}(\mathbf{Z}/p^n) \rightarrow \text{Spa } \mathbf{Z}_p$. Thus, after pullback along an orientable prism $(B, (d))$, it is given by

$$\text{colim}_{n \geq 1} \text{Spa}(((B//p^n)[1/d], B//p^n)).$$

Since this stack is obtained from $X^{\Delta, \blacksquare}$ by base change, the relevant results of [Kip26a] apply to it. See [Kip26a,  1.1] for cohomological smoothness and weak cohomological properness in six-functor formalisms.

Lemma 6.2. *Let $f : X \rightarrow Y$ be a morphism of analytic stacks, and suppose it is cohomologically smooth and weakly cohomologically proper. Then given a square*

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\iota_X} & X \\ \downarrow f_Z & & \downarrow f \\ Z & \xrightarrow{\iota_Y} & Y \end{array}$$

the map $f_Z : X \times_Y Z \rightarrow Z$ is cohomologically smooth and weakly cohomologically proper. Moreover, $f_Z^! \mathcal{O}_Z \simeq \iota_X^ f^! \mathcal{O}_Y$.*

Proof. The property of being weakly cohomologically proper is stable under arbitrary base change by [Kip26a, Remark 1.1.8], and cohomological smoothness is defined after arbitrary base change, so f_Z is again cohomologically smooth and weakly cohomologically proper. Put $\omega_f = f^! \mathcal{O}_Y$. For a cohomologically smooth map, ω_f is invertible, the natural map $\omega_f \otimes f^*(-) \rightarrow f^!(-)$ is an equivalence, and ω_f commutes with arbitrary base change; compare [Kip26b, Definition

1.2.1.1]. Hence the displayed square gives $\omega_{f_Z} \simeq \iota_X^* \omega_f$. Applying this to the unit object gives $f_Z^! \mathcal{O}_Z \simeq \omega_{f_Z} \simeq \iota_X^* f^! \mathcal{O}_Y$. $\square$

Lemma 6.3. *Let $f : X \rightarrow Y$ be a cohomologically smooth and weakly cohomologically proper morphism of analytic stacks, and put $\omega_f = f^! \mathcal{O}_Y$. Suppose further that*

$$Rf_! \simeq Rf_*$$

via the canonical map.

Then Rf_ preserves dualizable objects, and for every dualizable object $\mathcal{E} \in D_{\blacksquare}(X)$ there is a natural equivalence*

$$(Rf_* \mathcal{E})^\vee \simeq Rf_*(\mathcal{E}^\vee \otimes \omega_f).$$

Proof. For every $\mathcal{K} \in D_{\blacksquare}(Y)$, one has

$$\underline{R}\mathrm{Hom}_Y(Rf_* \mathcal{E}, \mathcal{K}) \simeq Rf_* \underline{R}\mathrm{Hom}_X(\mathcal{E}, f^! \mathcal{K}) \simeq Rf_*(\mathcal{E}^\vee \otimes \omega_f) \otimes \mathcal{K}.$$

The first isomorphism follows from $\underline{R}\mathrm{Hom}_Y(Rf_! \mathcal{E}, \mathcal{K}) \simeq Rf_* \underline{R}\mathrm{Hom}_X(\mathcal{E}, f^! \mathcal{K})$ (which one can check by showing maps into each are the same, and using the tensor-Hom adjunction and the projection formula) and $Rf_! \simeq Rf_*$, while the second is just using the formula $f^!(-) \simeq \omega_f \otimes Lf^*(-)$.

Thus $Rf_* \mathcal{E}$ is dualizable with dual $Rf_*(\mathcal{E}^\vee \otimes \omega_f)$. $\square$

Lemma 6.4. *Let $f : (X, \mathcal{M}_X) \rightarrow (Y, \mathcal{M}_Y)$ be a strict log smooth proper morphism of bounded fs log p -adic formal schemes of pure relative dimension d . Then the induced maps on $(X, \mathcal{M}_X)^{\Delta, \blacksquare}$ and $(X, \mathcal{M}_X)^{\mathcal{L}, \blacksquare}$ are cohomologically smooth and weakly cohomologically proper, and on both stacks one has*

$$f^! \mathcal{O} \simeq \mathcal{O}\{d\}[2d].$$

Moreover $Rf_! \simeq Rf_$ via the canonical map.*

Proof. The underlying map of formal schemes is smooth and proper. Theorem 4.1.3(C),(F) and Example 4.3.2 of [Kip26a] therefore show that the induced map

$$X^{\Delta, \blacksquare} \rightarrow Y^{\Delta, \blacksquare}$$

is cohomologically smooth and weakly cohomologically proper.⁸ We show in addition that there is a canonical equivalence $Rf_! \simeq Rf_*$.

We use $D_{\blacksquare}$ to denote the six functor formalism on analytic stacks.⁹ Let π_2 denote the second projection from $X^{\Delta, \blacksquare} \times_{Y^{\Delta, \blacksquare}} X^{\Delta, \blacksquare}$. Weak cohomological properness implies that f is $D_{\blacksquare}$ -prim,

⁸We note [Kip26a, Lemma 3.5.6] is false for a general connective perfect complex. In its applications to smooth proper morphisms with smooth source and target, the relevant cotangent complex is a vector bundle, in which case the argument is repairable.

⁹We abuse notation here somewhat, as of course in general $D(X)$ for an analytic stack has nothing to do with solid quasicoherent sheaves. However in all of our use cases this will be the case.

so [Mik25, Corollary 1.28] gives a codualizing object

$$\delta_f \simeq R\pi_{2,*}R\Delta_{f,!}\mathcal{O}$$

and a natural equivalence $Rf_!(\delta_f \otimes -) \xrightarrow{\sim} Rf_*$.

Thus it remains to trivialize δ_f canonically. Since $\pi_2 \circ \Delta_f = \text{id}$, it is enough to construct a canonical equivalence $R\Delta_{f,!} \simeq R\Delta_{f,*}$. Since $f : X \rightarrow Y$ is smooth and separated, the diagonal $X \rightarrow X \times_Y X$ on the formal schemes is a regular closed immersion and hence locally of finite presentation. The induced map Δ_f on analytifications of prismatic stacks is therefore $!$ -able by [Kip26a, Theorem 4.1.3(A) and Example 4.3.2]. Using [Mik25, Lemma 1.50] (see the footnote in [Mik25, Definition 1.10] for why it applies to analytic stacks), the desired equivalence holds provided that Δ_f is, universally $D_{\blacksquare}(-)^*$ -locally on its target, represented by morphisms in the class P of [Mik25, Definition 1.5(2)]. This class consists of affine morphisms $\text{AnSpec}(\mathcal{B}) \rightarrow \text{AnSpec}(\mathcal{A})$ for which the analytic structure on $\mathcal{B}$ is induced from that on $\mathcal{A}$, so it simply asks for the morphism to be affine proper (see [Kip26a, §2.1.10]) in our setting.

We verify this remaining hypothesis. Since the functor $X \mapsto X^{\Delta, \blacksquare}$ preserves finite limits by [Kip26a, Construction 3.7.1], the target of Δ_f is $(X \times_Y X)^{\Delta, \blacksquare}$. Work on the semiperfectoid atlas $\text{Spa}(\Delta_{R_0}, \Delta_{R_0})$ of this target from [Kip26a, Construction 3.7.1]. After a Zariski refinement, we can assume f is smooth of pure relative dimension d (as a map of formal schemes), so the diagonal is a regular immersion of codimension d . Thus, after this localization, the pullback of Δ_f on analytic stacks is induced by a derived quotient $S_0 = R_0 // (g_1, \dots, g_d)$. Choose a perfectoid ring R surjecting onto R_0 , lift the g_i to $f_i \in R$, and put $S = R // (f_1, \dots, f_d)$, again as a derived quotient. The setup is summarized by the following cartesian squares:

$$\begin{array}{ccc} X^{\Delta, \blacksquare} & \xrightarrow{\Delta_f} & X^{\Delta, \blacksquare} \times_{Y^{\Delta, \blacksquare}} X^{\Delta, \blacksquare} \\ \uparrow & & \uparrow \\ \text{Spa}(\Delta_{S_0}, \Delta_{S_0}) & \xrightarrow{\Delta_{f, R_0}} & \text{Spa}(\Delta_{R_0}, \Delta_{R_0}) \\ \downarrow & & \downarrow \\ \text{Spa}(\Delta_S, \Delta_S) & \longrightarrow & \text{Spa}(\Delta_R, \Delta_R). \end{array}$$

It is therefore enough to show that $\text{Spa}(\Delta_S, \Delta_S) \rightarrow \text{Spa}(\Delta_R, \Delta_R)$ is affine proper. Indeed, by construction $\text{Spa}(\Delta_{R_0}, \Delta_{R_0})$ is an effective epimorphism of analytic stacks, hence a universal $D_{\blacksquare}^*$ -cover. After a further Zariski refinement, orient the prismatic ideal as $I = (\tilde{d})$ and choose lifts $\tilde{f}_i \in \Delta_R$ of $f_i \in R \simeq \Delta_R/I$. One then has

$$\Delta_S \simeq \Delta_R \left\{ \frac{\tilde{f}_1}{\tilde{d}}, \dots, \frac{\tilde{f}_d}{\tilde{d}} \right\}_{\delta, (p, \tilde{d})}^{\wedge}.$$

Lemma A.14 and [Kip26a, Example 2.1.13] then show that $\text{Spa}(\Delta_S, \Delta_S) \rightarrow \text{Spa}(\Delta_R, \Delta_R)$ is affine proper.

Thus we have verified the hypotheses of [Mik25, Lemma 1.50] for Δ_f , and hence obtain a natural equivalence $R\Delta_{f,!} \simeq R\Delta_{f,*}$. It follows canonically that

$$\delta_f \simeq R\pi_{2,*}R\Delta_{f,!}\mathcal{O} \simeq R\pi_{2,*}R\Delta_{f,*}\mathcal{O} \simeq \mathcal{O},$$

where the last equivalence uses $\pi_2 \circ \Delta_f = \text{id}$. Substituting this identification in the preceding equivalence gives $Rf_! \simeq Rf_*$. By [Mik25, Lemma 1.50], this equivalence is independent of the chosen atlas; its compatibility with arbitrary base change follows from [Mik25, Remark 1.49 and Corollary 1.31(2)].

Theorem 4.1.3(F) of [Kip26a] identifies the dualizing complex with $\mathcal{O}\{d\}[2d]$. Strictness gives

$$(X, \mathcal{M}_X)^{\Delta, \blacksquare} \simeq X^{\Delta, \blacksquare} \times_{Y^{\Delta, \blacksquare}} (Y, \mathcal{M}_Y)^{\Delta, \blacksquare}.$$

The claimed result for $(X, \mathcal{M}_X)^{\Delta, \blacksquare}$ follows from Lemma 6.2 and base-change compatibility of $Rf_! \simeq Rf_*$. For the Laurent stack $(X, \mathcal{M}_X)^{\mathcal{L}, \blacksquare}$, the claim follows by base change. $\square$

The following proposition bridges the gap between the stacky approach and the site-theoretic approach we have taken thus far.

Proposition 6.5. *For every log formal scheme $(X, \mathcal{M})$ satisfying Assumption 3.1, there is a natural equivalence*

$$\text{an}_{(X, \mathcal{M})} : D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} D_{\blacksquare}((X, \mathcal{M})^{\mathcal{L}, \blacksquare}).$$

Proof. Locally we can give identical presentations of both categories. Using Proposition A.22 for a framed log affine we have (depending on a choice of log Breuil-Kisin prism) an identification

$$D_{\blacksquare}((X, \mathcal{M})^{\mathcal{L}, \blacksquare}) \simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A^{(m)}[1/I]_p^{\wedge}).$$

Theorem A.15 identifies this limit with $D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Both the source and target categories also satisfy strict-étale descent. Using [Kip26a, Theorem 2.3.3(A),(B) and Construction 3.7.1] we get étale descent for $X^{\Delta, \blacksquare}$ or that étale covers are sent to effective epimorphisms of analytic stacks. Passing to the Laurent stack $X^{\mathcal{L}, \blacksquare}$ we get étale descent as well, and therefore strict-étale descent for $(X, \mathcal{M})^{\mathcal{L}, \blacksquare}$. For the source, we use Corollary A.18.

This gives an identification depending on a choice of log Breuil-Kisin prism. However, this is induced by a canonical functor. For $(B, (d), \mathcal{N})^a$ in the strict log prismatic site, applying the functor $(-)^{\blacksquare}$ (Definition 6.1) to the map of Lemma A.4 gives

$$\text{Spa}(B, B) \longrightarrow (X, \mathcal{M})^{\Delta, \blacksquare}.$$

Base changing to Laurent stacks gives a pullback functor to $D_{\blacksquare}(B[1/d]_p^{\wedge})$ from $D_{\blacksquare}((X, \mathcal{M})^{\mathcal{L}, \blacksquare})$, and passing to the limit over all such log prisms gives a canonical functor

$$D_{\blacksquare}((X, \mathcal{M})^{\mathcal{L}, \blacksquare}) \rightarrow D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$$

which for a choice of log Breuil-Kisin prism induces the previous identification. The claim then follows; the functor an from the statement is the inverse. $\square$

Lemma 6.6. *For every $(X, \mathcal{M})$ satisfying Assumption 3.1, there is a functorial symmetric monoidal equivalence*

$$\mathrm{an}_{(X, \mathcal{M})} : \mathrm{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} \mathrm{Perf}((X, \mathcal{M})^{\mathcal{L}, \blacksquare}).$$

*It is compatible with pullback and $\underline{\mathrm{RHom}}$. If $(Y, \mathcal{N})$ also satisfies Assumption 3.1, $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{N})$ is strict log smooth and proper, and $f_{\mathcal{L}}$ is the induced map of Laurent analytic stacks, then the right adjoint $\mathrm{R}f_{\Delta, *}$ to pullback on crystals preserves perfect objects and*

$$\mathrm{an}_{(Y, \mathcal{N})} \mathrm{R}f_{\Delta, *} \simeq \mathrm{R}f_{\mathcal{L}, *} \mathrm{an}_{(X, \mathcal{M})}.$$

Proof. All functors in the construction of Proposition 6.5 are symmetric monoidal. Thus its equivalence restricts to dualizable objects and is compatible with $\underline{\mathrm{RHom}}$. Its compatibility with pullback gives

$$\begin{array}{ccc} \mathrm{D}_{\blacksquare}((Y, \mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) & \xrightarrow{\mathrm{an}_{(Y, \mathcal{N})}} & \mathrm{D}_{\blacksquare}((Y, \mathcal{N})^{\mathcal{L}, \blacksquare}) \\ \downarrow \mathrm{L}f^* & & \downarrow \mathrm{L}f_{\mathcal{L}}^* \\ \mathrm{D}_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) & \xrightarrow{\mathrm{an}_{(X, \mathcal{M})}} & \mathrm{D}_{\blacksquare}((X, \mathcal{M})^{\mathcal{L}, \blacksquare}). \end{array}$$

Taking right adjoints gives the asserted comparison of pushforwards on the ambient categories. Lemma 6.4 gives $\mathrm{R}f_{\mathcal{L}, !} \simeq \mathrm{R}f_{\mathcal{L}, *}$, so Lemma 6.3 shows that $\mathrm{R}f_{\mathcal{L}, *}$ preserves perfect objects. The comparison and the equivalence $\mathrm{an}_{(Y, \mathcal{N})}$ give the same assertion for $\mathrm{R}f_{\Delta, *}$. $\square$

We also obtain Poincaré duality.

Corollary 6.7. *Let $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{N})$ be as in Lemma 6.4, and suppose both source and target satisfy Assumption 3.1. Transport $f^!$ on perfect objects from the map $(X, \mathcal{M})^{\mathcal{L}, \blacksquare} \rightarrow (Y, \mathcal{N})^{\mathcal{L}, \blacksquare}$ through Lemma 6.6. Then $f^! \mathcal{G} \simeq f^* \mathcal{G}\{d\}[2d]$ for $\mathcal{G} \in \mathrm{Perf}((Y, \mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$, the right adjoint to pullback on crystals preserves perfect complexes, and*

$$\mathrm{R}f_* \mathcal{E} \simeq \underline{\mathrm{RHom}}_{\mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}}(\mathrm{R}f_*(\mathcal{E}^{\vee}\{d\}), \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}[-2d])$$

for every $\mathcal{E} \in \mathrm{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Proof. Transporting Lemma 6.4 through Lemma 6.6 gives $f^! \mathcal{G} \simeq f^* \mathcal{G}\{d\}[2d]$. Lemma 6.3, together with the pushforward comparison in Lemma 6.6, gives

$$(\mathrm{R}f_* \mathcal{E})^{\vee} \simeq \mathrm{R}f_*(\mathcal{E}^{\vee}\{d\})[2d].$$

Dualizing, this equivalence gives the claim. $\square$

We also note that the Poincaré duality pairing is compatible with Frobenius: the identification $f^! \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge} \simeq \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}\{d\}[2d]$ is Frobenius equivariant since it is upgraded to an identification of F -gauges in [Kip26a, Theorem 4.1.3(F)].

6.2. Pushforward compatibility. Let $f : (X, \mathcal{M}_X) \rightarrow (Y, \mathcal{M}_Y)$ be a strict log smooth proper morphism of pure relative dimension d between log p -adic formal schemes whose adic generic fibers have pure dimensions d_X and d_Y . We assume $(X, \mathcal{M}_X)$ and $(Y, \mathcal{M}_Y)$ satisfy Assumption 3.1 over one of the bases $\mathrm{Spf} W(k)[\zeta_{p^\ell}]$ covered in Corollary 5.30; closed strata are equipped with their pullback log structures.

Let $d = d_X - d_Y$. In such a setup, define

$$Rf_+(-) := Rf_*(-)\{d\}[d].$$

We will show that Rf_+ induces a functor

$$Rf_+ : D_{\mathrm{fgu}}((X, \mathcal{M}_X)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \rightarrow D_{\mathrm{fgu}}((Y, \mathcal{M}_Y)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge).$$

After this, the pushforward compatibility result is formal.

We first need some preliminaries on relative stacks. Let $(Y, \mathcal{M}_Y)$ be a framed log affine with associated logarithmic q -de Rham prism (A, I) , so $Y = \mathrm{Spf}(A/I)$, and let $(X, \mathcal{M}_X) \rightarrow (Y, \mathcal{M}_Y)$ be strict smooth. By Proposition A.22, the map $\mathrm{Spa}(A, A) \rightarrow (Y, \mathcal{M}_Y)^{\Delta, \blacksquare}$ is a representable proper effective epimorphism. Define $(X/A)^{\Delta, \blacksquare}$ via the cartesian square

$$\begin{array}{ccc} (X/A)^{\Delta, \blacksquare} & \longrightarrow & (X, \mathcal{M}_X)^{\Delta, \blacksquare} \\ \downarrow f_A & & \downarrow \\ \mathrm{Spa}(A, A) & \xrightarrow{\rho_A} & (Y, \mathcal{M}_Y)^{\Delta, \blacksquare} \end{array}$$

where the bottom map is the effective epimorphism of Proposition A.22. We define $(X/A)^{\mathcal{L}, \blacksquare}$ by further base change over $\mathrm{Spa}(A, A)$ to

$$(\mathrm{Spf}(A/I)/A)^{\mathcal{L}, \blacksquare} = \mathrm{colim}_n \mathrm{Spa}((A/p^n)[1/I], A/p^n).$$

Equivalently, we replace all terms in the cartesian square with their Laurent versions. The method of Proposition 6.5 identifies $\mathrm{Perf}((X/A)^{\mathcal{L}, \blacksquare})$ with $\mathrm{Perf}((X/A)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$, by locally using the Čech presentation of the stack induced by a Breuil-Kisin prism for X and identifying it with the site-theoretic one.

There is a natural Frobenius φ on $(X/A)^{\mathcal{L}, \blacksquare}$ over φ_A on $(\mathrm{Spf}(A/I)/A)^{\mathcal{L}, \blacksquare}$. The superscript (φ) denotes base change along this φ_A , also for charts and their coefficient rings. If f_A denotes the structure map, relative Frobenius is the induced map $\mathrm{Fr} = (\varphi, f_A) : (X/A)^{\mathcal{L}, \blacksquare} \rightarrow (X/A)^{\mathcal{L}, \blacksquare, (\varphi)}$. We can think of the following lemma as an analogue of Cartier descent; indeed there are some variants already existing, such as in [GLSQ22, §6], but they do not suffice for our applications.

Lemma 6.8. *In the above setting, relative Frobenius induces a fully faithful functor*

$$\mathrm{Fr}^* : \mathrm{Perf}((X/A)^{\mathcal{L}, \blacksquare, (\varphi)}) \longrightarrow \mathrm{Perf}((X/A)^{\mathcal{L}, \blacksquare}).$$

Proof. The projection formula immediately reduces the claim to proving that the unit $\mathcal{O} \rightarrow \mathrm{R}\mathrm{Fr}_* \mathcal{O}$ is an equivalence.

We may also work strict-étale locally by descent of the source and target: thus we assume that X is a framed log affine relative to (Y, M_Y) with relative coordinates $T_1, \dots, T_d$. More precisely, we have

$$(X, M_X) \rightarrow (Y, M_Y) \times_{\mathrm{Spf} W(k)[\zeta_{p^\ell}]} \widehat{\mathbf{G}}_m^d \rightarrow \mathbb{T}_r^{n+d}$$

where $\widehat{\mathbf{G}}_m^d$ has a trivial log structure. Thus (X, M_X) inherits a framing, and has a corresponding q -de Rham prism with ring B^+ and Laurentized ring $B = B^+[1/(q-1)]_p^\wedge$. A straightforward variant of Proposition 3.12 gives a proper morphism $\rho : [\mathrm{AnSpec} B/\Gamma] \rightarrow (X/A)^{\mathcal{L}, \blacksquare}$ with a fully faithful pullback where $\Gamma = \mathbf{Z}_p^d$ fixes A and $\gamma_i(T_j) = q^{p^\ell \delta_{ij}} T_j$. For $\ell = 0$, we use the same chart with $I = ([p]_q)$, which still gives a cover over $W(k)$. In this case $\gamma_i(T_j) = q^{p \delta_{ij}} T_j$ and $\varphi(I) = ([p]_{q^p})$; the proof is similar for this case so we focus on $\ell \geq 1$.

As in the strategy of Proposition 3.12, to verify $\mathcal{O} \rightarrow \mathrm{R}\mathrm{Fr}_* \mathcal{O}$ is an isomorphism we may base change by a proper effective epimorphism. One considers the cartesian square

$$\begin{array}{ccc} [\mathrm{AnSpec} C/\Gamma] & \longrightarrow & [\mathrm{AnSpec} B/\Gamma] \\ \downarrow & & \downarrow \rho \\ & & (X/A)^{\mathcal{L}, \blacksquare} \\ & & \downarrow \mathrm{Fr} \\ \mathrm{AnSpec} B^{(\varphi)} & \longrightarrow & (X/A)^{\mathcal{L}, \blacksquare, (\varphi)} \end{array}$$

By Proposition A.22, the bottom horizontal map is a proper effective epimorphism. Since ρ^* is fully faithful, $\mathcal{O} \rightarrow \mathrm{R}\rho_* \mathcal{O}$ is an equivalence, so it suffices to prove that $\mathcal{O} \rightarrow \mathrm{R}(\mathrm{Fr} \circ \rho)_* \mathcal{O}$ is an equivalence. To apply proper base change, we first prove properness of $\mathrm{Fr} \circ \rho$.

Using Lemma A.6 and finite limit compatibility as noted in Definition 6.1 we may identify C with the laurentization of

$$C^+ \simeq B^+[u_1^{\pm 1}, \dots, u_d^{\pm 1}] \left\{ \frac{u_1 - 1}{[p]_{q^{p^\ell}}}, \dots, \frac{u_d - 1}{[p]_{q^{p^\ell}}} \right\}_{\delta, (p, q-1)}^\wedge.$$

Here $\delta(u_i) = 0$ and $u_i = T_i^p / T_i^{(\varphi)}$, where $T_i^{(\varphi)}$ belongs to the fixed target copy of $B^{+, (\varphi)}$; thus $\gamma_i(u_j) = q^{p^{\ell+1} \delta_{ij}} u_j$.

Modulo $(p, q-1)$, one has $u_i = 1$, and the target map factors through the finite relative Frobenius $B^{+, (\varphi)} / (p, q-1) \rightarrow B^+ / (p, q-1)$. Lemma A.14 therefore shows that $C^+ / (p, q-1)$ is integral over $B^{+, (\varphi)} / (p, q-1)$. By [Kip26a, Example 2.1.13] and Laurent base change, $\mathrm{AnSpec} C \rightarrow \mathrm{AnSpec} B^{(\varphi)}$ is proper. Since Γ fixes the target, the left vertical map factors as

$$[\mathrm{AnSpec} C/\Gamma] \longrightarrow \mathrm{AnSpec} B^{(\varphi)} \times B\Gamma \longrightarrow \mathrm{AnSpec} B^{(\varphi)}.$$

The first map is proper by descent from $\mathrm{AnSpec} C \rightarrow \mathrm{AnSpec} B^{(\varphi)}$, and the second by properness of $B\Gamma$, as proved in Proposition 3.12. Properness is local on the target for the displayed cover, so $\mathrm{Fr} \circ \rho$ is proper. Proper base change now reduces the claim to the equivalence

$$B^{(\varphi)} \longrightarrow R\Gamma_{\blacksquare}(\Gamma, C).$$

We regard B^+ as a module over B^φ , where it decomposes naturally as

$$B^+ \simeq \bigoplus_{\beta \in \{0, \dots, p-1\}^d} B^{+,(\varphi)} T^\beta.$$

The ring C is p -completely nuclear over $B^{(\varphi)}$, so Lemma 3.9 and (1) compute its cohomology by the Koszul complex of $\gamma_1 - 1, \dots, \gamma_d - 1$. We use the toric η -basis from Definition 3.7 directly, as in our situation $\Gamma = \mathbf{Z}_p^d$ and this is easier to work with.

Explicitly, put $\eta_{i,m} = \prod_{j=0}^{m-1} (u_i - q^{p^{\ell+1}j}) / ([p]_{q^{p^\ell}}^m [m]_{q^{p^{\ell+1}}}!)$ and $\eta_\alpha = \prod_i \eta_{i,\alpha_i}$. Using Lemma 3.8, with q replaced by q^{p^ℓ} , we obtain

$$C^+ \simeq \widehat{\bigoplus_{\substack{\beta \in \{0, \dots, p-1\}^d \\ \alpha \in \mathbf{N}^d}} B^{+,(\varphi)} T^\beta \eta_\alpha}.$$

where the completion is $(p, q-1)$ -adic. This decomposition is Γ -equivariant after Laurentizing.

One checks $(\gamma_i - 1)\eta_\alpha = (q^{p^\ell} - 1)u_i\eta_{\alpha'}$, where α' is the same multi-index but with the i th position lowered by one, and negative indices mean zero. For $\beta = 0$ the normalized operators $\frac{\gamma_i - 1}{q^{p^\ell} - 1}$ are unit multiples of the lowering operators (i.e. lowering the i th index in the η_α basis); thus the $\beta = 0$ summand yields a copy of $B^{(\varphi)}$ in degree zero. When $\beta \neq 0$, choose i with $\beta_i \neq 0$. Since $\gamma_i \cdot T^\beta = q^{p^\ell \beta_i} T^\beta$, we get

$$\frac{\gamma_i - 1}{q^{p^\ell} - 1} (T^\beta \eta_\alpha) = T^\beta \left([\beta_i]_{q^{p^\ell}} \eta_\alpha + q^{p^\ell \beta_i} u_i \eta_{\alpha'} \right).$$

Before we invert $q-1$, this becomes modulo $(p, q-1)$ multiplication by β_i plus the lowering operator, which is then invertible. Thus every summand corresponding to $\beta \neq 0$ is acyclic for Γ . Using the Koszul complex it follows that $B^{(\varphi)} \xrightarrow{\sim} R\Gamma_{\blacksquare}(\Gamma, C)$, as desired. $\square$

Proposition 6.9. *Let $f : (X, \mathcal{M}_X) \rightarrow (Y, \mathcal{M}_Y)$ be a strict log smooth proper morphism between log formal schemes satisfying Assumption 3.1 over $W(k)[\zeta_{p^\ell}]$, with $\ell \geq 0$. For every $\mathcal{E} \in \mathrm{Perf}((X, \mathcal{M}_X)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]^\wedge_p)$, the canonical map*

$$\varphi^* Rf_* \mathcal{E} \longrightarrow Rf_* \varphi^* \mathcal{E}$$

is an isomorphism.

Proof. By Proposition 6.5 and proper base change from Lemma 6.4 we may reduce to the case where $(Y, \mathcal{M}_Y)$ is a framed log affine; note here we also use strict-étale descent from Corollary A.18.

We may evaluate the canonical comparison morphism on a logarithmic q -de Rham chart $(A, I, M_{\mathrm{Spf} A})$ of $(Y, \mathcal{M}_Y) = (\mathrm{Spf} A/I, M_{\mathrm{Spf} A/I})$. This evaluation is conservative, so we need only check the claim there. Strictness makes X smooth over $\mathrm{Spf} A/I$. Writing f_A for the structure map $(X/A)^{\mathcal{L}, \blacksquare} \rightarrow \mathrm{AnSpec} A[1/I]_p^\wedge = (\mathrm{Spf} A/I/A)^{\mathcal{L}, \blacksquare}$, we have a commutative diagram

$$\begin{array}{ccccc} (X/A)^{\mathcal{L}, \blacksquare} & \xrightarrow{\mathrm{Fr}} & (X/A)^{\mathcal{L}, \blacksquare, (\varphi)} & \xrightarrow{\tilde{\varphi}_A} & (X/A)^{\mathcal{L}, \blacksquare} \\ & \searrow f_A & \downarrow f_A^{(\varphi)} & & \downarrow f_A \\ & & \mathrm{AnSpec} A[1/I]_p^\wedge & \xrightarrow{\varphi_A} & \mathrm{AnSpec} A[1/I]_p^\wedge \end{array}$$

whose right square is cartesian, so $\tilde{\varphi}_A$ is the base change of φ_A along f_A , and $\tilde{\varphi}_A \circ \mathrm{Fr} = \varphi$. Put $\mathcal{E}^{(\varphi)} = \tilde{\varphi}_A^* \mathcal{E}$. Proper base change and Lemma 6.8 give

$$\begin{aligned} \varphi_A^* Rf_{A,*} \mathcal{E} &\simeq Rf_{A,*}^{(\varphi)} \mathcal{E}^{(\varphi)} \\ &\xrightarrow{\sim} Rf_{A,*}^{(\varphi)} R\mathrm{Fr}_* \mathrm{Fr}^* \mathcal{E}^{(\varphi)} \simeq Rf_{A,*} \varphi^* \mathcal{E}. \end{aligned}$$

Conservativity finishes the proof. $\square$

Theorem 6.10. *Let $f : (X, M_X) \rightarrow (Y, M_Y)$ be strict log smooth proper, where the source and target satisfy Assumption 3.1 over $\mathrm{Spf} W(k)[\zeta_{p^\ell}]$. Then Rf_+ induces a functor*

$$Rf_+ : D_{\mathrm{fgu}}((X, M_X)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \rightarrow D_{\mathrm{fgu}}((Y, M_Y)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge).$$

Proof. Since the finitely generated unit category is stable under shifts and twists we may test the assertion for Rf_* . We may also test the assertion modulo p by derived Nakayama, and further reduce to the case where the p -torsion input $\mathcal{E}$ lies in the heart. Then by Corollary 4.14, there is a coherent root $\mathcal{E}_0$ of $\mathcal{E}$. But then since $Rf_* \simeq Rf_!$ canonically in this situation on the Laurent stacks and Proposition 6.5, the pushforward commutes with colimits. In particular,

$$Rf_*(\mathrm{colim}_n \varphi^{*n} \mathcal{E}_0) \simeq \mathrm{colim}_n Rf_* \varphi^{*n} \mathcal{E}_0.$$

By Proposition 6.9 we see this is $\mathrm{colim}_n \varphi^{*n} Rf_* \mathcal{E}_0$. Using Lemma 6.6, we see $Rf_* \mathcal{E}_0$ is perfect, which proves the result is finitely generated unit. $\square$

Theorem 6.11. *Let $f : (X, M_X) \rightarrow (Y, M_Y)$ be a strict log smooth proper morphism of pure relative dimension d where both satisfy Assumption 3.1 over $\mathrm{Spf} W(k)[\zeta_{p^\ell}]$ whose log adic generic fibers are pure of dimensions d_X and d_Y respectively. Let D_X and D_Y be horizontal normal crossings divisors inducing M_X and M_Y . There is a commutative diagram*

$$\begin{array}{ccc} D_{\mathrm{fgu}}((X, M_X)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\mathrm{op}} & \xrightarrow{T_{\mathrm{ét}, X[d_X]}} & D_{\mathrm{cons}, D_X}^{(b)}((X, M_X)_\eta, \mathbf{Z}_p) \\ \downarrow Rf_* \{d\}[d] & & \downarrow Rf_{\eta,*} \\ D_{\mathrm{fgu}}((Y, M_Y)_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)^{\mathrm{op}} & \xrightarrow{T_{\mathrm{ét}, Y[d_Y]}} & D_{\mathrm{cons}, D_Y}^{(b)}((Y, M_Y)_\eta, \mathbf{Z}_p). \end{array}$$

Proof. Theorem 6.10 and Corollary 4.16 allow us to restrict the relevant functors to the finitely generated unit categories. For $\mathcal{E}$ on X and $\mathcal{G}$ on Y , transporting Lemma 6.4 through Proposition 6.5 and using the projection formula give

$$\begin{aligned} \mathrm{RHom}(\mathrm{R}f_+ \mathcal{E}[d], \mathcal{G}) &\simeq \mathrm{RHom}(\mathrm{R}f_*(\mathcal{E}\{d\}[2d]), \mathcal{G}) \\ &\simeq \mathrm{RHom}(\mathcal{E}\{d\}[2d], f^! \mathcal{G}) \simeq \mathrm{RHom}(\mathcal{E}, \mathrm{L}f^* \mathcal{G}). \end{aligned}$$

Thus $\mathrm{R}f_+[d]$ is left adjoint to $\mathrm{L}f^*$. The units and counits in this calculation are Frobenius compatible (by the compatibility noted after Corollary 6.7), so the adjunction restricts to the unit categories. The functor $\mathrm{R}f_{\eta,*}$ preserves p -Kummer constructible complexes, so the adjunction $\mathrm{L}f_{\eta}^* \rightleftarrows \mathrm{R}f_{\eta,*}$ restricts to these categories. Indeed, proper smooth pushforward preserves lissity on each logarithmic stratum. It is easy to check the p -Kummer condition is preserved as well (one may use a spectral sequence to understand the prime-to- p log inertia on the pushforward).

Pullback compatibility of $\mathrm{T}_{\mathrm{\acute{e}t}}$ carries the opposite of the first adjunction to an adjunction with left adjoint $\mathrm{L}f_{\eta}^*$. Uniqueness of right adjoints therefore gives

$$\mathrm{T}_{\mathrm{\acute{e}t},Y}(\mathrm{R}f_+ \mathcal{E}[d]) \simeq \mathrm{R}f_{\eta,*} \mathrm{T}_{\mathrm{\acute{e}t},X}(\mathcal{E}).$$

Since $d_X = d_Y + d$ this yields the asserted commutative diagram. $\square$

APPENDIX A. LOG PRISMATIC STACKS

A.1. The log prismatic stack. In this appendix by formal stacks we mean (accessible) fpqc sheaves in groupoids (or more generally spaces) on animated p -nilpotent rings.

We denote by Log the p -completion of Olsson's stack of fine log structures over $\mathbf{Z}_p$ [Ols03, Theorem 1.1]. For every fine monoid P , the toric stack

$$\mathcal{A}_P = [\mathrm{Spf} \mathbf{Z}_p[P] / \mathrm{Spf} \mathbf{Z}_p[P^{\mathrm{gp}}]]$$

classifies fine log structures equipped with a map from P which fpqc locally lifts to a chart by [Ols03, Proposition 5.14], and these chart stacks cover Log by [Ols03, Corollary 5.25]. One may check that these, and in fact Log , are formal δ -stacks in the sense of [AKN26, Construction 10.20]. We let Log^{Δ} be its prismaticization.

The general philosophy behind the log prismatic stack is easy to motivate. The data of a log formal scheme X with a fine log structure amounts to a map $X \rightarrow \mathrm{Log}$. Moreover in this interpretation many logarithmic cohomology theories $H_{\mathrm{log}}(-)$ can often be produced by taking a relative version $H(X/\mathrm{Log})$ using the ordinary cohomology theory $H(-)$. For example logarithmic de Rham cohomology can be recovered this way. In [AKN26] they define prismatic stacks relative to δ -rings. More generally, if $X \rightarrow \mathcal{S}$ is a morphism to a formal δ -stack, we write

$$(X/\mathcal{S})^{\Delta} := X^{\Delta} \times_{\mathcal{S}^{\Delta}} (\mathcal{S} \times \mathbf{Z}_p^{\Delta})$$

for its *relative prismaticization*. The point we explain next is what the map $\mathbf{Z}_p^{\Delta} \times \mathcal{S} \rightarrow \mathcal{S}^{\Delta}$ is.

We will use the same construction with $\mathcal{S} = \mathrm{Log}$, which we explain in more detail. We must specify a map $\mathrm{Log} \times \mathbf{Z}_p^{\Delta} \rightarrow \mathrm{Log}^{\Delta}$. Let R be a p -nilpotent ring and let $\mathcal{M}_R \in \mathrm{Log}(R)$. The

Witt vector adjunction and the δ -stack structure on Log give

$$\text{Log}(\mathbf{R}) \simeq \text{Map}_\delta(\text{Spf } W(\mathbf{R}), \text{Log}) \longrightarrow \text{Map}(\text{Spf } W(\mathbf{R}), \text{Log}).$$

Let $\mathcal{M}_{W(\mathbf{R})}$ be the resulting log structure. If $P \rightarrow \mathcal{M}_{\mathbf{R}}$ is a fine chart with structure map $\alpha : P \rightarrow \mathbf{R}$, then locally

$$\mathcal{M}_{W(\mathbf{R})} = \left(P \xrightarrow{m \mapsto [\alpha(m)]} W(\mathbf{R}) \right)^a.$$

We then define the map as

$$\pi_{\text{Log}}(\mathcal{M}_{\mathbf{R}}, I \rightarrow W(\mathbf{R})) = (I \rightarrow W(\mathbf{R}), \mathcal{M}_{W(\mathbf{R})}|_{\text{Spec}(W(\mathbf{R})/I)}).$$

Then fpqc sheafification yields $\pi_{\text{Log}} : \text{Log} \times \mathbf{Z}_p^\Delta \rightarrow \text{Log}^\Delta$. We remark here that $W(\mathbf{R})/I$ has a slightly different meaning, explained in [BL22, §2.1].

Definition A.1. Let $(X, \mathcal{M})$ be a bounded fs log formal scheme. Then there is a map $X \rightarrow \text{Log}$ induced by the log structure. We define $(X, \mathcal{M})^\Delta$ to be the fiber product

$$\begin{array}{ccc} (X, \mathcal{M})^\Delta & \longrightarrow & X^\Delta \\ \downarrow & & \downarrow \\ \text{Log} \times \mathbf{Z}_p^\Delta & \xrightarrow{\pi_{\text{Log}}} & \text{Log}^\Delta. \end{array}$$

This is the log prismatic stack introduced by Olsson in [Ols23].¹⁰

Throughout this appendix, we work in the situation of Assumption 3.1 over $\mathcal{O}_K$, where K is a complete discretely valued field of mixed characteristic $(0, p)$ with perfect residue field k . Strict-étale locally, every such log formal scheme is a framed log affine, for which there is a logarithmic Breuil-Kisin atlas. We define this terminology as follows.

Definition A.2. Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. A logarithmic Breuil-Kisin prism for $(X, \mathcal{M})$ is a log prism $(A, I, M_{\text{Spf } A})^a$ such that

$$(\text{Spf } A/I, M_{\text{Spf } A/I}) \rightarrow (X, \mathcal{M})$$

is a strict open immersion of log p -adic formal schemes.

A logarithmic Breuil-Kisin atlas for $(X, \mathcal{M})$ means this is an isomorphism. We recall that an ordinary Breuil-Kisin prism for a p -adic formal scheme X is a bounded prism (A, I) such that $\text{Spf } A/I \rightarrow X$ is an open immersion. It is an atlas if this map is an isomorphism. If $(X, \mathcal{M})$ is a framed log affine, $(A, I, M_{\text{Spf } A})^a$ is the logarithmic Breuil-Kisin prism associated with its framing by Lemma A.3, and $(B, (d), \mathcal{N})^a$ is a strict test prism, then Lemma A.6 constructs the coproduct and [Kos22, Proposition 4.13] makes

$$B \longrightarrow A \coprod_* B$$

¹⁰We note that the definition of this stack was also considered by Koshikawa.

completely faithfully flat. Here and below, $A \coprod_* B$ denotes the coproduct in the category of log prisms $(X, \mathcal{M})_{\Delta}^{\text{op}}$. Thus the sheaf represented by the atlas covers the final object of the strict log prismatic topos for the fpqc topology.

Lemma A.3. *Let $(X, \mathcal{M}) = (\text{Spf } R, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$. The construction below associates to its framing a logarithmic Breuil-Kisin prism $(A, I, M_{\text{Spf } A})^a$ which is an atlas for $(X, \mathcal{M})$.*

Proof. Write $R \simeq R_0 \widehat{\otimes}_{W(k)} \mathcal{O}_K$, coming from the framing $\square$. We first review what happens for a point. Choose a uniformizer π of $\mathcal{O}_K$ with Eisenstein polynomial $E(u)$, and give $\mathfrak{S} = W(k)[[u]]$ the δ structure with Frobenius $u \mapsto u^p$ (with the Witt vector Frobenius on $W(k)$). This makes $(\mathfrak{S}, (E(u)))$ a Breuil-Kisin prism for $\mathcal{O}_K$. When $\mathcal{O}_K = W(k)[\zeta_{p^\ell}]$ as in our applications, we may use the q -de Rham prism $(W(k)[[q-1]], (\varphi^{\ell-1}([p]_q)))$ with $q \mapsto q^p$; this is also a Breuil-Kisin prism under our definition. We discuss a general construction compatible with the first case, but a similar construction works in the second case.

For a framed log affine, give

$$W(k)\langle T_1, \dots, T_r, T_{r+1}^{\pm 1}, \dots, T_n^{\pm 1} \rangle$$

the Frobenius lift induced by $T_i \mapsto T_i^p$. Since R_0 is p -completely étale over this ring, we may lift Frobenius uniquely to R_0 . We then set

$$A = (R_0 \otimes_{W(k)} \mathfrak{S})_{(p, E(u))}^\wedge$$

with prismatic ideal $E(u)$ and the chart $\mathbf{N}^r \rightarrow A$ sending the i th generator to T_i and $\delta_{\log}(e_i) = 0$, giving a bounded fine log prism. One may check

$$(\text{Spf } A/I, M_{\text{Spf } A/I}) \simeq (X, \mathcal{M}).$$

□

A.2. Quasicoherent sheaves on the log prismatic stack. We show that quasicoherent sheaves on $(X, \mathcal{M})^\Delta$ recover log prismatic crystals. This ends up being slightly more subtle than the argument for X^Δ , because we must generally avoid using Δ_R for R quasiregular semiperfect. The basic issue is that in the log setting these will generally not come with fine log structures, so they will not admit maps to Log .

We first show strict log prisms give points of Olsson's log prismatic stack.

Lemma A.4. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$, and let $(A, I, \mathcal{N})^a$ be a bounded log prism with fine log structure and a strict map $(\text{Spf } A/I, \mathcal{N}|_{\text{Spf } A/I}) \rightarrow (X, \mathcal{M})$. Then there is a natural map*

$$\rho_{X,A} : \text{Spf } A \rightarrow (X, \mathcal{M})^\Delta.$$

If $(X, \mathcal{M})$ is a framed log affine and $(A, I, \mathcal{N})^a$ is the logarithmic Breuil-Kisin prism associated with its framing, then $\rho_{X,A}$ is an fpqc epimorphism.

Proof. Construction 8.14 of [BL22] gives $\mathrm{Spf} A \rightarrow X^\Delta$. By the fiber product definition of $(X, \mathcal{M})^\Delta$, it remains to complete the diagram

$$\begin{array}{ccc} \mathrm{Spf} A & \longrightarrow & X^\Delta \\ \downarrow & & \downarrow \\ \mathbf{Z}_p^\Delta \times \mathrm{Log} & \xrightarrow{\pi_{\mathrm{Log}}} & \mathrm{Log}^\Delta \end{array}$$

The log structure on the prism supplies the dashed arrow; unwinding definitions and using that we have a log prism one may check the composites to Log^Δ agree so we get the desired map.

Now suppose that $(X, \mathcal{M})$ is framed and A is the logarithmic Breuil-Kisin prism associated with its framing. After forgetting the log structure, $\rho_A : \mathrm{Spf} A \rightarrow X^\Delta$ is an fpqc epimorphism. It follows

$$\mathrm{Spf} A \times_{X^\Delta} (X, \mathcal{M})^\Delta \rightarrow (X, \mathcal{M})^\Delta$$

is an fpqc epimorphism. By étale descent of X^Δ , the stack $(X, \mathcal{M})^\Delta$ satisfies strict-étale descent. Thus when working locally we may choose a chart $P = \mathbf{N}^r$ for the log structure on $\mathrm{Spf} A$ and we may replace Log with a toric stack $\mathcal{A}_P := [\mathrm{Spf} \mathbf{Z}_p[P] / \mathrm{Spf} \mathbf{Z}_p[P^{\mathrm{gp}}]]$ in the fiber product defining $(X, \mathcal{M})^\Delta$. Lemma A.21 gives a fine log prism B_P and an fpqc epimorphism to $\mathrm{Spf} A \times_{\mathcal{A}_P} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P])$. Since $\mathrm{Spf} A \rightarrow X^\Delta$ and $\mathrm{Spf} \mathbf{Z}_p[P] \rightarrow \mathcal{A}_P$ are fpqc epimorphisms¹¹, one obtains an fpqc epimorphism

$$\rho_{X, B_P} : \mathrm{Spf} B_P \rightarrow (X, \mathcal{M})^\Delta.$$

We may check whether $\rho_{X, A} : \mathrm{Spf} A \rightarrow (X, \mathcal{M})^\Delta$ is an fpqc epimorphism after base change along one, and indeed we obtain

$$\begin{array}{ccc} \mathrm{Spf}(A \coprod_* B_P) & \longrightarrow & \mathrm{Spf} B_P \\ \downarrow & & \downarrow \\ \mathrm{Spf} A & \xrightarrow{\rho_{X, A}} & (X, \mathcal{M})^\Delta \end{array}$$

using Lemma A.6 (which only uses the first part of this lemma). Here $(A \coprod_* B_P)$ is the coproduct in the category of log prisms $(X, \mathcal{M})_{\Delta}^{\mathrm{op}}$ which exists by Lemma A.5. But one can directly check that

$$\mathrm{Spf}(A \coprod_* B_P) \rightarrow \mathrm{Spf} B_P$$

is (p, I) -completely fpqc as it is a prismatic envelope of a (p, I) -completely regular sequence, using [BS22, Proposition 3.13] (as constructed in Lemma A.5). This proves the claim. $\square$

¹¹Even better, both are flat covers of formal stacks. The latter is clear, while the former follows by noting $A \rightarrow A_{\mathrm{perf}}$ is p -completely faithfully flat (using Kunz's theorem) and applying the proof of [Pen25, Proposition 2.19].

Using these fpqc epimorphisms, we next want to relate fiber products $\mathrm{Spf} A \times_{(X, \mathcal{M})^\Delta} \mathrm{Spf} B$ to coproducts in the category of log prisms over $(X, \mathcal{M})$. However since [Kos22, Proposition 4.13] works with the relative rather than absolute site there is not a direct reference. We use a similar method to construct the envelope in our particular case.

Lemma A.5. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, let $(A, I, M_{\mathrm{Spf} A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $(B, (d), \mathcal{N})^a$ be an orientable bounded strict object of $(X, \mathcal{M})_\Delta$. The coproduct $A \coprod_* B$ exists.*

Proof. We recall that the construction in Lemma A.3 has two variants: the first uses the usual Breuil-Kisin prism with prismatic ideal $E(u)$ generated by an Eisenstein polynomial of a uniformizer π , and the second uses the q -de Rham prism. The argument in each case is very similar, so we will focus on the first case.

Write $R = R_0 \hat{\otimes}_{W(k)} \mathcal{O}_K$, with the strict p -completely étale framing

$$\square : (W(k) \langle T_1, \dots, T_r, T_{r+1}^{\pm 1}, \dots, T_n^{\pm 1} \rangle, \mathbf{N}^r) \longrightarrow (R_0, \mathbf{N}^r).$$

We keep $E(u)$ as a choice of Eisenstein polynomial for a choice of uniformizer π . From Lemma A.3, we construct $A = R_0[[u]]$ and $I = (E(u))$, where u, T_i are sent to 0 by δ ; it is equipped with a log structure with prelog chart $\mathbf{N}^r$ lifted along the framing.

For the other log prism $(B, (d), \mathcal{N})^a$, put $\bar{B} = B/d$. By strictness and [Kos22, Lemma 3.8] we get a chart $\mathbf{N}^r \rightarrow \Gamma(\mathrm{Spf} B, \mathcal{N})$ with $e_i \mapsto m_i \in \Gamma(\mathrm{Spf} B, \mathcal{N})$ (using the lemma to lift the sections on $\Gamma(\mathrm{Spf} B/d, \mathcal{N}|_{B/d})$ which is pulled back from $\mathcal{M}$ by strictness). Write $\alpha : \Gamma(\mathrm{Spf} B, \mathcal{N}) \rightarrow B$ for the log structure map on global sections. For A , we denote by e_i the analogous log sections $\mathbf{N}^r \rightarrow A$ which are sent modulo $E(u)$ to T_i .

To construct the desired coproduct, we must construct a log prism C with maps from A and B such that, for every bounded test log prism S over $(X, \mathcal{M})$,

$$\mathrm{Hom}_{(X, \mathcal{M})_\Delta^{\mathrm{op}}}(C, S) \simeq \mathrm{Hom}_{(X, \mathcal{M})_\Delta^{\mathrm{op}}}(A, S) \times \mathrm{Hom}_{(X, \mathcal{M})_\Delta^{\mathrm{op}}}(B, S).$$

For a pair $f : A \rightarrow S$ and $g : B \rightarrow S$ on the right, rigidity of prisms shows the prismatic ideal of S is dS . The log sections $g(m_i)$ and $f(e_i)$ have the same reduction. Lemma 3.8 of [Kos22] therefore gives a unique unit $\lambda_i \in 1 + dS$ such that $g(m_i) = \lambda_i f(e_i)$.

We do this directly – the construction is somewhat similar to what we did in Lemma 3.6. We adjoin variables v_i representing these units λ_i . Form

$$D_0^{\mathrm{ex}} = ((R_0[u] \otimes_{W(k)} B)[v_1^{\pm 1}, \dots, v_r^{\pm 1}] / (\alpha(m_i) - v_i T_i)_{i=1}^r)_{(p, d)}^\wedge.$$

Give it the log structure pulled back from B and set $\delta(v_i) = v_i^p \delta_{\log}(m_i)$. This is (similar to) the exactification of [Kos22, Construction 2.17], which works in the relative case. Using the maps $R_0[u] \rightarrow R \rightarrow \bar{B}$, $B \rightarrow \bar{B}$, and $v_i \mapsto 1$ we obtain

$$q_0 : D_0^{\mathrm{ex}} \rightarrow \bar{B}.$$

Using the étale framing and noting $T_i = \alpha(m_i)/v_i$ for $i \leq r$ to eliminate these variables we see D_0^{ex} is (p, d) -completely smooth over B of relative dimension $n + 1$. Locally $\ker(q_0) =$

$(d, x_1, \dots, x_{n+1})$, with the x_j a (p, d) -completely regular sequence relative to B (in the sense of [BS22, Definition 2.42]). We finally construct C as the prismatic envelope

$$C = (D_0^{\text{ex}} \{x_1/d, \dots, x_{n+1}/d\}_\delta)_{(p,d)}^\wedge,$$

with the naturally induced log structure from D_0^{ex} . The map $B \rightarrow C$ makes it so C is a strict log prism in $(X, \mathcal{M})_{\Delta}^{\text{op}}$. Since $E(u) \in \ker(q_0)$ maps into dC , the map $R_0[u] \rightarrow C$ extends to a log prism map $A \rightarrow C$.

One may verify the universal property directly similarly to Lemma 3.6 using the universal property of the prismatic envelope. $\square$

We now show how to recover these coproducts via the log prismatic stack.

Lemma A.6. *Let $(X, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, let $(A, I, M_{\text{Spf } A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $(B, (d), \mathcal{N})^a$ be an orientable bounded strict object of $(X, \mathcal{M})_{\Delta}$. The coproduct $A \coprod_* B$ exists by Lemma A.5, and*

$$\text{Spf}(A \coprod_* B) \simeq \text{Spf } A \times_{(X, \mathcal{M})_{\Delta}} \text{Spf } B.$$

If (A, I) is an ordinary Breuil-Kisin atlas for X obtained from a smooth lift over $W(k)[[u]]$ and $(B, (d))$ is an orientable bounded object of X_{Δ} , the analogous assertion holds without log structures.

Proof. Use the notation of Lemma A.5, with $\bar{B} = B/d$. The framing factors through $X \rightarrow \mathcal{A}_{\mathbf{N}^r} \rightarrow \text{Log}$, where $\mathcal{A}_{\mathbf{N}^r} \rightarrow \text{Log}$ is étale. Recall here $\mathcal{A}_P := [\text{Spf } \mathbf{Z}_p[P] / \text{Spf } \mathbf{Z}_p[P^{\text{gp}}]]$.

Thus $(X, \mathcal{M})_{\Delta} \simeq (X/\mathcal{A}_{\mathbf{N}^r})_{\Delta}$. The two maps to this stack come from the squares

$$\begin{array}{ccccc} X & \xrightarrow{\text{id}} & X & \longleftarrow & \text{Spf } \bar{B} \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spf } A & \longrightarrow & \mathcal{A}_{\mathbf{N}^r} & \longleftarrow & \text{Spf } B. \end{array}$$

Since $(X/\text{Spf } A)_{\Delta} \simeq \text{Spf } A$ and $(\text{Spf } \bar{B}/\text{Spf } B)_{\Delta} \simeq \text{Spf } B$, finite limit compatibility of relative prismaticization (for δ -pairs by [AKN26, Lemma 4.20]) gives

$$\text{Spf } A \times_{(X, \mathcal{M})_{\Delta}} \text{Spf } B \simeq (\text{Spf } \bar{B} / (\text{Spf } A \times_{\mathcal{A}_{\mathbf{N}^r}}^R \text{Spf } B))_{\Delta}^\wedge.$$

The chart presentation $\mathcal{A}_{\mathbf{N}^r} = [\text{Spf } \mathbf{Z}_p[\mathbf{N}^r] / \mathbf{G}_m^r]$ computes the denominator $\text{Spf } A \times_{\mathcal{A}_{\mathbf{N}^r}}^R \text{Spf } B$ explicitly as

$$\text{Spf} \left((A \otimes_{\mathbf{Z}_p}^L B)[v_1^{\pm 1}, \dots, v_r^{\pm 1}] // (\alpha(m_i) - v_i T_i)_{i=1}^r \right)_{(p, E, d)}^\wedge.$$

Here v_i are the coordinates on $\mathbf{G}_m^r$, and the map from $\text{Spf } \bar{B}$ is given by $v_i \mapsto 1$.

The completion $D^{\text{ex}} := (D_0^{\text{ex}})_{(p, d, E)}^\wedge$ of the exactification in Lemma A.5 has the presentation

$$D^{\text{ex}} = ((A \otimes_{W(k)}^L B)[v_1^{\pm 1}, \dots, v_r^{\pm 1}] // (\alpha(m_i) - v_i T_i)_{i=1}^r)_{(p, E, d)}^\wedge.$$

It has the same log prismatic envelope as D_0^{ex} : every map defining that envelope sends E into the target's prismatic ideal, so extends uniquely across this completion. The maps $W(k) \rightrightarrows \bar{B}$ induced by A and B agree. Since $W(k)$ is p -completely formally étale over $\mathbf{Z}_p$, the canonical map

$$(\bar{B}/D^{\text{ex}})^{\Delta} \xrightarrow{\sim} (\text{Spf } \bar{B}/(\text{Spf } A \times_{\mathcal{A}_{\mathbf{N}^r}}^R \text{Spf } B))^{\Delta}$$

is an equivalence (see [AKN26, Lemma 5.9]).

By [BL22, Theorem 7.17], this identifies

$$\text{Spf } A \times_{(X, \mathcal{M})^{\Delta}} \text{Spf } B \simeq \text{Spf } \Delta_{\bar{B}/D^{\text{ex}}}.$$

This may be computed directly as a prismatic envelope as $\bar{B}$ is locally cut out by a Koszul-regular sequence in D^{ex}/d , and as previously observed this prismatic envelope is the same as the one computed using D_0^{ex} . Thus in Lemma A.5 we have shown this is $\text{Spf}(A \coprod_* B)$ where the coproduct is in the category of prisms $(X, \mathcal{M})_{\Delta}^{\text{op}}$.

The ordinary assertion follows by a similar (far easier) argument. $\square$

Proposition A.7. *Let $(X, \mathcal{M}) = (\text{Spf } R, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, and let*

$$\rho_{X,A} : \text{Spf } A \rightarrow (X, \mathcal{M})^{\Delta}$$

be the fpqc epimorphism of Lemma A.4 associated with the logarithmic Breuil-Kisin prism of its framing. Then the Čech nerve of $\text{Spf } A$ identifies with $\text{Spf } A^{(n)}$ where $A^{(n)}$ are the iterated self-coproducts of $(A, I, M_{\text{Spf } A})^a$ in $(X, \mathcal{M})_{\Delta}^{\text{op}}$. In particular

$$(X, \mathcal{M})^{\Delta} \simeq |\text{Spf } A^{(\bullet)}|$$

in formal stacks.

Proof. Lemma A.6, applied with $B = A$, identifies the degree one term of the Čech nerve with $\text{Spf}(A \coprod_* A)$. The higher terms are the formal spectra of the iterated coproducts $A^{(n)}$, for both the site-theoretic Čech nerve and the Čech nerve of $\rho_{X,A} : \text{Spf } A \rightarrow (X, \mathcal{M})^{\Delta}$. Using Lemma A.4, this is an fpqc epimorphism so the claim follows. $\square$

Proposition A.8. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. There is a natural symmetric monoidal equivalence*

$$\iota_{(X, \mathcal{M})} : D((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}) \xrightarrow{\sim} D_{\text{qc}}((X, \mathcal{M})^{\Delta}).$$

Proof. We first note that the assertion can be checked strict-étale locally. Indeed for the crystal side by [Ino25, Lemma 5.9], crystals may be computed on the strict subsite; on a framed log affine, using the logarithmic Breuil-Kisin prism $(A, I, M_{\text{Spf } A})^a$ associated with its framing yields

$$\lim_{[n] \in \Delta} D_{(p, I^{(n)})}(A^{(n)}) \simeq D((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta})$$

where $A^{(n)}$ is the logarithmic Čech nerve. The proof of Corollary A.18 gives strict-étale descent of the crystal side.

The same holds for the stack $(X, \mathcal{M})^\Delta$ by étale descent for X^Δ in X [BL22]. We may therefore assume $(X, \mathcal{M}) = (\mathrm{Spf} R, \mathcal{M})$ is a framed log affine. The logarithmic Breuil-Kisin prism A associated with its framing covers the log prismatic stack by Lemma A.4.

Now we can use Proposition A.7 to identify

$$D_{\mathrm{qc}}((X, \mathcal{M})^\Delta) \simeq \lim_{[n] \in \Delta} D_{\mathrm{qc}}(\mathrm{Spf} A^{(n)}) \simeq D((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta).$$

Indeed, $\rho_{X,A} : \mathrm{Spf} A \rightarrow (X, \mathcal{M})^\Delta$ is an fpqc epimorphism and we have identified its Čech nerve with $\mathrm{Spf} A^{(n)}$, and thus D_{qc} satisfies descent for such maps. See the discussion after [Pen25, Definition 2.4] for more details. $\square$

For a strict morphism f , we write $Rf_{\Delta,*}$ and Rf_* for the right adjoints to pullback on the two categories of Proposition A.8, respectively.

Proposition A.9. *Let $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{N})$ be a strict morphism between pairs satisfying Assumption 3.1 over $\mathcal{O}_K$. For every $\mathcal{E} \in D((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta)$, there is a natural equivalence*

$$\iota_{(Y, \mathcal{N})} Rf_{\Delta,*} \mathcal{E} \simeq Rf_* \iota_{(X, \mathcal{M})} \mathcal{E}.$$

Proof. Compatibility of Proposition A.8 with strict pullback gives a commutative diagram

$$\begin{array}{ccc} D((Y, \mathcal{N})_\Delta, \mathcal{O}_\Delta) & \xrightarrow{\iota_{(Y, \mathcal{N})}} & D_{\mathrm{qc}}((Y, \mathcal{N})^\Delta) \\ \downarrow f^* & & \downarrow (f^\Delta)^* \\ D((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta) & \xrightarrow{\iota_{(X, \mathcal{M})}} & D_{\mathrm{qc}}((X, \mathcal{M})^\Delta). \end{array}$$

Both vertical pullback functors preserve colimits. Taking right adjoints of this square gives the asserted equivalence. $\square$

Remark A.10. Laurent F-crystals in [KY25, IKY26] use the saturated log prismatic site. By [Ino25, Lemma 5.9], the strict subsite is cofinal in the full log prismatic site. If $(X, \mathcal{M})$ is fs and saturated, a strict object has the same characteristic monoid as the pullback of $\overline{\mathcal{M}}$, and is therefore saturated. The strictification in [Ino25, Lemma 5.9] has this same characteristic monoid, so the strict subsite is also cofinal in the saturated log prismatic site. Thus the limits defining the crystal categories on the full and saturated sites agree.

We suppress this distinction for $D((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta)$ and its perfect and Laurent variants, but make no comparison between arbitrary sheaves on the underlying ringed topoi.

A.3. Solid Laurent crystals. In this subsection, we define a solid generalization of

$$\mathrm{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}).$$

We always use the analytic structure $\mathrm{Spa}(A, A)$.

Definition A.11. Let $(A, (d))$ be an orientable prism. Associated to the derived (p, d) -complete ring A is the analytic adic ring (A, A) in the sense of [AM24]. We define

$$D_{\blacksquare}(A[1/d]_p^{\wedge}) := \varprojlim_n \mathrm{Mod}_{(A//p^n)[1/d]} D_{\blacksquare}((A, A)),$$

where $(A//p^n)[1/d]$ is regarded as an algebra object of $D_{\blacksquare}((A, A))$. Equivalently, this is $D_{\blacksquare}$ of $\mathrm{colim}_n \mathrm{Spa}((A//p^n)[1/d], A//p^n)$, where we use Spa to denote the analytic stack attached to a Huber pair as in [Kip26a, Example 2.1.5].

Then

$$D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) := \lim_{(A, (d), M_{\mathrm{Spf} A})^a \in (X, \mathcal{M})_{\Delta}^{\mathrm{op}}} D_{\blacksquare}(A[1/d]_p^{\wedge}).$$

The definition is independent of the generator d , and one may (by ind-Zariski localization) assume that the prismatic ideal is principal.

In what follows we repeatedly use the following formal fact: if $\mathcal{C} \simeq \varprojlim_i \mathcal{C}_i$ is a limit of presentable symmetric monoidal stable ∞ -categories and an algebra $S \in \mathcal{C}$ has images S_i , then

$$\mathrm{Mod}_S(\mathcal{C}) \simeq \varprojlim_i \mathrm{Mod}_{S_i}(\mathcal{C}_i).$$

Indeed, algebra objects and their module objects are computed in limits.

We define perfect Laurent crystals by dualizability:

$$\mathrm{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) := D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})^{\mathrm{dual}}.$$

Lemma A.12. *Let $A \rightarrow B$ be a faithfully flat map of connective animated rings. If $\pi_0 B$ is countably presented as a $\pi_0 A$ -module, then $A \rightarrow B$ is descendable of index at most 2.*

Proof. See [Mat16, Corollary 3.33 and the proof of Proposition 3.32]. □

Lemma A.13. *Let $(X, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, let $(A, I, M_{\mathrm{Spf} A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $(B, (d), \mathcal{N})^a$ be an orientable bounded strict object of the same log prismatic site. Put*

$$C = A \coprod_{*} B$$

for the coproduct in $(X, \mathcal{M})_{\Delta}^{\mathrm{op}}$. Then C exists, and

$$B \longrightarrow C$$

is completely descendable for the (p, d) -adic topologies in the sense of [Kip26b, Definition 2.4.1.7], and the induced algebra is descendable in $D_{\blacksquare}((B, B))$.

Moreover, $\mathrm{Spa}(C, C) \rightarrow \mathrm{Spa}(B, B)$ is a proper descendable cover.

Note that as a degenerate case the same assertions hold when X is smooth over $\mathcal{O}_K$, (A, I) is an ordinary Breuil-Kisin atlas obtained from a smooth lift over $W(k)[[u]]$, $(B, (d))$ is an orientable bounded object of X_Δ , and $C = A \coprod_* B$.

Proof. We prove the logarithmic assertion; the ordinary case is identical.

By the construction in Lemma A.5, Zariski locally we have

$$(6) \quad C \simeq (D^{\mathrm{ex}}\{x_1/d, \dots, x_m/d\}_\delta)_{(p,d)}^\wedge,$$

where $\ker(D^{\mathrm{ex}} \rightarrow \bar{B}) = (d, x_1, \dots, x_m)$. Using [Mao24, Proposition 5.52, and Remark 5.54] we see that $B \rightarrow C$ is (p, d) -completely faithfully flat. Its derived reductions modulo (p^k, d^k) are faithfully flat and countably presented on π_0 as modules, since we adjoin finitely many variables as a δ ring. By Lemma A.12, these reductions are descendable of index ≤ 2 .

Rigidity makes $B \rightarrow C$ adic. Put $J = \mathrm{fib}(B \rightarrow C)$ in the derived (p, d) -complete category and $J_k = J // (p^k, d^k)$. Derived completeness gives

$$\mathrm{Map}_B(J \widehat{\otimes}_B J, B) \simeq \mathrm{R} \lim_k \mathrm{Map}_{B // (p^k, d^k)}(J_k^{\otimes 2}, B // (p^k, d^k)).$$

The canonical map $J \widehat{\otimes}_B J \rightarrow B$ is null at every level. After this trick one may still apply the argument of [BS17, Lemma 11.22] (despite that it applies to $\mathbf{N}$ -indexed colimits) to show that $(J \widehat{\otimes}_B J)^{\otimes 2} \rightarrow B$ is null and C is descendable of index ≤ 4 in the derived (p, d) -complete category.

This is called complete descendability in [Kip26b, Definition 2.4.1.7]. Under $D(\mathrm{Spf} B) \simeq D_{(p,d)\text{-comp}}(B)$, and the lax symmetric monoidal analytification functor used in the proof of [Kip26a, Theorem 2.3.3(A)] preserves descendable algebras. Hence the algebra induced by C in $D_\bullet((B, B))$ is descendable.

It remains to prove properness. Example 2.1.13 of [Kip26a] says that a map $\mathrm{Spa}(C, C) \rightarrow \mathrm{Spa}(B, B)$ is proper if and only if C is integral over the subring generated by B and topologically nilpotent elements of C . In the present situation, it is therefore enough to prove that

$$\pi_0(B // (p, d)) \longrightarrow \pi_0(C // (p, d))$$

is integral. Indeed, if a monic polynomial f over $\pi_0 B$ annihilates c modulo (p, d) , then $f(t) - f(c)$ is a monic polynomial over $\pi_0 B + (p, d)\pi_0 C$ annihilating c .

Applying Lemma A.14 to (6), using $D^{\mathrm{ex}}/(d, x_1, \dots, x_m) \simeq B/d$, gives the required integrality and hence properness. The same argument applies to the ordinary case. $\square$

Lemma A.14. *Let A be an animated δ -ring, let $d \in \pi_0 A$ be distinguished, and let $\{f_i\}$ be any (arbitrarily large) family of elements of $\pi_0 A$. Suppose A is further derived (p, d) complete. Then the morphism*

$$\mathrm{Spf}(A\{f_i/d\}_\delta)_{(p,d)}^\wedge \rightarrow \mathrm{Spf} A$$

induced by the structure map is integral in the sense of [Kip26a, 2.2.5]. In other words, on rings it is true in the ordinary sense modulo (p, d) .

Proof. As we discuss integrality on formal schemes, we must show

$$\pi_0(A// (p, d)) \rightarrow \pi_0 \left((A\{f_i/d\}_\delta)^\wedge_{(p, d)} // (p, d) \right)$$

is integral. It suffices to check all of the iterates $\delta^n(f_i/d)$ satisfy monic polynomials in $\pi_0(A// (p, d))[T]$ modulo (p, d) . We will show that $\delta^n(f_i/d)$ satisfies a monic polynomial in $\pi_0(A// (p, d))[(f_i/d), \delta(f_i/d), \dots, \delta^{n-1}(f_i/d)]$.

The key identity for us will be

$$0 = \delta^n(d(f_i/d) - f_i) = \varphi^n(d)\delta^n(f_i/d) + F_n((f_i/d), \delta(f_i/d), \dots, \delta^{n-1}(f_i/d))$$

where F_n is a polynomial with coefficients in A . Moreover, for $n \geq 1$, modulo (p, d) , F_n has degree p in $\delta^{n-1}(f_i/d)$, with leading coefficient $\delta(\varphi^{n-1}(d))$. We will drop the argument of F_n from here on.

The identity holds for $n = 0$ with $F_0 = -f_i$. Applying δ to the relation, one sees that $\delta^{n+1}(d(f_i/d) - f_i)$ equals

$$\delta(\varphi^n(d)\delta^n(f_i/d)) + \delta(F_n) - \sum_{j=1}^{p-1} \frac{1}{p} \binom{p}{j} (\varphi^n(d)\delta^n(f_i/d))^j F_n^{p-j}.$$

One computes $\delta(\varphi^n(d)\delta^n(f_i/d)) = \varphi^{n+1}(d)\delta^{n+1}(f_i/d) + \delta(\varphi^n(d))(\delta^n(f_i/d))^p$ as well. Collecting the remaining terms to form F_{n+1} gives the first part of the claim. For the second part, if we work modulo (p, d) , the previous display becomes $\delta(\varphi^n(d)\delta^n(f_i/d)) + \delta(F_n)$, as the factor we subtract has all terms divisible by $\varphi^n(d) \in (p, d)$. Since F_n involves only the preceding iterates, $\delta(F_n)$ has degree at most one in $\delta^n(f_i/d)$ modulo p . Thus the degree- p term of F_{n+1} has the asserted coefficient.

To deduce the desired claim for $\delta^n(f_i/d)$, one then reduces this relation for the index $n + 1$ modulo (p, d) : this kills the term $\varphi^{n+1}(d)\delta^{n+1}(f_i/d)$, leaving only the remaining terms $F_{n+1}((f_i/d), \delta(f_i/d), \dots, \delta^n(f_i/d))$. We know the leading coefficient in $\delta^n(f_i/d)$ for F_{n+1} modulo (p, d) was $\delta(\varphi^n(d))$ attached to $(\delta^n(f_i/d))^p$, but φ and δ commute so this is a unit (since d is distinguished; i.e. $\delta(d)$ is a unit). Thus after we divide by this unit, we obtain a monic polynomial in $\pi_0(A// (p, d))[(f_i/d), \delta(f_i/d), \dots, \delta^{n-1}(f_i/d)]$ that $\delta^n(f_i/d)$ satisfies modulo (p, d) . The claim follows. $\square$

Theorem A.15. *Let $(X, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, and let $(A, I, M_{\text{Spf } A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing. If $(A^{(m)}, I, M^{(m)})^a$ is the degree m term of its Čech nerve in the strict log prismatic site, then*

$$D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]^\wedge_p) \simeq \varprojlim_{[m] \in \Delta} D_\bullet(A^{(m)}[1/I]^\wedge_p).$$

Proof. We work on the strict log prismatic site, where orientable objects form a Zariski basis. For an orientable strict test prism $(B, (d), \mathcal{N})^a$, Lemma A.6 identifies the pullback of the atlas with

$$B \longrightarrow A \coprod_* B.$$

The result of Lemma A.13 gives

$$D_{\blacksquare}(B[1/d]_p^\wedge) \simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}((A^{(m)} \coprod_* B)[1/d]_p^\wedge)$$

because modulo every p^n , the degree m term of the Čech nerve of $B \rightarrow A \coprod_* B$ after Laurentization identifies with $((A^{(m)} \coprod_* B) // p^n)[1/d]$, and forming module objects commutes with limits in symmetric monoidal presentable categories. This shows the assertion that $(B, (d), M_{\mathrm{Spf} B})^a \mapsto D_{\blacksquare}(B[1/d]_p^\wedge)$ satisfies descent for $h_A \rightarrow *$ in the log prismatic topos of $(X, \mathcal{M})_\Delta$; here we use h for the Yoneda embedding on a log prism. Indeed, it suffices to check this on all representable sheaves, which we have just done. Thus, taking global sections one obtains

$$D_{\blacksquare}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A^{(m)}[1/I]_p^\wedge).$$

□

Lemma A.16. *Let $(A, (d))$ be an orientable transversal prism, i.e. A/d is p -torsion-free. There are functorial equivalences*

$$D_{\blacksquare}(A[1/d]_p^\wedge)^{\mathrm{dual}} \simeq \varprojlim_n \mathrm{Perf}((A/p^n)[1/d]) \simeq \mathrm{Perf}(A[1/d]_p^\wedge).$$

In the setting of Theorem A.15, these give

$$\mathrm{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \simeq \varprojlim_{[m] \in \Delta} \mathrm{Perf}(A^{(m)}[1/I]_p^\wedge).$$

Proof. This follows by taking dualizable objects in Theorem A.15 and applying [Sch26b, Theorem 14.10], noting dualizables can be detected termwise in a limit diagram. □

Lemma A.17. *Let $(\mathrm{Spf} R, \mathcal{M})$ be a framed log affine over $\mathrm{Spf} W(k)[\zeta_p]$. Let $((A^+)^{(m)}, [p]_q, \mathcal{M}^{(m)})^a$ be the degree m term of the log q -de Rham Čech nerve in $(\mathrm{Spf} R, \mathcal{M})_\Delta$, and put $A^{(m)} = (A^+)^{(m)}[1/[p]_q]_p^\wedge$. Then*

$$D_{\blacksquare}((\mathrm{Spf} R, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge) \simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A^{(m)}).$$

Moreover, the Γ action on the degree zero term gives a conservative evaluation functor to $D_{\blacksquare}(A)^{h\Gamma}$.

Proof. In Lemma A.3, the log q -de Rham prism is a valid choice of logarithmic Breuil-Kisin atlas. Theorem A.15 therefore gives the needed Čech description. The Γ action arises from log prism automorphisms, and the crystal property therefore gives the action on degree zero

evaluation. The analytic maps $A^{(s)} \rightarrow C(\Gamma^s, A)$ constructed in Lemma 3.8 identify the coface maps and show that this action is continuous (a priori the map lands in $\text{Map}(\Gamma^s, A)$). The evaluation on the degree zero term is conservative. $\square$

Corollary A.18. *On log formal schemes satisfying Assumption 3.1 over $\mathcal{O}_K$, the assignment*

$$(X, \mathcal{M}) \mapsto D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$$

is a stack for the strict-étale topology. The same holds for $\text{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Proof. Framed log affines form a basis for the strict-étale topology by Lemma 3.4. It is therefore enough to consider an affine strict-étale cover $U \rightarrow X$ by framed log affines with X framed. Give every term $U^{(n)}$ of its Čech nerve the framing induced from X . Let A_X be the logarithmic Breuil-Kisin prism associated with the framing of X . The étale lifting property identifies its pullback to each $U^{(n)}$ with the logarithmic Breuil-Kisin prism $A_{U^{(n)}}$ associated with the induced framing. Write $A_X^{(m)}$ and $A_{U^{(n)}}^{(m)}$ for the Laurent rings in degree m of their Čech nerves. They form the (augmented) bicosimplicial diagram

$$\begin{array}{ccccc} D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) & \longrightarrow & D_{\blacksquare}(A_X^{(0)}) & \rightrightarrows & D_{\blacksquare}(A_X^{(1)}) \rightrightarrows \cdots \\ \downarrow & & \downarrow & & \downarrow \\ D_{\blacksquare}((U, \mathcal{M}|_U)_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) & \longrightarrow & D_{\blacksquare}(A_U^{(0)}) & \rightrightarrows & D_{\blacksquare}(A_U^{(1)}) \rightrightarrows \cdots \\ \Downarrow & & \Downarrow & & \Downarrow \\ D_{\blacksquare}((U^{(1)}, \mathcal{M}|_{U^{(1)}})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) & \longrightarrow & D_{\blacksquare}(A_{U^{(1)}}^{(0)}) & \rightrightarrows & D_{\blacksquare}(A_{U^{(1)}}^{(1)}) \rightrightarrows \cdots \\ \Downarrow & & \Downarrow & & \Downarrow \\ \vdots & & \vdots & & \vdots \end{array}$$

For fixed n , Theorem A.15 identifies the limit of the n th row with the (solid) Laurent crystal category of $U^{(n)}$. For fixed m , compatibility with strict-étale localization identifies the m th column with the Čech nerve of a completely faithfully étale map. Such maps are completely descendable, so the limit of this column is $D_{\blacksquare}(A_X^{(m)})$. Since limits commute with limits, we obtain

$$\begin{aligned} \varprojlim_{[n] \in \Delta} D_{\blacksquare}((U^{(n)}, \mathcal{M}|_{U^{(n)}})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) &\simeq \varprojlim_{[n] \in \Delta} \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A_{U^{(n)}}^{(m)}) \\ &\simeq \varprojlim_{[m] \in \Delta} \varprojlim_{[n] \in \Delta} D_{\blacksquare}(A_{U^{(n)}}^{(m)}) \\ &\simeq \varprojlim_{[m] \in \Delta} D_{\blacksquare}(A_X^{(m)}) \\ &\simeq D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}). \end{aligned}$$

All equivalences are induced by the augmentations, so their composite is the strict-étale descent equivalence.

The descent equivalence is symmetric monoidal, and dualizable objects are detected termwise in limits. It therefore restricts to the asserted descent equivalence for $\mathbf{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$. $\square$

The canonical t -structure on $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ is defined by taking the connective objects to be the crystals whose evaluations on every prism are connective, and taking the coconnective objects to be the right orthogonal of the connective objects shifted by one. Extension of scalars preserves connective objects, so this defines an accessible t -structure.

Corollary A.19. *Let $(X, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. The canonical t -structure on $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ restricts to $\mathbf{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$.*

Proof. Using a covering by framed log affines via Lemma 3.4, we may use their associated logarithmic Breuil-Kisin prisms $(A, I, M_{\mathrm{Spf} A})^a$ from Lemma A.3 to produce a covering family (by Theorem A.15). The argument ends up being slightly technical since there is a subtlety that it is not clear the maps in the Breuil-Kisin Čech nerve are t -exact for $D_\bullet(-)$.

First we simply check that there exists a t -structure on $\mathbf{Perf}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ regarded as ordinary Laurent crystals, defined via orthogonality from the Breuil-Kisin presentation of $\mathbf{Perf}^{\leq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$: we ask for the connective objects to be those which are connective on each Čech term $A^{(n)}[1/I]_p^\wedge$. This follows by the abstract argument of [Bha22, Footnote 70], noting that we actually only need the maps $A[1/I]_p^\wedge \rightarrow A^{(n)}[1/I]_p^\wedge$ to be flat (which they are in the ordinary sense since A is Noetherian, using that (p, I) -complete flatness upgrades to ordinary flatness with a Noetherian source). By Lemma A.16 this also yields a t -structure on the dualizable objects in $D_\bullet((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ as dualizability is detected termwise in a limit.

One may check $\mathcal{E} \in \mathbf{Perf}^{\leq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ is also in $D_\bullet^{\leq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$. We must check all evaluations on log prisms are connective, as we define the canonical t -structure on $D_\bullet^{\leq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ this way by orthogonality. By construction of $\mathbf{Perf}^{\leq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ we see the evaluation on $(A \coprod_* B)[1/I]^\wedge$ is connective (as it is obtained by pullback from the value on $(A, I, M_{\mathrm{Spf} A})^a$, which by definition is connective). In Lemma A.13 we argued $(A \coprod_* B)[1/I]^\wedge$ is a descendable algebra in $D_\bullet(B[1/J]_p^\wedge)$. In particular, the base change functor $B[1/J]_p^\wedge \rightarrow (A \coprod_* B)[1/I]^\wedge$ is conservative: in this situation base change detects connectivity on perfect complexes.

One may also check that $\mathcal{E} \in \mathbf{Perf}^{\geq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ is also in $D_\bullet^{\geq 0}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge)$ for its canonical t -structure. Indeed this is defined as the right orthogonal of

$$D_\bullet^{\leq -1}((X, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/I_\Delta]_p^\wedge).$$

For $\mathcal{E}' \in D_{\blacksquare}^{\leq -1}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$, every evaluation on Čech terms $A^{(n)}[1/I]_p^{\wedge}$ lies in the ≤ -1 part of the solid t -structure. Writing $\mathcal{E}_n$ and $\mathcal{E}'_n$ for these evaluations, we get

$$\mathrm{Map}_{D_{\blacksquare}(A^{(n)}[1/I]_p^{\wedge})}(\mathcal{E}'_n, \mathcal{E}_n) = 0$$

using that $\mathcal{E}_n$ are in the ≥ 0 part (using that it is perfect here to relate ordinary and solid t -structures). But mapping spaces are computed by the limit over the Čech nerve. Thus the first claim of the corollary follows. $\square$

We denote its heart by $\mathrm{Coh}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. Corollary A.18 gives strict-étale descent for its heart.

A.4. Analytifications. We compare Breuil-Kisin presentations with the analytification of prismatic stacks.

Proposition A.20. *Let X be an affine smooth p -adic formal scheme over $\mathcal{O}_K$, and let (A, I) be an ordinary Breuil-Kisin atlas obtained from a smooth lift over $W(k)[[u]]$. The induced map*

$$\mathrm{Spa}(A, A) \longrightarrow X^{\Delta, \blacksquare}$$

is a representable proper effective epimorphism. If $A^{(n)}$ denotes the n th iterated self-coproduct in the category of prisms over X , then

$$X^{\Delta, \blacksquare} \simeq |\mathrm{Spa}(A^{(\bullet)}, A^{(\bullet)})|$$

in analytic stacks.

Proof. Choose a classical regular semiperfectoid cover of X in the infinite-root étale topology. Construction 3.7.1 and Corollary 3.6.3 of [Kip26a] show that

$$\coprod_i \mathrm{Spa}(B_i, B_i) \longrightarrow X^{\Delta, \blacksquare}$$

is an effective epimorphism; after a Zariski refinement, the B_i are orientable bounded prisms over X . It therefore suffices to test the first assertion after pullback to each $\mathrm{Spa}(B_i, B_i)$. Lemma A.6 and finite limit preservation of analytic realization $(-)^{\blacksquare}$ (Definition 6.1) identify this pullback with

$$\mathrm{Spa}(A \coprod_{*} B_i, A \coprod_{*} B_i) \longrightarrow \mathrm{Spa}(B_i, B_i),$$

which is a proper descendable cover by Lemma A.13. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; effective epimorphisms are local on the target in any ∞ -topos. This proves the first assertion. Applying Lemma A.6 iteratively with $B = A$ identifies the Čech terms with $\mathrm{Spa}(A^{(n)}, A^{(n)})$. The resulting Čech nerve realizes to $X^{\Delta, \blacksquare}$ because $\mathrm{Spa}(A, A) \rightarrow X^{\Delta, \blacksquare}$ is an effective epimorphism. $\square$

For a fine monoid P , let $\mathcal{A}_P = [\mathrm{Spf} \mathbf{Z}_p[P] / \mathrm{Spf} \mathbf{Z}_p[P^{\mathrm{gp}}]]$ be its Artin cone. Its toric atlas is $\mathrm{Spf} \mathbf{Z}_p[P] \rightarrow \mathcal{A}_P$. The universal log structure induces an étale map $\mathcal{A}_P \rightarrow \mathrm{Log}$ by [Ols03, Corollary 5.24].

Lemma A.21. *Let $(X, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, let $(A, I, M_{\mathrm{Spf} A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $P = \mathbf{N}^r$ be its framing chart. There is an orientable bounded fine log prism $(B_P, I, \mathcal{N}_P)^a$ with a strict map*

$$(\mathrm{Spf}(B_P/I), \mathcal{N}_P|_{\mathrm{Spf}(B_P/I)}) \longrightarrow (X, \mathcal{M})$$

and a commutative diagram

$$\begin{array}{ccc} \mathrm{Spf} B_P & \xrightarrow{\mathrm{fpqc}} & \mathrm{Spf} A \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P]) \\ & \searrow \rho_{X, B_P} & \downarrow \\ & & (X, \mathcal{M})^\Delta. \end{array}$$

The right vertical arrow is the natural map. After applying the functor of Definition 6.1, the induced map

$$\mathrm{Spa}(B_P, B_P) \longrightarrow \mathrm{Spa}(A, A) \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P])^\blacksquare$$

is a proper effective epimorphism.

Proof. Put

$$X_P = \mathrm{Spf}(A/I) \times_{\mathcal{A}_P} \mathrm{Spf} \mathbf{Z}_p[P].$$

We equip X_P with the pullback log structure. Let $T_i \in A$ be the images of the framing chart generators. This chart trivializes the torus torsor $X_P \rightarrow \mathrm{Spf}(A/I)$, giving

$$X_P \simeq \mathrm{Spf}((A/I)[u_1^{\pm 1}, \dots, u_r^{\pm 1}]_p^\wedge).$$

Choose the units u_i so that the projection $X_P \rightarrow \mathrm{Spf} \mathbf{Z}_p[P]$ pulls back the monomial corresponding to $e_i \in P$ to $u_i T_i$. Put $T_P = A \widehat{\otimes}_{\mathbf{Z}_p} \mathbf{Z}_p[P]$, with toric coordinates x_i and $\delta(x_i) = 0$. Relative base change and finite limit preservation of the relative prismaticization yield

$$\mathrm{Spf} A \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P]) \simeq (X_P/A)^\Delta \times_{(\mathrm{Spf}(T_P/I)/A)^\Delta} \mathrm{Spf} T_P.$$

Here we use [BL22, Construction 3.11, Remark 8.9, Remark 8.11]. This can be further simplified to $(X_P)^\Delta \times_{(\mathrm{Spf}(T_P/I))^\Delta} \mathrm{Spf} T_P$.

Set

$$C_P = A[u_1^{\pm 1}, \dots, u_r^{\pm 1}]_{(p, I)}^\wedge,$$

with $\varphi(u_i) = u_i^p$. There is an fpqc epimorphism $\mathrm{Spf} C_P \rightarrow X_P^\Delta$; upon base change, we obtain

$$\mathrm{Spf} C_P \times_{(\mathrm{Spf}(T_P/I))^\Delta} \mathrm{Spf} T_P \rightarrow \mathrm{Spf} A \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P])$$

The ordinary case of Lemma A.6 identifies the source with $\mathrm{Spf} B_P$, where B_P is an orientable bounded prism. Thus we obtain the desired fpqc epimorphism by base change. By finite limit compatibility of $(-)^{\blacksquare}$ from Definition 6.1 and Proposition A.20 we similarly obtain that

$$\mathrm{Spa}(B_P, B_P) \longrightarrow \mathrm{Spa}(A, A) \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P])^\blacksquare$$

is a proper effective epimorphism.

We give B_P the log structure $\mathcal{N}_P$ pulled back from T_P , with chart $e_i \mapsto x_i$ and $\delta_{\log}(e_i) = 0$; this allows us to define ρ_{X, B_P} , as one may check that modulo I the map to $(X, \mathcal{M})$ is strict.

To prove commutativity of the triangle in the lemma statement, consider the following diagram.

$$\begin{array}{ccccc} \mathrm{Spf}(B_P/I) & \longrightarrow & X_P & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spf} B_P & \longrightarrow & \mathrm{Spf} T_P & \longrightarrow & \mathcal{A}_P. \end{array}$$

Functoriality of the relative prismatization in δ -pairs uses the outer square to induce ρ_{X, B_P} (similar to Lemma A.6), and the relative prismatization $(X_P / \mathrm{Spf} T_P)^\Delta$ identifies with $\mathrm{Spf} A \times_{\mathcal{A}_P} (\mathbf{Z}_p^\Delta \times \mathrm{Spf} \mathbf{Z}_p[P])$. Thus commutativity of this diagram shows the factorization in the triangle. $\square$

Proposition A.22. *Let $(X, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, and let $(A, I, M_{\mathrm{Spf} A})^a$ be the logarithmic Breuil-Kisin prism associated with its framing. The induced map*

$$\mathrm{Spa}(A, A) \longrightarrow (X, \mathcal{M})^{\Delta, \blacksquare}$$

is a representable proper effective epimorphism. If $A^{(n)}$ denotes the n th iterated self-coproduct in $(X, \mathcal{M})_{\Delta}^{\mathrm{op}}$, then

$$(X, \mathcal{M})^{\Delta, \blacksquare} \simeq |\mathrm{Spa}(A^{(\bullet)}, A^{(\bullet)})|$$

in analytic stacks.

Proof. Let $P = \mathbf{N}^r$ be the framing chart; its reduction is a chart for $\mathcal{M}$ on $X = \mathrm{Spf} A/I$. This induces a map $\mathrm{Spf} A \rightarrow \mathcal{A}_P^\Delta \rightarrow \mathrm{Log}^\Delta$. By Definition A.1, the fiber product $\mathrm{Spf} A \times_{X^\Delta} (X, \mathcal{M})^\Delta$ is $\mathrm{Spf} A \times_{\mathrm{Log}^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Log})$. Since $\mathcal{A}_P \rightarrow \mathrm{Log}$ is étale, [BL22, Remark 3.9 and Construction 3.11] gives

$$\mathrm{Spf} A \times_{\mathrm{Log}^\Delta} (\mathbf{Z}_p^\Delta \times \mathrm{Log}) \simeq \mathrm{Spf} A \times_{\mathcal{A}_P^\Delta} (\mathbf{Z}_p^\Delta \times \mathcal{A}_P).$$

The toric atlas $\mathrm{Spf} \mathbf{Z}_P[P] \rightarrow \mathcal{A}_P$ and Lemma A.21 give an effective epimorphism

$$\mathrm{Spa}(B_P, B_P) \longrightarrow \mathrm{Spa}(A, A) \times_{X^{\Delta, \blacksquare}} (X, \mathcal{M})^{\Delta, \blacksquare}.$$

The projection from the fiber product on the right to $(X, \mathcal{M})^{\Delta, \blacksquare}$ is the base change of $\mathrm{Spa}(A, A) \rightarrow X^{\Delta, \blacksquare}$, so it is an effective epimorphism by Proposition A.20. Thus the analytic realization of ρ_{X, B_P} is an effective epimorphism. Lemma A.6 identifies the pullback of $\mathrm{Spa}(A, A)$ along this map with

$$\mathrm{Spa}(A \coprod_{*} B_P, A \coprod_{*} B_P) \longrightarrow \mathrm{Spa}(B_P, B_P).$$

This is a proper descendable cover by Lemma A.13. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; the covering property is local in any ∞ -topos. This proves the first assertion. Lemma A.6 and finite limit preservation of $(-)^{\blacksquare}$ (as in Definition 6.1) identify the n th Čech term with $\mathrm{Spa}(A^{(n)}, A^{(n)})$. Since the map from $\mathrm{Spa}(A, A)$ is an effective epimorphism, its Čech nerve realizes to $(X, \mathcal{M})^{\Delta, \blacksquare}$. $\square$

REFERENCES

- [AB26] Abhinandan and Denis Benois, *Log-crystalline representations and (φ, Γ) -modules*, Preprint, arXiv:2609.11266, 2026.
- [AI08] Fabrizio Andreatta and Adrian Iovita, *Global applications of relative (φ, Γ) -Modules I*, Représentations p -adiques de groupes p -adiques I: représentations galoisiennes et (φ, Γ) -modules (Laurent Berger, Christophe Breuil, and Pierre Colmez, eds.), Astérisque, no. 319, Société mathématique de France, 2008, pp. 339–420.
- [AKN26] Benjamin Antieau, Achim Krause, and Thomas Nikolaus, *Prismatic cohomology relative to δ -rings*, Preprint, arXiv:2310.12770, 2026.
- [AM24] Johannes Anschütz and Lucas Mann, *Descent for solid quasi-coherent sheaves on perfectoid spaces*, Preprint, arXiv:2403.01951v2, 2024.
- [ATJLS10] Leovigildo Alonso Tarrío, Ana Jeremías López, and Manuel Saorín, *Classifying compactly generated t -structures on the derived category of a Noetherian ring*, Journal of Algebra **324** (2010), no. 3, 313–346.
- [BC09] Olivier Brinon and Brian Conrad, *CMI summer school notes on p -adic hodge theory*, Preliminary version, June 24, 2009, 2009.
- [BGR84] Siegfried Bosch, Ulrich Güntzer, and Reinhold Remmert, *Non-archimedean analysis: A systematic approach to rigid analytic geometry*, Grundlehren der mathematischen Wissenschaften, vol. 261, Springer-Verlag, Berlin, 1984.
- [BH22] Bhargav Bhatt and David Hansen, *The six functors for Zariski-constructible sheaves in rigid geometry*, Compositio Mathematica **158** (2022), no. 2, 437–482.
- [Bha20] Bhargav Bhatt, *Cohen–macaulayness of absolute integral closures*, Preprint, arXiv:2008.08070, 2020.
- [Bha22] ———, *Prismatic F -gauges*, <https://www.math.ias.edu/~bhatt/teaching/mat549f22/lectures.pdf>, 2022.
- [Bha25] ———, *Aspects of p -adic Hodge theory*, Lecture notes, Princeton University, Fall 2025, 2025.
- [BL19] Bhargav Bhatt and Jacob Lurie, *A Riemann–Hilbert correspondence in positive characteristic*, Cambridge Journal of Mathematics **7** (2019), no. 1–2, 71–217.
- [BL22] ———, *The prismatization of p -adic formal schemes*, Preprint, arXiv:2201.06124, 2022.
- [Bli04] Manuel Blickle, *The intersection homology D -module in finite characteristic*, Mathematische Annalen **328** (2004), no. 3, 425–450.
- [Bli08] ———, *Minimal γ -sheaves*, Algebra & Number Theory **2** (2008), no. 3, 347–368.
- [Bos23] Guido Bosco, *On the p -adic pro-étale cohomology of Drinfeld symmetric spaces*, Preprint, arXiv:2110.10683, 2023.
- [BR25] Laurent Berger and Sandra Rozensztajn, *Super-Hölder vectors and the field of norms*, Algebra & Number Theory **19** (2025), no. 1, 195–211.
- [BS15] Bhargav Bhatt and Peter Scholze, *The pro-étale topology for schemes*, Astérisque (2015), no. 369, 99–201.
- [BS17] ———, *Projectivity of the Witt vector affine Grassmannian*, Inventiones Mathematicae **209** (2017), no. 2, 329–423.
- [BS22] ———, *Prisms and prismatic cohomology*, Annals of Mathematics **196** (2022), no. 3, 1135–1275.
- [BS23] ———, *Prismatic F -crystals and crystalline Galois representations*, Cambridge Journal of Mathematics **11** (2023), no. 2, 507–562.
- [CS23] Dustin Clausen and Peter Scholze, *Analytic stacks*, Lecture course, Institut des Hautes Études Scientifiques and Max Planck Institute for Mathematics, 2023.
- [DLLZ23] Hansheng Diao, Kai-Wen Lan, Ruochuan Liu, and Xinwen Zhu, *Logarithmic adic spaces: some foundational results, p -adic Hodge Theory, Singular Varieties, and Non-Abelian Aspects* (Bhargav Bhatt and Martin Olsson, eds.), Simons Symposia, Springer, Cham, 2023, pp. 65–182.
- [DLMS24] Heng Du, Tong Liu, Yong Suk Moon, and Koji Shimizu, *Log prismatic F -crystals and purity*, Preprint, arXiv:2404.19603v2, 2024.
- [EK04a] Matthew Emerton and Mark Kisin, *An introduction to the Riemann–Hilbert correspondence for unit F -crystals*, Geometric Aspects of Dwork Theory (Alan Adolphson, Francesco Baldassarri, Pierre Berthelot, Nicholas Katz, and François Loeser, eds.), Walter de Gruyter, Berlin, 2004, pp. 677–700.

- [EK04b] ———, *The Riemann–Hilbert correspondence for unit F -crystals*, *Astérisque*, no. 293, Société mathématique de France, 2004. MR 2071510
- [Fon90] Jean-Marc Fontaine, *Représentations p -adiques des corps locaux. I*, The Grothendieck Festschrift, Vol. II, Progress in Mathematics, vol. 87, Birkhäuser Boston, Boston, MA, 1990, pp. 249–309.
- [Gab04] Ofer Gabber, *Notes on some t -structures*, Geometric Aspects of Dwork Theory (Alan Adolphson, Francesco Baldassarri, Pierre Berthelot, Nicholas Katz, and François Loeser, eds.), vol. II, Walter de Gruyter, Berlin, 2004, pp. 711–734.
- [GL25] Haoyang Guo and Shizhang Li, *Frobenius height of prismatic cohomology with coefficients*, Preprint, arXiv:2309.06663v2, 2025.
- [GLSQ22] Michel Gros, Bernard Le Stum, and Adolfo Quirós, *Twisted differential operators of negative level and prismatic crystals*, *Tunisian Journal of Mathematics* **4** (2022), no. 1, 19–53.
- [GLSQ23] ———, *Absolute calculus and prismatic crystals on cyclotomic rings*, Preprint, arXiv:2310.13790, 2023.
- [Hau24] Maximilian Hauck, *A stacky approach to p -adic Hodge theory*, Preprint, arXiv:2409.10557, 2024.
- [HM24] Claudius Heyer and Lucas Mann, *6-Functor Formalisms and Smooth Representations*, Preprint, arXiv:2410.13038, 2024.
- [HRS23] Tamir Hemo, Timo Richarz, and Jakob Scholbach, *Constructible sheaves on schemes*, *Advances in Mathematics* **429** (2023), 109179.
- [IKY26] Kentaro Inoue, Teruhisa Koshikawa, and Zijian Yao, *Corrigendum to “Logarithmic prismatic cohomology II”*, 2026, Corrigendum, available at <https://researchmap.jp/7000017226/misc/53305261>.
- [Ino25] Kentaro Inoue, *Log prismatic F -crystals and realization functors*, Preprint, arXiv:2505.01084, 2025.
- [Kat73] Nicholas M. Katz, *p -adic properties of modular schemes and modular forms*, *Modular Functions of One Variable III* (Willem Kuyk and Jean-Pierre Serre, eds.), *Lecture Notes in Mathematics*, vol. 350, Springer, Berlin, 1973, pp. 69–190. MR 0447119
- [Kip26a] Niklas Kipp, *A six-functor formalism for syntomic cohomology*, Preprint, arXiv:2608.15744, 2026.
- [Kip26b] ———, *A six-functor formalism for syntomic cohomology of p -adic formal schemes*, Ph.D. thesis, Universität Regensburg, 2026.
- [KL15] Kiran S. Kedlaya and Ruochuan Liu, *Relative p -adic Hodge theory: Foundations*, *Astérisque*, no. 371, Société mathématique de France, 2015.
- [KL16] ———, *Relative p -adic Hodge theory, II: Imperfect period rings*, Preprint, arXiv:1602.06899, 2016.
- [Kos22] Teruhisa Koshikawa, *Logarithmic prismatic cohomology I*, Preprint, arXiv:2007.14037v3, 2022.
- [KY25] Teruhisa Koshikawa and Zijian Yao, *Logarithmic prismatic cohomology II*, *Advances in Mathematics* **479** (2025), 110446.
- [Man22] Lucas Mann, *A p -adic 6-functor formalism in rigid-analytic geometry*, Preprint, arXiv:2206.02022, 2022.
- [Mao24] Zhouhang Mao, *Revisiting derived crystalline cohomology*, *Bulletin de la Société Mathématique de France* **152** (2024), no. 4, 659–784.
- [Mat16] Akhil Mathew, *The Galois group of a stable homotopy theory*, *Advances in Mathematics* **291** (2016), 403–541.
- [Mat22] ———, *Faithfully flat descent of almost perfect complexes in rigid geometry*, *Journal of Pure and Applied Algebra* **226** (2022), no. 5, 106938.
- [Mat23] ———, *The mod- p Riemann–Hilbert correspondence and the perfect site*, *Tunisian Journal of Mathematics* **5** (2023), no. 2, 369–403.
- [Mik25] Yutaro Mikami, *Finiteness and duality of cohomology of (φ, Γ) -modules and the six-functor formalism of locally analytic representations*, Preprint, arXiv:2504.01780, 2025.
- [Mik26] ———, *(φ, Γ) -modules over relatively discrete algebras*, Preprint, arXiv:2409.14145v3, 2026.
- [Ols03] Martin C. Olsson, *Logarithmic geometry and algebraic stacks*, *Annales scientifiques de l’École Normale Supérieure* **36** (2003), no. 5, 747–791.
- [Ols23] ———, *Stacks and log prismatic cohomology*, Video lecture, Workshop on p -adic Arithmetic Geometry, Institute for Advanced Study, <https://www.ias.edu/video/stacks-and-log-prismatic-cohomology>, 2023, Lecture delivered November 13, 2023.
- [Pen25] Dylan Pentland, *Syntomification and crystalline local systems*, Preprint, arXiv:2510.16961, 2025.

- [Por25] Gal Porat, *Locally analytic vectors and decompletion in mixed characteristic*, Forum of Mathematics, Sigma **13** (2025), 1–26.
- [RC24] Juan Esteban Rodríguez Camargo, *Notes on solid geometry*, Seminar notes, 2024.
- [Sch26a] Peter Scholze, *Étale cohomology of diamonds*, Astérisque **466** (2026), 439–674.
- [Sch26b] ———, *Lectures on analytic geometry*, Lecture notes, arXiv:2605.03655, 2026.
- [Sta26] Stacks Project Authors, *Stacks project*, <https://stacks.math.columbia.edu>, 2026.
- [SV11] Peter Schneider and Marie-France Vignéras, *A functor from smooth $\mathfrak{o}$ -torsion representations to (φ, Γ) -modules*, On Certain L -Functions, Clay Mathematics Proceedings, vol. 13, American Mathematical Society, Providence, RI, 2011, pp. 525–601.
- [Tsu99] Takeshi Tsuji, *Poincaré duality for logarithmic crystalline cohomology*, Compositio Mathematica **118** (1999), no. 1, 11–41.
- [Wat26] Takumi Watanabe, *A remark on an integral structure of the imperfect coefficient ring of (φ, Γ) -modules*, Preprint, arXiv:2604.17559; Appendix 2 by Dylan Pentland, 2026.
- [Wig73] David Wigner, *Algebraic cohomology of topological groups*, Transactions of the American Mathematical Society **178** (1973), 83–93.

Email address: `dpentland@g.harvard.edu`